# CS1231S · Exam Condensed Addendum

Lectures 7–13, Tutorials 6–11 and the AY25/26 final: functions, induction and recursion, cardinality, counting and probability, graphs and trees. This is the part of the Exam Condensed that the Midterm Condensed does not contain; use it beside a printed Midterm Condensed.

**Numbering:** sections keep their Exam Condensed numbers (§26–52 for content; §56, §58, §60, §62 and §65 for the playbook pieces). A reference to §1–25 points to the same section of the Midterm Condensed. Appendix A is unchanged and is not repeated. **Final exam format (AY25/26):** 2 hours, open book, 100 marks: 23 MCQ (46) + 3 structured questions (54).

**Conventions:** as in the Midterm Condensed. $\mathbb{N} = \{0, 1, 2, ...\}$. Blue boxes define terms; grey boxes contain proofs and worked examples; green boxes state results; amber boxes solve tutorial and past-paper questions. Theorem labels follow the lecture slides (L = lecture, s = slide).

## Part VI · Functions

### 26 · Functions, images and preimages

DEFINITION **Function (L7 s7)**

A **function** $f$ from a set $X$ to a set $Y$, written $f : X \to Y$, is a relation $f \subseteq X \times Y$ satisfying

**(F1)** $\forall x \in X\, \exists y \in Y\, ((x, y) \in f)$; every input has an output.

**(F2)** $\forall x \in X\, \forall y_1, y_2 \in Y\, (((x, y_1) \in f \wedge (x, y_2) \in f) \to y_1 = y_2)$; the output is unique.

Equivalently, $\forall x \in X\, \exists! y \in Y\, ((x, y) \in f)$. On an arrow diagram: every element of $X$ has exactly one arrow leaving it.

DEFINITION **Image, preimage (L7 s11) · Setwise image and preimage (L7 s12) · Domain, co-domain, range (L7 s16)**

Write $f(x) = y$ iff $(x, y) \in f$; $f$ **maps** $x$ to $y$ ($f : x \mapsto y$); $x$ is the **argument**, $f(x)$ the **image** of $x$, and $x$ is **a preimage** of $y$.

For $A \subseteq X$ and $B \subseteq Y$:

$$f(A) = \{f(x) : x \in A\} \quad \text{(setwise image)}, \qquad f^{-1}(B) = \{x \in X : f(x) \in B\} \quad \text{(setwise preimage)}.$$

$X$ is the **domain**, $Y$ the **co-domain**, and the **range** is $f(X) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$. Always range $\subseteq$ co-domain.

TRAP **Two meanings of $f^{-1}$ (L7 s12, s15)**

- $f^{-1}(B)$ with $B$ a **subset** of the co-domain is the setwise preimage. It exists for every function and is a set.
- $f^{-1}(y)$ with $y$ an **element** of the co-domain is the inverse function (§28). It exists only when $f$ is a bijection.
- For the preimage of a single element write $f^{-1}(\{b\})$, never $f^{-1}(b)$, unless $f$ is known to be bijective.

WORKED EXAMPLE **L7 Ex. 1, 4, 6 · T6 D1, Q1, Q2**

- **Not a function.** "$f(x)$ is the real $y$ with $x^2 + y^2 = 1$" on $\mathbb{R}$: $x = 2$ has no $y$ (F1 fails); $x = 0$ has $y = \pm 1$ (F2 fails).
- **T6 D1.** $f(z) = 1$ if $2 \mid z$, 2 if $3 \mid z$: not a function ($f(6)$ has two values, $f(1)$ has none). $f : \mathbb{R} \to \mathbb{Z}$, $f(x) = 2x$: not a function ($f(0.3) = 0.6 \notin \mathbb{Z}$). $f : \mathbb{Z} \to \mathbb{R}$, $f(x) = 2x$: a function.
- **T6 Q1** on $\mathbb{N}$. $xR_1y \leftrightarrow x^2 = y^2$: a function (F1: take $y = x$; F2: $y_1^2 = y_2^2$ with $y_1, y_2 \geq 0$ gives $y_1 = y_2$). $xR_2y \leftrightarrow y \mid x$: not a function ($6R_2 2$ and $6R_2 3$). $xR_3y \leftrightarrow y = x + 1$: a function.
- **Ex. 4.** $g : \mathbb{Z} \to \mathbb{Z}$, $g(x) = x^2$: $g(\{-1, 0, 1\}) = \{0, 1\}$ and $g^{-1}(\{0, 1, 2\}) = \{-1, 0, 1\}$.
- **Ex. 6.** $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 2x + 1$: domain $\mathbb{Z}$, co-domain $\mathbb{Z}$, range = the odd integers.
- **T6 Q2.** $\text{len} : \{s, u\}^* \to \mathbb{Z}_{\geq 0}$: $\text{len}(\{\varepsilon, ss, uu, ssss\}) = \{0, 2, 4\}$; $\text{len}^{-1}(\{3\})$ is the set of all 8 strings of length 3; the inverse function $\text{len}^{-1}$ does not exist, since $\text{len}(s) = \text{len}(u)$.

PROOF **T6 Q9 · $X \subseteq f^{-1}(f(X))$ and $f(f^{-1}(Y)) \subseteq Y$; the reverse inclusions can fail**

Let $f : A \to B$, $X \subseteq A$, $Y \subseteq B$.

1. ($X \subseteq f^{-1}(f(X))$) Let $x \in X$. Then $f(x) \in f(X)$ *(definition of $f(X)$)*, so $x \in f^{-1}(f(X))$ *(definition of setwise preimage)*.
2. ($f(f^{-1}(Y)) \subseteq Y$) Let $y \in f(f^{-1}(Y))$. Then $y = f(x)$ for some $x \in f^{-1}(Y)$ *(definition of setwise image)*, and $f(x) \in Y$ *(definition of setwise preimage)*. So $y \in Y$. ■

**Counterexamples for the reverse.** $f:\{-1,1\}\to\{0\}$, $X=\{1\}$: $f^{-1}(f(X))=\{-1,1\}\not\subseteq X$.
$f:\{0\}\to\{-1,1\}$ with $f(0)=1$, $Y=\{-1\}$: $f(f^{-1}(Y))=\varnothing$, so $Y\not\subseteq f(f^{-1}(Y))$.
Equality holds in the first when $f$ is injective and in the second when $f$ is surjective.

## 27 · Sequences, strings and function equality

DEFINITION **Sequence (L7 s18) · Fibonacci sequence (L7 s19) · String (L7 s20)**
A **sequence** $a_0, a_1, a_2, \ldots$ is represented by a function $a$ with domain $\mathbb{Z}_{\geq 0}$ and $a(n)=a_n$; any function with domain $\mathbb{Z}_{\geq m}$ represents a sequence.
**Fibonacci:** $F_0=0$, $F_1=1$, $F_{n+2}=F_{n+1}+F_n$: $0,1,1,2,3,5,8,13,21,34,55,\ldots$
A **string** (word) over a set $A$ is an expression $a_0a_1\cdots a_{l-1}$ with $l\in\mathbb{Z}_{\geq 0}$ and each $a_i\in A$; $l$ is its **length**; the **empty string** $\varepsilon$ has length 0; $A^*$ is the set of all strings over $A$. Two sequences (strings of the same length) are equal iff they agree at every index.

DEFINITION **Floor, ceiling (T6 D3 · AY23/24 final)**
For a real number $x$: the **floor** $\lfloor x\rfloor$ is the unique integer $n$ with $n\leq x<n+1$, i.e. $\lfloor x\rfloor = \max\{m\in\mathbb{Z}: m\leq x\}$; the **ceiling** $\lceil x\rceil$ is the unique integer $n$ with $n-1<x\leq n$, i.e. $\lceil x\rceil = \min\{m\in\mathbb{Z}: m\geq x\}$.
$\lfloor 2.7\rfloor=2$, $\lceil 2.7\rceil=3$, $\lfloor -2.7\rfloor=-3$, $\lceil -2.7\rceil=-2$, $\lfloor 5\rfloor=\lceil 5\rceil=5$. Read off the definitions:
- $\lfloor x\rfloor\leq x\leq\lceil x\rceil$; $x\in\mathbb{Z}\Leftrightarrow\lfloor x\rfloor=x\Leftrightarrow\lceil x\rceil=x$; $x\notin\mathbb{Z}\Rightarrow\lceil x\rceil=\lfloor x\rfloor+1$.
- For $n\in\mathbb{Z}$: $\lfloor n/2\rfloor=n/2$ if $n$ is even and $(n-1)/2$ if $n$ is odd; $\lceil n/2\rceil=n/2$ if $n$ is even and $(n+1)/2$ if $n$ is odd.

THEOREM **Theorem 7.1.1 · Function equality (L7 s22)**
$f:A\to B$ and $g:C\to D$ are equal iff (i) $A=C$ and $B=D$, and (ii) $f(x)=g(x)$ for all $x\in A$.

WORKED EXAMPLE **L7 Ex. 11–13 · T6 D3**
- On $X=\{0,1,2\}$, $f(x)=(x^2+x+1)\bmod 3$ and $g(x)=(x+2)^2\bmod 3$ both give $1,0,1$, so $f=g$.
- $f,g:\{0,2\}\to\mathbb{Z}$ with $f(x)=2x$, $g(x)=x^2$: equal (same domain, co-domain, values $0,4$).
- $f:\mathbb{Z}\to\mathbb{Z}$ and $g:\mathbb{Z}\to\mathbb{Q}$ with $f(x)=x^3=g(x)$: **not** equal, since the co-domains differ.
- **T6 D3.** $f,g:\mathbb{Q}\to\mathbb{Q}$, $f(x)=\lfloor x\rfloor+1$, $g(x)=\lceil x\rceil$: both have range $\mathbb{Z}$, but $f(0)=1\neq 0=g(0)$, so $f\neq g$ (they agree exactly off the integers).

## 28 · Injections, surjections, bijections, inverses

DEFINITION **Injection (L7 s25) · Surjection (L7 s27) · Bijection (L7 s29)**
Let $f:X\to Y$.
- $f$ is **injective** (one-to-one) iff $\forall x_1,x_2\in X\,(f(x_1)=f(x_2)\Rightarrow x_1=x_2)$. Contrapositive: $x_1\neq x_2\Rightarrow f(x_1)\neq f(x_2)$.
- $f$ is **surjective** (onto) iff $\forall y\in Y\,\exists x\in X\,(y=f(x))$: range = co-domain.
- $f$ is **bijective** (a one-to-one correspondence) iff it is injective and surjective: $\forall y\in Y\,\exists! x\in X\,(y=f(x))$.

Arrows **into** each element of the co-domain: at most one (injective), at least one (surjective), exactly one (bijective).

METHOD **Negations (to disprove)**
- **Not injective:** $\exists x_1,x_2\in X\,(f(x_1)=f(x_2)\wedge x_1\neq x_2)$. Name two inputs.
- **Not surjective:** $\exists y\in Y\,\forall x\in X\,(y\neq f(x))$. Name one $y$ and show no $x$ reaches it.

METHOD **Proof templates**
- **Injective:** "Let $x_1,x_2\in X$ with $f(x_1)=f(x_2)$" ... "so $x_1=x_2$."
- **Surjective:** "Take any $y\in Y$. Let $x=\ldots$" (solve $y=f(x)$ on scratch paper). Check $x\in X$ and $f(x)=y$.
- **Bijective:** both, or exhibit an inverse (Theorem 7.2.3).

PROOF **L7 Ex. 14, 16 · $f:\mathbb{Q}\to\mathbb{Q}$, $f(x)=3x+1$ is a bijection** **Ex. 15, 17 · $g:\mathbb{Z}\to\mathbb{Z}$, $g(x)=x^2$ is neither**
1. (Injective) Let $x_1,x_2\in\mathbb{Q}$ with $f(x_1)=f(x_2)$. Then $3x_1+1=3x_2+1$, so $x_1=x_2$ *(basic algebra)*.

2. (Surjective) Take any $y \in \mathbb{Q}$. Let $x = (y-1)/3$. Then $x \in \mathbb{Q}$ *(closure of $\mathbb{Q}$)* and $f(x) = 3 \cdot \frac{y-1}{3} + 1 = y$. ■

$g$ is not injective: $g(1) = 1 = g(-1)$ but $1 \neq -1$. $g$ is not surjective: $g(x) = x^2 \geq 0 > -1$ for all $x \in \mathbb{Z}$, so $-1$ has no preimage.

**The domain matters:** $x \mapsto 3x+1$ on $\mathbb{Z}$ is injective but not surjective ($y = 0$ would need $x = -1/3 \notin \mathbb{Z}$).

WORKED EXAMPLE **T6 Q4 · Classify four functions**

**(a)** $f : \mathbb{Q} \to \mathbb{Q}$, $x \mapsto 12x + 31$: bijective, with $f^{-1}(y) = (y-31)/12$, because $y = 12x + 31 \Leftrightarrow x = (y-31)/12$ for all $x, y \in \mathbb{Q}$.

**(b)** $g : \text{Bool}^2 \to \text{Bool}$, $(p, q) \mapsto p \wedge \sim q$: not injective ($g(F,T) = F = g(F,F)$); surjective ($g(T,F) = T$).

**(c)** $h : \text{Bool}^2 \to \text{Bool}^2$, $(p, q) \mapsto (p \wedge q, p \vee q)$: not injective ($h(T,F) = (F,T) = h(F,T)$); not surjective: if $p \wedge q = T$ then $p \vee q = T$, so $(T, F)$ is not in the range.

**(d)** $k : \mathbb{Z} \to \mathbb{Z}$, $k(x) = x$ if $x$ is even, $2x - 1$ if $x$ is odd.

1. $k$ preserves parity: $x$ even gives $k(x) = x$ even; $x$ odd gives $k(x) = 2(x-1) + 1$ odd. So $x$ is even iff $k(x)$ is even.
2. (Injective) Let $k(x_1) = k(x_2)$. If this value is even, both $x_i$ are even and $x_1 = k(x_1) = k(x_2) = x_2$. If it is odd, both are odd and $2x_1 - 1 = 2x_2 - 1$, so $x_1 = x_2$.
3. (Not surjective) Suppose $k(x) = 3$. Then $x$ is odd *(line 1)*, so $2x - 1 = 3$ and $x = 2$, which is even: contradiction. So 3 has no preimage. ■

DEFINITION **Inverse function (L7 s31)**

Let $f : X \to Y$. A function $g : Y \to X$ is an **inverse** of $f$ iff

$$\forall x \in X \, \forall y \in Y \, (y = f(x) \Leftrightarrow x = g(y)).$$

The inverse, when it exists, is unique and is written $f^{-1}$. Example: $f(x) = 3x + 1$ on $\mathbb{Q}$ has $f^{-1}(y) = (y-1)/3$.

PROOF **Proposition · Uniqueness of inverses (L7 s34)** **Theorem 7.2.3 · $f$ is bijective iff $f$ has an inverse; then $f^{-1}$ is a bijection (L7 s35–36)**

**Uniqueness.** If $g_1, g_2 : Y \to X$ are inverses of $f$, then for all $x \in X$, $y \in Y$: $x = g_1(y) \Leftrightarrow y = f(x) \Leftrightarrow x = g_2(y)$. So $g_1 = g_2$ *(Theorem 7.1.1)*.

**Theorem 7.2.3.**

1. ("if") Suppose $f$ has an inverse $g : Y \to X$.
   1.1. (Injective) Let $f(x_1) = f(x_2) = y$. Then $x_1 = g(y)$ and $x_2 = g(y)$ *($g$ is an inverse)*, so $x_1 = x_2$.
   1.2. (Surjective) Let $y \in Y$ and $x = g(y)$. Then $y = f(x)$ *($g$ is an inverse)*.
2. ("only if") Suppose $f$ is bijective. Then $\forall y \in Y \, \exists! x \in X \, (y = f(x))$ *(definition of bijection)*. Define $g(y)$ to be that unique $x$. Then $g$ is well defined and $y = f(x) \Leftrightarrow x = g(y)$, so $g$ is an inverse of $f$.
3. ($f^{-1}$ is a bijection) $f$ is an inverse of $f^{-1}$, since the defining condition is symmetric in $f$ and $g$. So $f^{-1}$ is bijective by line 1. ■

THEOREM **T6 Q5 (AY22/23 S2 exam) · Left and right inverses**

$g : B \to A$ is a **left inverse** of $f : A \to B$ iff $g(f(a)) = a$ for all $a \in A$; $h : B \to A$ is a **right inverse** iff $f(h(b)) = b$ for all $b \in B$.

***f* is injective iff it has a left inverse; *f* is surjective iff it has a right inverse** (for $A \neq \varnothing$). Neither kind of inverse implies the other. In T6 Q4: left inverses for $f, k$; right inverses for $f, g$.

## 29 · Composition of functions

DEFINITION **Composition (L7 s39) · Identity function (L7 s41)**

For $f : X \to Y$ and $g : Y \to Z$, the **composition** $g \circ f : X \to Z$ is defined by $(g \circ f)(x) = g(f(x))$ for all $x \in X$. ***f* is applied first.**

The **identity function** on $X$ is $\text{id}_X : X \to X$ with $\text{id}_X(x) = x$.

THEOREM **Theorems on composition (L7 s44–52)**

- **7.3.1.** For $f : X \to Y$: $f \circ \text{id}_X = f$ and $\text{id}_Y \circ f = f$.
- **7.3.2.** If $f : X \to Y$ is a bijection: $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$.
- **Associativity.** For $f : A \to B$, $g : B \to C$, $h : C \to D$: $(h \circ g) \circ f = h \circ (g \circ f)$.
- **Not commutative.** $f(x) = 3x$, $g(x) = x + 1$ on $\mathbb{Z}$: $(g \circ f)(0) = 1 \neq 3 = (f \circ g)(0)$.
- **7.3.3.** $f, g$ injective $\Rightarrow g \circ f$ injective. **7.3.4.** $f, g$ surjective $\Rightarrow g \circ f$ surjective. Hence bijections compose to a bijection.

PROOF **Associativity (L7 s47) · Theorem 7.3.3 (L7 s50) · Theorem 7.3.4 (L7 s52)**
**Associativity.** Both sides have domain $A$ and co-domain $D$, and for every $x \in A$: $((h \circ g) \circ f)(x)$ $= (h \circ g)(f(x)) = h(g(f(x))) = h((g \circ f)(x)) = (h \circ (g \circ f))(x)$. Equal by Theorem 7.1.1.
**7.3.3.**
1. Let $f : X \to Y$ and $g : Y \to Z$ be injective, and let $x_1, x_2 \in X$ with $(g \circ f)(x_1) = (g \circ f)(x_2)$.
2. Then $g(f(x_1)) = g(f(x_2))$ *(definition of composition)*.
3. So $f(x_1) = f(x_2)$ *($g$ is injective)*, and then $x_1 = x_2$ *($f$ is injective)*. ■

**7.3.4.**
1. Let $f : X \to Y$ and $g : Y \to Z$ be surjective, and let $z \in Z$.
2. There is $y \in Y$ with $g(y) = z$ *($g$ is surjective)*, and $x \in X$ with $f(x) = y$ *($f$ is surjective)*.
3. Then $(g \circ f)(x) = g(f(x)) = g(y) = z$. ■

TRAP **T6 D4 · A wrong surjectivity proof**
"$\forall y \exists x\, f(x) = y$, and $\forall z \exists y\, g(y) = z$, so $(g \circ f)(x) = g(y) = z$." The $x$, $y$, $z$ in the last line were never fixed: the proof must **start** with an arbitrary $z \in Z$, **then** obtain $y$ from $g$, **then** obtain $x$ from $f$ for that particular $y$.

PROOF **T6 Q3 · For bijections $f : A \to B$, $g : B \to C$: $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$**
1. $f^{-1}, g^{-1}$ are bijections *(Theorem 7.2.3)* $g \circ f$ and $f^{-1} \circ g^{-1}$ are bijections *(Theorems 7.3.3, 7.3.4)* so $(g \circ f)^{-1}$ exists.
2. Both $(g \circ f)^{-1}$ and $f^{-1} \circ g^{-1}$ have domain $C$ and co-domain $A$.
3. Let $z \in C$. Let $y = g^{-1}(z)$, so $z = g(y)$; let $x = f^{-1}(y)$, so $y = f(x)$ *(definition of inverse)*.
4. Then $(g \circ f)(x) = g(f(x)) = g(y) = z$, so $(g \circ f)^{-1}(z) = x$ *(definition of inverse)*.
5. Also $(f^{-1} \circ g^{-1})(z) = f^{-1}(g^{-1}(z)) = f^{-1}(y) = x$.
6. The two functions agree at every $z \in C$, so they are equal *(Theorem 7.1.1)*. ■

PROOF **T6 Q6 · If $g \circ f$ is injective, then $f$ is injective · If $g \circ f$ is surjective, then $g$ is surjective**
1. Let $f : B \to C$ and let $g$ have domain $C$ with $g \circ f$ injective. Let $x_1, x_2 \in B$ with $f(x_1) = f(x_2)$.
2. Then $(g \circ f)(x_1) = g(f(x_1)) = g(f(x_2)) = (g \circ f)(x_2)$, so $x_1 = x_2$ *($g \circ f$ is injective)*. ■

For the second: given $z$, surjectivity of $g \circ f$ gives $x$ with $g(f(x)) = z$, and $y = f(x)$ is a preimage of $z$ under $g$.
**The other halves fail:** $X = Z = \{0\}$, $Y = \{0, 1\}$, $f(0) = 0$, $g(0) = g(1) = 0$. Then $g \circ f = \mathrm{id}_{\{0\}}$ is bijective, yet $g$ is not injective and $f$ is not surjective.

PAST PAPER **Order of a bijection · T6 Q7, Q8 · T8 Q10(c) · AY25 final Q11, Q12, Q26(b)**
The **order** of a bijection $f : A \to A$ is the least $n \in \mathbb{Z}^+$ with $f \circ f \circ \cdots \circ f = \mathrm{id}_A$ ($n$ copies). The **order of an input** $x$ is the least $m \in \mathbb{Z}^+$ with $f^{(m)}(x) = x$.
**T6 Q7.** On $\{1, 2, 3\}$, $g$ swaps 1, 2 and $h$ swaps 2, 3: orders 2 and 2. $g \circ h$ sends $1 \mapsto 2 \mapsto 3 \mapsto 1$: order 3; likewise $h \circ g$.
**T6 Q8.** $g(x) = 3x \bmod 5$: $g^{(3)}(21) = g^{(2)}(3) = g(4) = 2$. Order of 3: $3 \mapsto 4 \mapsto 2 \mapsto 1 \mapsto 3$, so 4. Orders of $0, 1, 2, 3, 4$ are $1, 4, 4, 4, 4$: classes $\{0\}$ and $\{1, 2, 3, 4\}$.
**T8 Q10(c).** $f = \{(1, 2), (2, 4), (3, 1), (4, 3)\}$, $g = \{(1, 4), (2, 1), (3, 2), (4, 3)\}$: orders 4 and 4; $f^{-1} \circ g$ sends $1 \mapsto 2 \mapsto 3 \mapsto 1$, $4 \mapsto 4$: order 3.
**AY25 Q11, Q12.** $f(p, q) = (p, p \oplus q)$ and $g(p, q) = (p \oplus q, q)$ on "Bool"$^2$. $f \circ f = \mathrm{id}$, so $f = f^{-1}$; but $f^{-1} \circ f$ $= \mathrm{id} \neq f$. $(g \circ f)(p, q) = (q, p \oplus q)$ has order 3.
**AY25 Q26(b).** $z : \mathbb{Z}_5 \to \mathbb{Z}_5$, $z(x) = 2x + 1$: $z^{-1}([3]) = [1]$; $z^{(2)}(x) = 4x + 3$; $[3] \mapsto [2] \mapsto [0] \mapsto [1] \mapsto [3]$, so the order of $[3]$ is 4.

## 30 · Addition and multiplication on $\mathbb{Z}_n$

DEFINITION **$\mathbb{Z}_n$ (L7 s54–55) · Addition and multiplication on $\mathbb{Z}_n$ (L7 s56)**
$\mathbb{Z}_n$ denotes the quotient $\mathbb{Z}/\underset{n}{\sim}$, where $\underset{n}{\sim}$ is congruence mod $n$: $\mathbb{Z}_n = \{[0], [1], \ldots, [n-1]\}$ with $[r]$ $= \{nk + r : k \in \mathbb{Z}\}$.
For $[x], [y] \in \mathbb{Z}_n$: $[x] + [y] = [x + y]$ and $[x] \cdot [y] = [x \cdot y]$. In $\mathbb{Z}_3$: $[0] + [1] = [1]$, $[2] \cdot [2] = [4] = [1]$.

TRAP **$\mathbb{Z}_n$ has two meanings in the slides**
Lecture 7 uses $\mathbb{Z}_n = \mathbb{Z}/\underset{n}{\sim}$ (a set of $n$ classes). Lecture 9, slide 9 writes $\mathbb{Z}_n = \{1, 2, \ldots, n\}$ when defining finite sets. Read the definition in the question.

PROOF **Propositions · + and · on $\mathbb{Z}_n$ are well defined (L7 s57–58)**
Claim: for all $[x_1], [y_1], [x_2], [y_2] \in \mathbb{Z}_n$, if $[x_1] = [x_2]$ and $[y_1] = [y_2]$ then $[x_1] + [y_1] = [x_2] + [y_2]$ and $[x_1] \cdot [y_1]$ $= [x_2] \cdot [y_2]$.

1. Let $[x_1]=[x_2]$ and $[y_1]=[y_2]$. Then $x_1 \equiv x_2$ and $y_1 \equiv y_2 \pmod n$ *(Lemma Rel.1)*.
2. So $x_1 - x_2 = nk$ and $y_1 - y_2 = nl$ for some $k, l \in \mathbb{Z}$ *(definition of congruence)*.
3. $(x_1+y_1)-(x_2+y_2) = n(k+l)$, so $x_1+y_1 \equiv x_2+y_2 \pmod n$, hence $[x_1+y_1]=[x_2+y_2]$ *(Lemma Rel.1)*.
4. $x_1y_1 - x_2y_2 = (nk+x_2)(nl+y_2) - x_2y_2 = n(nkl+ky_2+lx_2)$ with $nkl+ky_2+lx_2 \in \mathbb{Z}$ *(closure)*, hence $[x_1y_1]=[x_2y_2]$. ■

The representative-level version of this proof is in §23.

METHOD **Is a rule on classes well defined?**

A rule $F([x]) = \ldots$ defines a function on $A/\sim$ only if the output does not depend on the representative: **show $x \sim x' \Rightarrow$ same output**. To disprove, name two representatives of one class with different outputs.

- $[x]_3 \mapsto [x]_2$ from $\mathbb{Z}_3$ to $\mathbb{Z}_2$: **not** well defined ($[0]_3=[3]_3$ but $[0]_2 \neq [3]_2$). From $\mathbb{Z}_4$ to $\mathbb{Z}_2$ it is well defined, as $4 \mid (x-x') \Rightarrow 2 \mid (x-x')$.
- **T6 Q10.** On $\mathbb{Q}/\sim$ with $x \sim y \leftrightarrow x-y \in \mathbb{Z}$: $+$ is well defined ($(x_1+y_1)-(x_2+y_2) = k+l \in \mathbb{Z}$); $\cdot$ is **not**: $[1/2]=[-1/2]$, but $[1/2]\cdot[1/2] = [1/4] \neq [-1/4] = [1/2]\cdot[-1/2]$, as $1/4-(-1/4) = 1/2 \notin \mathbb{Z}$.
- **T8 Q10 (AY22/23 S2).** If $g : X \to Y$ satisfies $g(a)=g(b) \Leftrightarrow a \sim b$, then $f([x]) = g(x)$ is well defined ($[x]=[x'] \Rightarrow x \sim x' \Rightarrow g(x)=g(x')$) **and** injective ($f([x])=f([x']) \Rightarrow g(x)=g(x') \Rightarrow x \sim x' \Rightarrow [x]=[x']$).

# Part VII · Induction and Recursion

## 31 · Sequences, sums and products

DEFINITION **Sequence, term, explicit formula (L8 s5–6) · Summation (L8 s7) · Product (L8 s11)**

A **sequence** is an ordered list $a_m, a_{m+1}, \ldots, a_n$ (or infinite) of **terms**; $k$ in $a_k$ is the **index**. An **explicit formula** gives $a_k$ in terms of $k$. For integers $m \le n$:

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + \cdots + a_n, \qquad \prod_{k=m}^{n} a_k = a_m \cdot a_{m+1} \cdot \cdots \cdot a_n.$$

$k$ is the **index**, $m$ the **lower limit**, $n$ the **upper limit**. **Recursive form:** $\sum_{k=m}^{m} a_k = a_m$ and $\sum_{k=m}^{n} a_k = \left(\sum_{k=m}^{n-1} a_k\right) + a_n$ for $n > m$; likewise for $\prod$. An **empty sum** is 0; an **empty product** is 1.

THEOREM **Theorem 5.1.1 (L8 s14) · Standard sequences (L8 s18–20)**

For real sequences $a_k, b_k$, real $c$ and $n \ge m$:

$$\sum_{k=m}^{n} a_k + \sum_{k=m}^{n} b_k = \sum_{k=m}^{n} (a_k + b_k), \quad c \cdot \sum_{k=m}^{n} a_k = \sum_{k=m}^{n} c \cdot a_k, \quad \left(\prod_{k=m}^{n} a_k\right) \cdot \left(\prod_{k=m}^{n} b_k\right) = \prod_{k=m}^{n} (a_k \cdot b_k).$$

**Arithmetic:** $a_k = a_{k-1} + d$, so $a_n = a_0 + dn$ and $\sum_{k=0}^{n-1} a_k = \frac{n}{2}(2a_0 + (n-1)d)$.

**Geometric:** $a_k = ra_{k-1}$, so $a_n = a_0 r^n$ and $\sum_{k=0}^{n-1} a_k = a_0 \frac{1-r^n}{1-r}$ for $r \neq 1$.

Squares $1, 4, 9, 16, \ldots$; triangle numbers $1, 3, 6, 10, \ldots$ $(n(n+1)/2)$; lazy caterer $1, 2, 4, 7, 11, 16, \ldots$ $(n(n+1)/2+1)$.

WORKED EXAMPLE **L8 Ex. 4, 5, 7 · Telescoping, a product, change of variable**

- **Telescoping.** $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, so $\sum_{k=1}^{n} \frac{1}{k(k+1)} = \left(1-\frac{1}{2}\right) + \left(\frac{1}{2}-\frac{1}{3}\right) + \cdots + \left(\frac{1}{n}-\frac{1}{n+1}\right) = 1 - \frac{1}{n+1}$.
- $\prod_{k=1}^{5}(k+2) = 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 = 2520$.
- **Change of variable.** The index is a dummy: $\sum_{k=1}^{3} k^2 = \sum_{i=1}^{3} i^2 = \sum_{k=3}^{5} (k-2)^2$. Put $j = k-1$: $\sum_{k=1}^{n+1} \frac{k}{n+k} = \sum_{j=0}^{n} \frac{j+1}{n+j+1}$. Change the limits and every occurrence of the index together.

## 32 · Mathematical induction

DEFINITION **Principle of Mathematical Induction, PMI (L8 s24) · Closed form (L8 s27)**

Let $P(n)$ be a property defined for integers $n$ and let $a$ be a fixed integer. Suppose

1. $P(a)$ is true (**basis step**)
2. for all integers $k \ge a$, if $P(k)$ is true then $P(k+1)$ is true. (**inductive step**; $P(k)$ is the **inductive hypothesis**)

Then $P(n)$ is true for all integers $n \ge a$. PMI is taken as an axiom.

A sum is in **closed form** when it is written without an ellipsis or a summation symbol.

METHOD **The format used in class (L8 s26) · Mistakes to avoid (T7 Q1, D2, D3)**

1. Let $P(n) \equiv (...)$ for $n \geq a$. (set up the predicate)
2. Basis step: verify $P(a)$.
3. Assume $P(k)$ is true for an arbitrary $k \geq a$. (inductive hypothesis)
4. Inductive step: derive $P(k+1)$, marking where $P(k)$ is used.
5. Therefore $P(n)$ is true for all $n \geq a$ by MI.

- $P(k)$ is a **statement**, not a number: never write "$(k+1) + P(k)$".
- Do not start from $P(k+1)$ and reduce it to a true identity; start from one side of $P(k+1)$ and reach the other.
- Do not assume "$P(k)$ for all $k$"; that is the conclusion.
- Check the **basis value matches the claim** (T7 D2: the claim is for $n \in \mathbb{N}$, so the basis must be $n = 0$, and the step must be shown, not asserted).
- The inductive step must cover an **arbitrary** object of size $k+1$ (T7 D3: start from any group of $k+1$ people and remove one, rather than building up from a group of $k$).

TRAP **Nine common mistakes in induction proofs (Week 7 deck)**

1. **Assuming what is to be proved.** The hypothesis is $P(k)$, not $P(k+1)$; "by induction, $P(k+1)$ is true" is circular.
2. **Never using the hypothesis.** A derivation of $P(k+1)$ that does not use $P(k)$ is a direct proof, not a proof by induction.
3. **Proving the wrong implication.** The step is $P(k) \Rightarrow P(k+1)$; the converse $P(k+1) \Rightarrow P(k)$ builds no chain.
4. **Treating $k$ as a particular number.** The step must work for an arbitrary $k$ in the required range.
5. **Using $n$ in place of $k$ in the step:** writing $P(n) \Rightarrow P(n+1)$ while proving $P(n)$ for all $n \geq 1$.
6. **"Assume $P(k)$ is true for all $k \geq 1$."** That assumes the whole theorem. Write "let $k \geq 1$ be arbitrary and assume $P(k)$".
7. **An unjustified algebraic or inequality step** that forces $P(k+1)$: reaching $2^{k+1} < 2k^2$ and then claiming $2k^2 < (k+1)^2$ without checking when it holds. The structure looks perfect, yet there is a gap in the row of dominoes.
8. **Using more than the hypothesis provides.** In 1PI, assuming $P(k)$ does not give $P(k-1)$, $P(k-2)$; if earlier cases are needed, use 2PI.
9. **Too few basis cases for 2PI.** Proving $P(k) \Rightarrow P(k+2)$ with only $P(1)$ in the basis gives $P(1), P(3), P(5), ...$ and none of $P(2), P(4), P(6), ...$

PROOF **Theorem 5.2.2 (5th ed. 5.2.1) · $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ for all $n \geq 1$ (L8 s26)**

1. Let $P(n) \equiv (1 + 2 + \cdots + n = n(n+1)/2)$ for $n \in \mathbb{Z}^+$.
2. Basis step: $1 = 1(1+1)/2$, so $P(1)$ is true.
3. Assume $P(k)$ is true for an arbitrary $k \geq 1$: $1 + 2 + \cdots + k = k(k+1)/2$.
4. Inductive step: $1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1)$ *(inductive hypothesis)* $= \frac{(k+1)(k+2)}{2} = \frac{(k+1)((k+1)+1)}{2}$. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for all $n \in \mathbb{Z}^+$. ■

PROOF **Theorem 5.2.3 (5th ed. 5.2.2) · $\sum_{i=0}^{n} r^i = \frac{r^{n+1}-1}{r-1}$ for real $r \neq 1$ and $n \geq 0$ (L8 s28)**

1. Let $P(n) \equiv \left(\sum_{i=0}^{n} r^i = (r^{n+1} - 1)/(r-1)\right)$ for $n \geq 0$.
2. Basis step: $r^0 = 1 = (r^1 - 1)/(r-1)$, so $P(0)$ is true.
3. Assume $P(k)$ for an arbitrary $k \geq 0$.
4. Inductive step: $\sum_{i=0}^{k+1} r^i = \sum_{i=0}^{k} r^i + r^{k+1} = \frac{r^{k+1}-1}{r-1} + r^{k+1}$ *(inductive hypothesis)* $= \frac{r^{k+1}-1+r^{k+1}(r-1)}{r-1} = \frac{r^{(k+1)+1}-1}{r-1}$.
5. Therefore $P(n)$ is true for all $n \geq 0$. ■

PROOF **Proposition 5.3.1 (5th ed. 5.3.2) · $3 \mid (2^{2n} - 1)$ for all $n \geq 0$ (L8 s29)**

1. Let $P(n) \equiv (3 \mid (2^{2n} - 1))$ for $n \geq 0$.
2. Basis step: $2^0 - 1 = 0 = 3 \cdot 0$, so $P(0)$ is true.
3. Assume $P(k)$ for an arbitrary $k \geq 0$: $2^{2k} - 1 = 3r$ for some $r \in \mathbb{Z}$ *(definition of divisibility)*.
4. Inductive step: $2^{2(k+1)} - 1 = 2^{2k} \cdot 4 - 1 = 2^{2k} \cdot 3 + \left(2^{2k} - 1\right) = 2^{2k} \cdot 3 + 3r = 3\left(2^{2k} + r\right)$, and $2^{2k} + r \in \mathbb{Z}$ *(closure)*. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for all $n \geq 0$. ■

PROOF **Proposition 5.3.2 (5th ed. 5.3.3) · $2n + 1 < 2^n$ for all $n \geq 3$ (L8 s30)**

1. Let $P(n) \equiv (2n + 1 < 2^n)$ for $n \geq 3$.
2. Basis step: $2 \cdot 3 + 1 = 7 < 8 = 2^3$, so $P(3)$ is true.
3. Assume $P(k)$ for an arbitrary $k \geq 3$.
4. Inductive step: $2(k+1) + 1 = (2k+1) + 2 < 2^k + 2$ *(inductive hypothesis)* $< 2^k + 2^k$ *($2 < 2^k$ for $k \geq 2$)* $= 2^{k+1}$.
5. Therefore $P(n)$ is true for all $n \geq 3$. ■

The basis cannot be lowered: $n=1$ gives $3<2$ and $n=2$ gives $5<4$, both false.

TRAP **L8 Ex. 12 · "All cows have the same colour"**
The step "cows 1 to $k$ share a colour, cows 2 to $k+1$ share a colour, and cows 2 to $k$ are common to both" fails for $k=1$: the overlap $\{2,\dots,k\}$ is **empty**, so nothing links cow 1 to cow 2. $P(1)\Rightarrow P(2)$ is false, and the chain breaks at its first link. The inductive step must work for **every** $k\geq a$.

WORKED EXAMPLE **L8 Ex. 13 · Tiling by L-trominoes**
$P(n)$: every $2^n\times 2^n$ board with one square removed can be tiled by L-trominoes. $P(1)$: the board is one L-tromino. Step: split a $2^{k+1}\times 2^{k+1}$ board into four $2^k\times 2^k$ quadrants; place one tromino at the centre covering one square of each quadrant not containing the removed square. Each quadrant now has one square missing and is tiled by $P(k)$.

PROOF **T7 Q2 · $1^2+2^2+\cdots+n^2=n(n+1)\frac{2n+1}{6}$** **T7 Q3 · Bernoulli: $1+nx\leq(1+x)^n$ for real $x\geq -1$**
**Q2.** $P(1)$: $1=(1\cdot 2\cdot 3)/6$. Step: $1^2+\cdots+k^2+(k+1)^2=\frac{k(k+1)(2k+1)}{6}+(k+1)^2$ *(IH)* $=\frac{(k+1)(2k^2+7k+6)}{6}$ $=\frac{(k+1)(k+2)(2k+3)}{6}$, which is $P(k+1)$.
**Q3.** $P(1)$: $1+x=(1+x)^1$. Step: $(1+x)^{k+1}=(1+x)^k(1+x)\geq(1+kx)(1+x)$ *(IH, and $1+x\geq 0$)* $=1+(k+1)x+kx^2\geq 1+(k+1)x$ *($kx^2\geq 0$).* ■

PROOF **T7 Q4 · For odd $a$: $2^{n+2}\mid(a^{2^n}-1)$ for all $n\in\mathbb{Z}^+$**
1. Let $P(n)\equiv(2^{n+2}\mid(a^{2^n}-1))$.
2. Basis step: $a=2p+1$ for some $p\in\mathbb{Z}$, so $a^2-1=(a-1)(a+1)=4p(p+1)=8m$ for some $m\in\mathbb{Z}$ *(a product of two consecutive integers is even)*. So $P(1)$ is true.
3. Assume $P(k)$: $a^{2^k}-1=2^{k+2}m$ for some $m\in\mathbb{Z}$.
4. Inductive step: $a^{2^{k+1}}-1=\left(a^{2^k}\right)^2-1=\left(a^{2^k}-1\right)\left(a^{2^k}+1\right)=2^{k+2}m\left(2^{k+2}m+2\right)=2^{k+3}m\left(2^{k+1}m+1\right)$. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for all $n\in\mathbb{Z}^+$ by MI. ■

PAST PAPER **AY22/23 Q21 · For the Fibonacci sequence $f_0,f_1,f_2,\dots=0,1,1,2,3,\dots$ and all $n\in\mathbb{N}$: $\sum_{i=0}^{n}f_i^2=f_nf_{n+1}$**
1. For each $n\in\mathbb{N}$ let $P(n)$ be the proposition $\sum_{i=0}^{n}f_i^2=f_nf_{n+1}$.
2. (Basis step) $f_0^2=0^2=0=0\cdot 1=f_0f_1$, so $P(0)$ is true.
3. (Induction step) Let $k\in\mathbb{N}$ such that $P(k)$ is true. Then $\sum_{i=0}^{k+1}f_i^2=\sum_{i=0}^{k}f_i^2+f_{k+1}^2=f_kf_{k+1}+f_{k+1}^2$ *(induction hypothesis)* $=f_{k+1}(f_k+f_{k+1})$ *(basic algebra)* $=f_{k+1}f_{k+2}$ *(definition of Fibonacci)*. So $P(k+1)$ is true.
4. Therefore $\forall n\in\mathbb{N}\,P(n)$ is true by MI. ■

Check with $n=4$: $0+1+1+4+9=15=3\cdot 5=f_4f_5$. 1PI is enough: the step uses only $P(k)$ and the recurrence.

## 33 · Strong induction and well-ordering

DEFINITION **1PI and 2PI (L8 s37)**
Let $P(n)$ be a property of integers $n\geq a$.
- **Weak (regular) induction, 1PI:** if $P(a)$ holds, and $P(k)\Rightarrow P(k+1)$ for every $k\geq a$, then $P(n)$ holds for all $n\geq a$.
- **Strong induction, 2PI:** if $P(a)$ holds, and $(P(a)\wedge P(a+1)\wedge\cdots\wedge P(k))\Rightarrow P(k+1)$ for every $k\geq a$, then $P(n)$ holds for all $n\geq a$.
- **2PI, variant with several basis cases:** if $P(a),P(a+1),\dots,P(b)$ hold, and $P(k)\Rightarrow P(k+b-a+1)$ for every $k\geq a$, then $P(n)$ holds for all $n\geq a$.

1PI, 2PI and the well-ordering principle are equivalent in power.

METHOD **Choosing the form**
Use 2PI when $P(k+1)$ depends on an earlier case that is not $P(k)$: a factor of $k+1$, the case $k-3$, or two or three previous terms of a recurrence. **Prove as many basis cases as the step reaches back**: a step using $P(k-3)$ needs four consecutive basis cases; a recurrence $a_{n+3}=a_{n+2}+a_{n+1}+a_n$ needs three.

PROOF **L8 Ex. 14 · Every integer greater than 1 is divisible by a prime (2PI, L8 s39)**
1. Let $P(n)\equiv(n$ is divisible by a prime$)$ for $n>1$.
2. Basis step: $2\mid 2$ and 2 is prime, so $P(2)$ is true.
3. Inductive step: let $k\geq 2$ and assume $P(i)$ for all integers $i$ with $2\leq i\leq k$.

3.1. Case 1 ($k+1$ is prime): $k+1$ is divisible by the prime $k+1$.
3.2. Case 2 ($k+1$ is not prime): $k+1 = ab$ with $1 < a < k+1$ and $1 < b < k+1$. Then $2 \le a \le k$, so a prime $p$ divides $a$ *(inductive hypothesis)* and $a \mid (k+1)$. So $p \mid (k+1)$ *(Theorem 4.3.3)*.
4. Therefore every integer greater than 1 is divisible by a prime. ■

The well-ordering proof of the same result is in §12.

PROOF **L8 Ex. 15, 16 · Every amount $n \ge 12$ is $4a+5b$ with $a, b \in \mathbb{N}$ (1PI and 2PI, L8 s40–41) T7 Q5 · $n \ge 8$ is $3x+5y$**

**By 1PI.** $P(12)$: $12 = 4 \cdot 3$. Step: suppose $k = 4a+5b$. Case 1 ($a \ge 1$): replace one 4 by a 5: $k+1 = 4(a-1)+5(b+1)$. Case 2 ($a = 0$): then $k = 5b \ge 12$ forces $b \ge 3$; replace three 5s by four 4s: $k+1 = 4 \cdot 4 + 5(b-3)$.

**By 2PI.**
1. Let $P(n) \equiv (n = 4a+5b \text{ for some } a, b \in \mathbb{N})$ for $n \ge 12$.
2. Basis step: $12 = 4 \cdot 3 + 5 \cdot 0$; $13 = 4 \cdot 2 + 5 \cdot 1$; $14 = 4 \cdot 1 + 5 \cdot 2$; $15 = 4 \cdot 0 + 5 \cdot 3$.
3. Assume $P(i)$ for $12 \le i \le k$, for an arbitrary $k \ge 15$.
4. Inductive step: $12 \le k-3 \le k$, so $k-3 = 4a+5b$ for some $a, b \in \mathbb{N}$ *(inductive hypothesis)* then $k+1 = 4(a+1)+5b$.
5. Therefore $P(n)$ is true for all $n \ge 12$. ■

**T7 Q5.** Same pattern: basis $8 = 3+5$, $9 = 3 \cdot 3$, $10 = 5 \cdot 2$; step $k+3 = 3(x+1)+5y$. By 1PI instead: if $y > 0$, $k+1 = 3(x+2)+5(y-1)$; if $y = 0$, then $x \ge 3$ and $k+1 = 3(x-3)+5 \cdot 2$. (11 is not $4a+5b$ and 7 is not $3x+5y$, so the bounds are sharp.)

PROOF **T7 Q6 · Every positive integer is a sum of distinct powers of 2 (2PI) T7 Q7 · $a_n < 3^n$**

**Q6.** $P(1)$: $1 = 2^0$. Let $k \ge 1$ and assume $P(1), \ldots, P(k)$. Write $k+1 = 2m$ or $2m+1$ with $m \in \mathbb{Z}$; then $1 \le m \le k$, so $m = 2^{i_1} + \cdots + 2^{i_l}$ with $i_1 < \cdots < i_l$ *(inductive hypothesis)*. If $k+1 = 2m$, then $k+1 = 2^{i_1+1} + \cdots + 2^{i_l+1}$. If $k+1 = 2m+1$, then $k+1 = 2^0 + 2^{i_1+1} + \cdots + 2^{i_l+1}$, and $0 < i_1 + 1$. Exponents stay distinct.

**Q7.** $a_0 = 0, a_1 = 2, a_2 = 7$, $a_{n+3} = a_{n+2} + a_{n+1} + a_n$. Basis: $0 < 1$, $2 < 3$, $7 < 9$. Step: if $P(k), P(k+1), P(k+2)$ hold, then $a_{k+3} < 3^{k+2} + 3^{k+1} + 3^k < 3 \cdot 3^{k+2} = 3^{k+3}$. ■

PAST PAPER **T7 Q8 (AY23/24 S1 exam) · T8 Q9 (AY22/23 S2 exam) · T7 Q9 · Fibonacci and Hanoi**

**T7 Q8.** $P(a,b) \equiv F(a+b) = F(a+1)F(b) + F(a)F(b-1)$. Basis: $P(0,b)$: $F(b) = 1 \cdot F(b) + 0 \cdot F(b-1)$; $P(1,b)$: $F(b+1) = F(b) + F(b-1) = F(2)F(b) + F(1)F(b-1)$. Step: $F(k+1+b) = F(k+b) + F(k-1+b)$ *(definition of F)* $= (F(k+1)F(b) + F(k)F(b-1)) + (F(k)F(b) + F(k-1)F(b-1))$ *($P(k,b)$, $P(k-1,b)$)* $= (F(k+1) + F(k))F(b) + (F(k) + F(k-1))F(b-1)$ *(distributive law, twice)* $= F(k+2)F(b) + F(k+1)F(b-1)$.

**T8 Q9.** $\text{Even}(F_n) \Leftrightarrow \text{Even}(F_{n+3})$. Basis $P(0)$: $F_0 = 0$, $F_3 = 2$; $P(1)$: $F_1 = 1$, $F_4 = 3$. Step ($k \ge 1$): $\text{Even}(F_{k+1}) \Leftrightarrow (\text{Even}(F_k) \Leftrightarrow \text{Even}(F_{k-1}))$ *(Fact 1)* $\Leftrightarrow (\text{Even}(F_{k+3}) \Leftrightarrow \text{Even}(F_{k+2}))$ *($P(k)$, $P(k-1)$)* $\Leftrightarrow \text{Even}(F_{k+4})$ *(Fact 1)*.

**T7 Q9, Tower of Hanoi.** $T(1) = 1$. To move $k+1$ disks: move the top $k$ to the spare peg ($2^k - 1$ moves, IH), move the largest (1 move), move the $k$ back on top ($2^k - 1$ moves). $T(k+1) = 2(2^k - 1) + 1 = 2^{k+1} - 1$.

DEFINITION **Well-Ordering Principle for the integers (L8 s43)**

**Every non-empty subset of $\mathbb{Z}_{\ge 0}$ has a smallest element.** (Epp's version allows any set of integers bounded below; the course uses the statement above.)

PROOF **Well-Ordering Principle, proved by 2PI (L8 s44) · L8 Ex. 17**
1. Suppose not: let $S \subseteq \mathbb{Z}_{\ge 0}$ be non-empty with no smallest element.
2. For $n \in \mathbb{Z}_{\ge 0}$ let $P(n) \equiv (n \notin S)$.
3. Let $k \in \mathbb{Z}_{\ge 0}$ with $P(0), \ldots, P(k-1)$ true, i.e. $0, 1, \ldots, k-1 \notin S$. If $k \in S$, then $k$ is the smallest element of $S$, contradicting line 1. So $k \notin S$ and $P(k)$ is true. (For $k = 0$ the hypothesis is empty, so this also gives the basis.)
4. Hence $P(n)$ is true for all $n \in \mathbb{Z}_{\ge 0}$ *(2PI)*, so $S = \varnothing$, contradicting line 1. ■

**Ex. 17.** (a) The positive reals have no least element ($x/2 < x$); no violation, as WOP concerns sets of integers. (b) $\{n \in \mathbb{Z}_{\ge 0} : n^2 < n\} = \varnothing$; no violation, as WOP concerns non-empty sets. (c) Non-negative integers of the form $46 - 7k$: $\ldots, 18, 11, 4$; least element 4.

PROOF **WOP by infinite descent · PMI derived from WOP (Week 7 deck)**

**WOP, alternative proof.**
1. Suppose not: let $S \subseteq \mathbb{Z}_{\ge 0}$ be non-empty with no smallest element. Choose some $a \in S$.

2. $a$ is not the smallest element, so there is $a_1 \in S$ with $a_1 < a$. But $a_1$ is not the smallest either, so there is $a_2 \in S$ with $a_2 < a_1$.
3. Continuing, we get an infinite descending sequence $a > a_1 > a_2 > \cdots$, which cannot occur in $\mathbb{Z}_{\geq 0}$ (only $a$ non-negative integers lie below $a$). Hence $S$ must have a smallest element. ■

**Induction from WOP (smallest counterexample).**

1. Suppose $P(0)$ is true and $P(k) \Rightarrow P(k+1)$ for all $k \geq 0$. Suppose some $P(k)$ is false.
2. Then the set of counterexamples $C = \{k \in \mathbb{Z}_{\geq 0} : P(k) \text{ is false}\}$ is non-empty, so $C$ has a smallest element $m$ *(WOP)*.
3. $m > 0$ *($P(0)$ is true)*, and $P(m-1)$ is true *(minimality of $m$)*.
4. Then $P(m)$ is true *($P(m-1) \Rightarrow P(m)$)*, contradicting $m \in C$. Hence $P(n)$ is true for all $n \geq 0$. ■

Dominoes: if some domino does not fall, there is a **first** one that does not; it is not domino 1 (basis), so the domino before it falls, and then it falls too.

**Template for $\forall n \geq a\, P(n)$ by WOP:** suppose the set of counterexamples is non-empty; take its least element $m$; show $m \neq a$; use the truth of $P$ below $m$ to get $P(m)$ or a smaller counterexample: contradiction. Mathematical induction and WOP are equivalent: each proves the other.

## 34 · Recursion and structural induction

**DEFINITION Recurrence relation, initial conditions (L8 s48)**

A **recurrence relation** for $a_0, a_1, a_2, \ldots$ is a formula relating each term $a_k$ to certain predecessors $a_{k-1}, a_{k-2}, \ldots, a_{k-i}$. The **initial conditions** give the first terms $a_0, \ldots, a_{i-1}$.

Examples: $F_n = F_{n-1} + F_{n-2}$ with $F_0 = 0, F_1 = 1$; $0! = 1$, $n! = n \cdot (n-1)!$; $a^0 = 1$, $a^n = a^{n-1} \cdot a$.

**PROOF L8 Ex. 19, 20 · For real $a_1, \ldots, a_n$, $b_1, \ldots, b_n$ and every $n \in \mathbb{Z}^+$: $\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$**

1. Let $P(n) \equiv \left(\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i\right)$ for $n \geq 1$.
2. Basis step: $\sum_{i=1}^{1}(a_i + b_i) = a_1 + b_1 = \sum_{i=1}^{1} a_i + \sum_{i=1}^{1} b_i$, so $P(1)$ is true.
3. Assume $P(k)$ is true for an arbitrary $k \geq 1$.
4. Inductive step: $\sum_{i=1}^{k+1}(a_i + b_i) = \sum_{i=1}^{k}(a_i + b_i) + (a_{k+1} + b_{k+1})$ *(definition of $\sum$)* $= \sum_{i=1}^{k} a_i + \sum_{i=1}^{k} b_i + (a_{k+1} + b_{k+1})$ *(inductive hypothesis)* $= \left(\sum_{i=1}^{k} a_i + a_{k+1}\right) + \left(\sum_{i=1}^{k} b_i + b_{k+1}\right)$ *(associative and commutative laws)* $= \sum_{i=1}^{k+1} a_i + \sum_{i=1}^{k+1} b_i$ *(definition of $\sum$)*. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for every positive integer $n$. ■

The recursive form of $\sum$ (§31) is what lets a statement about a sum of any length be proved by induction.

**DEFINITION Recursively defined set (L8 s56, s58) · Structural induction (L8 s58)**

A recursive definition of a set $S$ has three clauses:

- **(base clause)** certain elements, the **founders**, are in $S$;
- **(recursion clause)** $S$ is closed under certain functions, the **constructors**: if $f$ is a constructor and $x \in S$, then $f(x) \in S$;
- **(minimality clause)** membership of $S$ can always be demonstrated by finitely many successive applications of the clauses above.

**Structural induction over $S$.** To prove $\forall x \in S\, P(x)$, it suffices to show **(basis step)** $P(c)$ for every founder $c$; and **(induction step)** $\forall x \in S\,(P(x) \Rightarrow P(f(x)))$ for every constructor $f$.

**WORKED EXAMPLE L8 Ex. 21–24 · Recursively defined sets**

- $\mathbb{Z}_{\geq 0}$: founder 0; constructor $x \mapsto x + 1$. $2\mathbb{Z}$: founder 0; constructors $x \mapsto x - 2$ and $x \mapsto x + 2$.
- **Legal parentheses $P$:** base $() \in P$; if $E \in P$ then $(E) \in P$; if $E, F \in P$ then $EF \in P$. Derivation of $(())()$: $()$ by the base; $(())$ by rule (a) with $E = ()$; $(())()$ by rule (b) with $E = (())$, $F = ()$.
- **Ex. 24.** $1 \in S$; $x \in S \Rightarrow 2x \in S \wedge 3x \in S$. Then $S = \{2^a 3^b : a, b \in \mathbb{N}\}$: $9, 12 \in S$ and $10, 11, 13 \notin S$ (each has a prime factor other than 2 and 3).

**METHOD Membership questions on a recursive set**

**To show $x \in S$:** write a derivation, one clause per line.

**To show $x \notin S$:** find an **invariant** $P$ true of every founder and preserved by every constructor, with $P(x)$ false; prove $\forall y \in S\, P(y)$ by structural induction.

PROOF **T7 Q10 · $2 \in S$; $x \in S \Rightarrow 3x \in S \wedge x^2 \in S$. Which of $0, 6, 15, 16, 36$ are in $S$?**
$6 = 3 \cdot 2 \in S$; $36 = 6^2 \in S$; $4 = 2^2 \in S$, so $16 = 4^2 \in S$.
**$0 \notin S$:** every $n \in S$ satisfies $n \geq 2$.
1. Let $P(n) \equiv (n \geq 2)$ for $n \in S$.
2. (Basis step) $2 \geq 2$, so $P(2)$ is true.
3. (Induction step) Let $x \in S$ with $x \geq 2$. Then $3x \geq 6 \geq 2$ and $x^2 \geq 4 \geq 2$, so $P(3x)$ and $P(x^2)$ are true.
4. Hence $\forall n \in S\, P(n)$ by structural induction over $S$. ■

**$15 \notin S$:** every element of $S$ is even (2 is even; if $x$ is even so are $3x$ and $x^2$), by the same argument.

PAST PAPER **AY20/21 S2 Q11 (Week 7 deck) · $T \subseteq \{\mathrm{s}, \mathrm{u}\}^*$ with $\mathrm{su} \in T$; if $\sigma, \tau \in T$ then $\mathrm{s}\ \sigma\ \mathrm{u} \in T$ and $\mathrm{s}\ \sigma\tau\ \mathrm{u} \in T$**
Which are in $T$? A. ssssusuuusuu B. ssussusususuuu C. ssussuusssuu D. ussususussuu
**A $\in T$:** su (base); $\mathrm{s} \cdot \mathrm{su} \cdot \mathrm{su} \cdot \mathrm{u} = \mathrm{ssusuu}$; $\mathrm{s} \cdot \mathrm{ssusuu} \cdot \mathrm{u} = \mathrm{sssusuuu}$; $\mathrm{s} \cdot \mathrm{sssusuuu} \cdot \mathrm{su} \cdot \mathrm{u} = \mathrm{ssssusuuusuu}$.
**Invariants,** each proved by structural induction over $T$ (founder su; constructors $\sigma \mapsto \mathrm{s}\ \sigma\ \mathrm{u}$ and $(\sigma, \tau) \mapsto \mathrm{s}\ \sigma\tau\ \mathrm{u}$; a constructor with two arguments assumes $P(\sigma)$ **and** $P(\tau)$):
- (i) every element starts with s and ends with u; (ii) every element of length at least 3 starts with ss and ends with uu;
- (iii) no element contains usus; (iv) every element has as many s's as u's.

**Proof of (iv).**
1. Let $P(\sigma) \equiv$ (the number of s's in $\sigma$ equals the number of u's) for $\sigma \in T$.
2. (Basis step) su has one s and one u, so $P(\mathrm{su})$ is true.
3. (Induction step) Let $\sigma, \tau \in T$ with $P(\sigma)$ and $P(\tau)$ true, having $a$ and $b$ s's respectively. Then $\mathrm{s}\ \sigma\ \mathrm{u}$ has $a+1$ s's and $a+1$ u's, and $\mathrm{s}\ \sigma\tau\ \mathrm{u}$ has $a+b+1$ of each. So $P(\mathrm{s}\ \sigma\ \mathrm{u})$ and $P(\mathrm{s}\ \sigma\tau\ \mathrm{u})$ are true.
4. Hence $\forall \sigma \in T\, P(\sigma)$ by structural induction over $T$. ■

(i) is immediate from the shape of the clauses, and (ii) follows from (i): in $\mathrm{s}\ \sigma\ \mathrm{u}$ and $\mathrm{s}\ \sigma\tau\ \mathrm{u}$ the inner strings start with s and end with u. For (iii): an occurrence of usus in a new string avoids the outer s and u, so it lies inside $\sigma$ or $\tau$ (excluded by the hypothesis) or straddles the join in $\mathrm{s}\ \sigma\tau\ \mathrm{u}$, which needs $\sigma$ to end in s, or $\mathrm{s}\ \sigma$ to end in usu, or $\tau\ \mathrm{u}$ to start with sus; (i) and (ii) exclude all three.
**B** contains usus (letters 6 to 9), against (iii). **C** has 7 s's and 5 u's, against (iv). **D** starts with u, against (i).
**Only A.**

PAST PAPER **AY22/23 Q14 · $1 \in S$ and $2 \in S$; if $x, y \in S$ then $x/y \in S$**
**Claim: $S = \{2^n : n \in \mathbb{Z}\} = \{\ldots, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, \ldots\}$.**
1. ($\subseteq$) Let $P(x) \equiv (x = 2^n$ for some $n \in \mathbb{Z})$ for $x \in S$. (Basis step) $1 = 2^0$ and $2 = 2^1$. (Induction step) Let $x, y \in S$ with $x = 2^a$ and $y = 2^b$, $a, b \in \mathbb{Z}$. Then $x/y = 2^{a-b}$ and $a - b \in \mathbb{Z}$, so $P(x/y)$ is true. Hence $\forall x \in S\, P(x)$ by structural induction over $S$.
2. ($\supseteq$) $2^{-1} = 1/2 \in S$. If $2^n \in S$ then $2^{n+1} = 2^n / 2^{-1} \in S$, so $2^n \in S$ for all $n \geq 0$ by MI; and then $2^{-n} = 1/2^n \in S$. ■

A. $\forall x \in S\, (x \in \mathbb{Z} \Rightarrow x$ is even): false, $x = 1$. B. $\forall x \in S\, (x \in \mathbb{Q} \Rightarrow x < 1)$: false, $x = 2$.
C. $\forall x, y \in S\, (x + y \geq 1)$: false, $\frac{1}{2} + \frac{1}{4} = \frac{3}{4} < 1$. D. $|S| = |\mathbb{Z}|$: **true**, since $f : \mathbb{Z} \to S$, $f(n) = 2^n$ is a bijection.

PAST PAPER **T7 Q11 · AY25 final Q14–Q16 · Recursive sets of sets and strings**
**T7 Q11.** $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 3, 5, 7, 9\}$ are founders; constructors $\cap$, $\cup$, $\setminus$. $C = \{2, 4, 7, 9\}$ $= (A \setminus B) \cup (B \setminus A) \in S$. $D = \{2, 3, 4, 5\} \notin S$: the invariant "$1 \in X \Leftrightarrow 3 \in X$" holds for $A$ and $B$ and is preserved by $\cap$, $\cup$, $\setminus$ (check the four cases of $1, 3 \in X$ or not, $1, 3 \in Y$ or not), but fails for $D$.
**AY25 Q14, Q15.** $a \in S$; $x \in S \Rightarrow xa \in S$; $x \in S \Rightarrow bxb \in S$. Elements of length at most 4: *a*; *aa*; *aaa*, *bab*; *aaaa*, *baba*, *baab*: **7**. A string of length $n$ arises from one of length $n-1$ (append $a$) or $n-2$ (wrap in $b$), in exactly one way, so $G_n = G_{n-1} + G_{n-2}$ with $G_1 = G_2 = 1$: $G_n = F_n$.
**AY25 Q16.** $S_0 = \{1, 2, 3\}$, $S_{n+1} = (S_n \cup \{x + 3 : x \in S_n\}) \setminus \{2, 3\}$. $S_1 = \{1, 4, 5, 6\}$, $S_2 = \{1, 4, 5, 6, 7, 8, 9\}$, and so on: the union of all $S_n$ is $\mathbb{Z}^+$ (2 and 3 come from $S_0$).

# Part VIII · Cardinality

## 35 · Pigeonhole principle and finite sets

THEOREM **Pigeonhole Principle · Dual Pigeonhole Principle (L9 s6)**
Let $A$ and $B$ be **finite** sets.
- **PHP.** If there is an injection $f : A \to B$, then $|A| \leq |B|$. Contrapositive: if $m > n$ pigeons are put into $n$ pigeonholes, some pigeonhole holds at least two pigeons.
- **Dual PHP.** If there is a surjection $f : A \to B$, then $|A| \geq |B|$. Contrapositive: if $m < n$ pigeons are put into $n$ pigeonholes, some pigeonhole is empty.

PROOF **PHP and Dual PHP (L9 s7–8)**
**PHP.**
1. Let $A = \{a_1, a_2, ..., a_m\}$ with $m = |A|$.
2. If $a_i \neq a_j$ then $f(a_i) \neq f(a_j)$ *(f is injective)*, so $f(a_1), ..., f(a_m)$ are $m$ different elements of $B$.
3. Hence $|B| \geq m = |A|$. ■

**Dual PHP.**
1. Let $B = \{b_1, b_2, ..., b_n\}$ with $n = |B|$. For each $b_i$ pick $a_i \in A$ with $f(a_i) = b_i$ *(f is surjective)*.
2. If $b_i \neq b_j$ then $f(a_i) \neq f(a_j)$, so $a_i \neq a_j$ *(f is a function)*.
3. So $a_1, ..., a_n$ are $n$ different elements of $A$, and $|A| \geq n = |B|$. ■

DEFINITION **Finite, infinite (L9 s9) · Cardinality of a finite set (L9 s10)**
Let $\mathbb{Z}_n = \{1, 2, ..., n\}$ (in this lecture only). A set $S$ is **finite** iff $S = \varnothing$ or there is a bijection from $S$ to $\mathbb{Z}_n$ for some $n \in \mathbb{Z}^+$; otherwise $S$ is **infinite**.
The **cardinality** of a finite set $S$ is $|S| = 0$ if $S = \varnothing$, and $|S| = n$ if there is a bijection $f : S \to \mathbb{Z}_n$.

THEOREM **Finite sets (L9 s10–11)**
- **Equality of cardinality.** For finite $A$, $B$: $|A| = |B|$ iff there is a bijection $f : A \to B$. ($\Leftarrow$: PHP and Dual PHP. $\Rightarrow$: map $a_i \mapsto b_i$.)
- **Theorem Cardinality.1.** Any subset of a finite set is finite. Contrapositive: if $A \subseteq B$ and $A$ is infinite, then $B$ is infinite.

PAST PAPER **T8 Q8 (AY22/23 S1 exam) · $|A| \leq |R|$ for a relation $R$ on a finite set $A$?**
**(i) Reflexive: true.** Define $f : A \to R$ by $f(x) = (x, x)$; this is a function into $R$ because $R$ is reflexive. If $f(x_1) = f(x_2)$ then $(x_1, x_1) = (x_2, x_2)$, so $x_1 = x_2$ *(equality of ordered pairs)*. $f$ is an injection, so $|A| \leq |R|$ *(PHP)*.
**(ii) Symmetric, (iii) transitive: false.** The empty relation on a non-empty $A$ is symmetric and transitive, with $|A| > 0 = |\varnothing|$.

## 36 · Same cardinality

DEFINITION **Same cardinality (Cantor, L9 s12)**
For **any** sets $A$ and $B$: $A$ has the **same cardinality** as $B$, written $|A| = |B|$, iff there is a bijection $f : A \to B$.
This defines "$|A| = |B|$" without defining $|A|$ for infinite sets.

PROOF **Theorem 7.4.1 · Same-cardinality is an equivalence relation (L9 s14–16)**
1. (Reflexive) $\mathrm{id}_A : A \to A$ is injective ($\mathrm{id}_A(x_1) = \mathrm{id}_A(x_2)$ gives $x_1 = x_2$) and surjective ($\mathrm{id}_A(x) = x$). So $|A| = |A|$.
2. (Symmetric) Let $|A| = |B|$, with a bijection $f : A \to B$ *(Cantor's definition)*. Then $f^{-1} : B \to A$ is a bijection *(Theorem 7.2.3)*. So $|B| = |A|$.
3. (Transitive) Let $|A| = |B|$ and $|B| = |C|$, with bijections $f : A \to B$ and $g : B \to C$. Then $g \circ f$ has the inverse $f^{-1} \circ g^{-1}$ *(T6 Q3, §29)*, so $g \circ f : A \to C$ is a bijection *(Theorem 7.2.3)*. So $|A| = |C|$. ■

PROOF **$|2\mathbb{Z}| = |\mathbb{Z}|$ (L9 s17–18) · L9 Ex. 1, 2**
Define $H : \mathbb{Z} \to 2\mathbb{Z}$ by $H(n) = 2n$.
1. (Injective) If $H(n_1) = H(n_2)$ then $2n_1 = 2n_2$, so $n_1 = n_2$.
2. (Surjective) Let $m \in 2\mathbb{Z}$. Then $m = 2k$ for some $k \in \mathbb{Z}$ *(definition of even)*, and $H(k) = 2k = m$.
3. So $H$ is a bijection and $|\mathbb{Z}| = |2\mathbb{Z}|$ *(Cantor's definition)*. ■

Likewise $|\mathbb{N}| = |\mathbb{N} \setminus \{0\}|$ via $x \mapsto x + 1$, and $|\mathbb{N}| = |\mathbb{N} \setminus \{1, 3, 5, ...\}|$ via $x \mapsto 2x$.
**An infinite set can have the same cardinality as a proper subset of itself.** For a finite $A$ and a proper

subset $B$, always $|B| < |A|$.
**Alternative definition of infinite (Week 8 deck).** Some mathematicians use this as the definition: $A$ is infinite iff there is a set $B$ with $B \subseteq A \wedge B \neq A \wedge |B| = |A|$.
Week 8 deck checks: $|\mathbb{Z}^-| = |\mathbb{Z}^+|$, $|\mathbb{Z} \cup \{a,b,c\}| = |\mathbb{Z}|$, $|\mathbb{Z}| = |\mathbb{Z}^+|$, $|\mathbb{Z}| = |2\mathbb{Z}|$, $|\mathbb{Q}| = |\mathbb{Z}|$ are all true; $|\{a,b,c\}| = |\{a,b\}|$ is false.

## 37 · Countable sets

**DEFINITION $\aleph_0$ · Countably infinite · Countable, uncountable (L9 s23)**
$\aleph_0 = |\mathbb{Z}^+|$ (some authors use $\mathbb{N}$; either may be used). A set $S$ is **countably infinite** iff $|S| = \aleph_0$, i.e. there is a bijection between $\mathbb{Z}^+$ (or $\mathbb{N}$) and $S$.
A set is **countable** iff it is finite or countably infinite; otherwise it is **uncountable**.

**THEOREM Proposition 9.1 · Lemma 9.2 · Countability via sequences (L9 s33–34)**
For an **infinite** set $B$:
- **Proposition 9.1.** $B$ is countable iff there is a sequence $b_0, b_1, b_2, ... \in B$ in which every element of $B$ appears **exactly once**.
- **Lemma 9.2.** $B$ is countable iff there is a sequence $b_0, b_1, b_2, ...$ in which every element of $B$ **appears**.

A bijection $f : \mathbb{Z}_{\geq 0} \to B$ is the sequence $f(0), f(1), ...$: surjective = every element appears at least once; injective = at most once. For Lemma 9.2 ("if"): delete the terms not in $B$ and all but the first appearance of each element; what remains lists $B$ exactly once.

**WORKED EXAMPLE Standard countable sets (L9 s24–30, s35) · T8 Q1, D1**
- $\mathbb{Z}$**:** list $0, 1, -1, 2, -2, ...$ Formula on $\mathbb{Z}^+$: $f(n) = n/2$ if $n$ is even, $-(n-1)/2$ if $n$ is odd. On $\mathbb{N}$ (T8 Q1): $g(n) = (-1)^n \left\lfloor \frac{n+1}{2} \right\rfloor$, giving $0, -1, 1, -2, 2, ...$
- $\mathbb{Z}^+ \times \mathbb{Z}^+$**:** count along the diagonals $(1,1), (1,2), (2,1), (1,3), (2,2), (3,1), ...$ The bijection is $f(x,y) = \frac{(x+y-2)(x+y-1)}{2} + x$.
- $\mathbb{Q}^+$**:** put $p/q$ in a grid, walk the diagonals, and skip values already counted: $\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{3}{1}, \frac{1}{3}, ...$ Skipping gives injectivity; every grid point is reached, giving surjectivity.
- $\mathbb{Q}$**:** $x \mapsto -x$ is a bijection $\mathbb{Q}^+ \to \mathbb{Q}^-$, so $\mathbb{Q}^-$ is countably infinite. List $\mathbb{Q}^+$ as $q_1, q_2, q_3, ...$; then $0, q_1, -q_1, q_2, -q_2, ...$ contains every element of $\mathbb{Q} = \mathbb{Q}^- \cup \{0\} \cup \mathbb{Q}^+$ exactly once, so $\mathbb{Q}$ is countable (Proposition 9.1) and $|\mathbb{Q}| = |\mathbb{Z}|$. Equivalently: Lemma 9.4 for $\mathbb{Q}^- \cup \mathbb{Q}^+$, then T8 Q2 for $\{0\}$.
- **Perfect squares** $\{0, 1, 4, 9, ...\}$ (T8 D1): $n \mapsto n^2$ is a bijection from $\mathbb{N}$; also a subset of $\mathbb{N}$ (Theorem 7.4.3).

**PROOF T8 Q1 · $g : \mathbb{N} \to \mathbb{Z}$, $g(n) = (-1)^n \left\lfloor \frac{n+1}{2} \right\rfloor$ is a bijection (so $\mathbb{Z}$ is countable with $\aleph_0 = |\mathbb{N}|$)**
For even $n$: $g(n) = \lfloor (n+1)/2 \rfloor = n/2 \geq 0$. For odd $n$: $g(n) = -(n+1)/2 < 0$.
1. (Injective) Let $a, b \in \mathbb{N}$ with $g(a) = g(b)$. Then $g(a), g(b)$ are both non-negative or both negative.
   1.1. Case 1 (both non-negative): $a$ and $b$ are even, so $a/2 = b/2$ and $a = b$.
   1.2. Case 2 (both negative): $a$ and $b$ are odd, so $-(a+1)/2 = -(b+1)/2$ and $a = b$.
2. (Surjective) Let $m \in \mathbb{Z}$.
   2.1. Case 1 ($m \geq 0$): let $n = 2m$. Then $n \in \mathbb{N}$ and $g(n) = (-1)^{2m} \lfloor (2m+1)/2 \rfloor = m$.
   2.2. Case 2 ($m < 0$): let $n = -2m - 1$. Then $n \in \mathbb{N}$ is odd and $g(n) = (-1)^{-2m-1} \lfloor (-2m)/2 \rfloor = -(-m) = m$.
3. Therefore $g$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}$. ■

**THEOREM Closure theorems (L9 s31, s41–45)**
- **Cartesian product.** If $A$ and $B$ are countably infinite, so is $A \times B$; likewise $A_1 \times \cdots \times A_n$ for $n \geq 2$.
- **Unions.** The union of countably many countable sets is countable: if $A_1, A_2, ...$ are countable, so is $\bigcup_{i=1}^{\infty} A_i$.
- **Theorem 7.4.3.** Any subset of a countable set is countable. **Corollary 7.4.4.** Any set with an uncountable subset is uncountable.
- **Proposition 9.3.** Every infinite set has a countably infinite subset.
- **Lemma 9.4.** If $A$ and $B$ are countably infinite, then $A \cup B$ is countable.

**PROOF Lemma 9.4 (L9 s45) · Proposition 9.3 (L9 s44) · Theorem 7.4.3 (L9 s41–42)**
**Lemma 9.4.** Take sequences $a_0, a_1, ...$ and $b_0, b_1, ...$ in which every element of $A$, respectively $B$, appears *(Lemma 9.2)*. Then $a_0, b_0, a_1, b_1, ...$ is a sequence in which every element of $A \cup B$ appears, so $A \cup B$ is countable *(Lemma 9.2)*.
**Proposition 9.3.** Let $B$ be infinite. Choose $b_0, b_1, b_2, ...$ from $B$ one at a time, each different from all earlier choices; this is always possible, since otherwise $B = \{b_0, ..., b_{n-1}\}$ would be finite. Then $\{b_0, b_1, ...\} \subseteq B$ is countably infinite.
**Theorem 7.4.3.** Let $A$ be countable and $B \subseteq A$. If $B$ is finite it is countable. Otherwise $A$ is countably infinite;

take a bijection $f : \mathbb{Z}^+ \to A$ and let $M = f^{-1}(B)$. Define $g(1) = f(i_1)$ with $i_1 = \min M$ and $g(k) = f(i_k)$ with $i_k = \min\{m \in M : m > i_{k-1}\}$. Then $g : \mathbb{Z}^+ \to B$ lists the elements of $B$ in order of their indices, each exactly once, so it is a bijection. ■

PROOF **T8 Q2 · $B$ countably infinite, $C$ finite ⇒ $B \cup C$ countable** **T8 Q5 · A countable union of countably infinite sets is countable**

**Q2.** Take a sequence $b_0, b_1, ...$ in which every element of $B$ appears *(Lemma 9.2)* and write $C = \{c_0, ..., c_{n-1}\}$. Then $c_0, ..., c_{n-1}, b_0, b_1, ...$ contains every element of $B \cup C$, so $B \cup C$ is countable *(Lemma 9.2)*. (As a bijection: with $C \setminus B = \{c_0, ..., c_{k-1}\}$ and a bijection $f : \mathbb{N} \to B$, set $g(x) = c_x$ if $x < k$ and $f(x - k)$ otherwise.)

**Q5.**

1. $\mathbb{Z}^+ \times \mathbb{Z}^+$ is countable, so there is a bijection $f : \mathbb{Z}^+ \to \mathbb{Z}^+ \times \mathbb{Z}^+$.
2. For each $i \in \mathbb{Z}^+$ take a sequence $b_{i,1}, b_{i,2}, ...$ in which every element of $S_i$ appears *(Lemma 9.2)*.
3. Define $c_k = b_{i,j}$ where $(i, j) = f(k)$.
4. Let $x \in \bigcup_{i \in \mathbb{Z}^+} S_i$. Then $x \in S_i$ for some $i$, so $x = b_{i,j}$ for some $j$; and $f(k) = (i, j)$ for some $k$ *(f is surjective)*. So $x = c_k$.
5. Every element of the union appears in $c_1, c_2, ...$, so the union is countable *(Lemma 9.2)*. ■

PROOF **T8 Q4(a) · If $A_1, A_2, A_3, ...$ are countable, then $\bigcup_{i=1}^n A_i$ is countable for every $n \in \mathbb{Z}^+$**

1. Let $P(n) \equiv \left(\bigcup_{i=1}^n A_i \text{ is countable}\right)$ for $n \in \mathbb{Z}^+$.
2. Basis step: $\bigcup_{i=1}^1 A_i = A_1$ is countable, so $P(1)$ is true.
3. Inductive step: suppose $P(k)$ is true for an arbitrary $k \geq 1$. Write $\bigcup_{i=1}^{k+1} A_i = \left(\bigcup_{i=1}^k A_i\right) \cup A_{k+1} = U \cup A_{k+1}$.
   3.1. Case 1 ($U$ and $A_{k+1}$ both finite): $U \cup A_{k+1}$ is finite, hence countable.
   3.2. Case 2 (exactly one is finite): the other is countably infinite, so $U \cup A_{k+1}$ is countable *(T8 Q2)*.
   3.3. Case 3 (neither is finite): both are countably infinite, so $U \cup A_{k+1}$ is countable *(Lemma 9.4)*.
4. Therefore $\bigcup_{i=1}^n A_i$ is countable for every $n \in \mathbb{Z}^+$ by MI. ■

(b) This does not prove that $\bigcup_{i=1}^\infty A_i$ is countable; see the trap below and T8 Q5.

TRAP **T8 Q3, Q4 · Induction reaches every finite $n$, never infinity**

- "$\bigcup_{i=1}^n A_i$ is finite (or countable) for every $n$" is proved by induction on $n$. It does **not** follow that $\bigcup_{i=1}^\infty A_i$ is finite: $A_i = \{i\}$ gives $\mathbb{Z}^+$. The infinite union of countable sets **is** countable, but that needs the argument of T8 Q5.
- In an inductive step the new set $A_{k+1}$ must be **arbitrary**; choosing $A_{k+1} = \varnothing$ proves nothing.
- T8 D2: "uncountable and infinite" is redundant, since every uncountable set is infinite.

## 38 · Uncountable sets

PROOF **Theorem 7.4.2 (Cantor) · The interval $(0, 1) = \{x \in \mathbb{R} : 0 < x < 1\}$ is uncountable (L9 s38–40)**

1. Suppose $(0, 1)$ is countable. It is not finite, so it is countably infinite.
2. List its elements as a sequence $x_1, x_2, x_3, ...$ with decimal expansions $x_n = 0.a_{n1}a_{n2}a_{n3}...$, each $a_{ij} \in \{0, 1, ..., 9\}$ (use 0.5000... and not 0.4999...).
3. Construct $d = 0.d_1d_2d_3...$ with $d_n = 1$ if $a_{nn} \neq 1$, and $d_n = 2$ if $a_{nn} = 1$.
4. For every $n \in \mathbb{Z}^+$, $d_n \neq a_{nn}$, so $d \neq x_n$.
5. But $d \in (0, 1)$, so $d$ should be in the list: contradiction. Therefore $(0, 1)$ is uncountable. ■

This is **Cantor's diagonalization argument**: build an object that differs from the $n$-th listed object at position $n$.

TRAP **Week 8 deck FAQ · Why diagonalization does not make $\mathbb{Z}^+$ uncountable**

List the positive integers in decimal, padded with leading zeros (...000001, ...000002, ...), and change one digit of each along a diagonal. The resulting object typically has **infinitely many non-zero digits** extending to the left, like ...3141592653. An integer has a finite decimal representation $d_nd_{n-1}\cdots d_2d_1$, so this object is **not a positive integer**: it is missing from the list, but it was never required to be on it. A real in $(0, 1)$ may have infinitely many digits $0.d_1d_2d_3...$, so there the diagonal object **is** in the set.

Same contrast: the set of all **finite** binary strings $\{\varepsilon, 0, 1, 00, 01, ...\}$ is countable; the set of all **infinite** binary strings $b_1b_2b_3...$ is uncountable.

THEOREM **Consequences (L9 s43, s46–47)**

- $\mathbb{R}$ is uncountable, since $(0, 1) \subseteq \mathbb{R}$ *(Corollary 7.4.4)*.
- $|\mathbb{R}| = |(0, 1)|$: bend $(0, 1)$ into a circle resting on the number line at 0 and project each point from the top of the circle onto the line; this projection is a bijection. (One formula: $x \mapsto \tan(\pi(x - 1/2))$.)
- The irrationals $\mathbb{R} \setminus \mathbb{Q}$ are uncountable: otherwise $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be countable.

PROOF **T8 Q7 · If $A$ is countably infinite, then $\mathcal{P}(A)$ is uncountable**

1. Suppose $\mathcal{P}(A)$ is countable. It is infinite, since $\{a\} \in \mathcal{P}(A)$ for every $a \in A$.
2. There is a sequence $a_0, a_1, a_2, \ldots$ in which every element of $A$ appears exactly once, and a sequence $B_0, B_1, B_2, \ldots$ in which every element of $\mathcal{P}(A)$ appears exactly once *(Proposition 9.1)*.
3. Define $B = \{a_i : a_i \notin B_i\}$. Then $B \in \mathcal{P}(A)$.
4. Let $i \in \mathbb{N}$. If $a_i \notin B_i$ then $a_i \in B$; if $a_i \in B_i$ then $a_i \notin B$ (each $a_i$ appears once only). Either way $B \neq B_i$.
5. So $B$ is not in the sequence, contradicting line 2. Therefore $\mathcal{P}(A)$ is uncountable. ■

PROOF **T8 Q6 · For $X$ infinite and $Y$ finite there is a bijection $X \cup Y \to X$**

1. Take a countably infinite $X_0 \subseteq X$ *(Proposition 9.3)* and let $Y_0 = Y \setminus X$, a finite set.
2. $X_0 \cup Y_0$ is countably infinite *(T8 Q2)*, so there is a bijection $f : X_0 \cup Y_0 \to X_0$.
3. Define $g : X \cup Y \to X$ by $g(z) = f(z)$ if $z \in X_0 \cup Y_0$, and $g(z) = z$ otherwise. Then $g$ is a bijection. ■

METHOD **Countable or uncountable? (T8 D3 · AY25 final Q13, Q17, Q18)**

**Show countable:** exhibit a sequence containing every element (Lemma 9.2), or build the set from countable sets by subset, finite product, or countable union.

**Show uncountable:** diagonalize, or find an uncountable subset, or biject with $\mathcal{P}(\mathbb{N})$ or $\mathbb{R}$.

| Set | | Reason |
|---|---|---|
| $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Z}^+ \times \mathbb{Z}^+$, $\mathbb{Q} \times \mathbb{Q}$ (points with rational coordinates) | countable | sequences above; product theorem |
| all functions $\{0,1\} \to \mathbb{N}$ | countable | same as $\mathbb{N} \times \mathbb{N}$ |
| all 2-element subsets of $\mathbb{N}$; all finite strings over a finite alphabet | countable | inject into $\mathbb{N} \times \mathbb{N}$; countable union of finite sets |
| any subset of $\mathbb{Q}$; any partition of $\mathbb{N}$ (as a set of components) | countable | Theorem 7.4.3; send each component to its least element |
| $\mathbb{Z}^*$: all finite strings over the countably infinite alphabet $\mathbb{Z}$ (AY22/23 Q17) | countable | $\mathbb{Z}^* = \mathbb{Z}^0 \cup \mathbb{Z}^1 \cup \mathbb{Z}^2 \cup \cdots$; each $\mathbb{Z}^n$ is countable (general product); countable union (T8 Q5) |
| $\mathbb{Q}$; the rationals strictly between 0 and 1 (AY22/23 Q17) | countable | $\mathbb{Q}$ from $\mathbb{Q}^+$ (§37); subset of $\mathbb{Q}$ |
| $(0,1)$, $\mathbb{R}$, $\mathbb{R} \setminus \mathbb{Q}$ | uncountable | Theorem 7.4.2 and Corollary 7.4.4 |
| $\mathcal{P}(\mathbb{N})$; all functions $\mathbb{N} \to \{0,1\}$; all infinite sequences of integers | uncountable | T8 Q7; diagonalization |
| all relations on $\mathbb{Z}$ (AY22/23 Q17) | uncountable | a relation on $\mathbb{Z}$ is a subset of $\mathbb{Z} \times \mathbb{Z}$, so the set is $\mathcal{P}(\mathbb{Z} \times \mathbb{Z})$ with $\mathbb{Z} \times \mathbb{Z}$ countably infinite (T8 Q7) |
| the set of all partitions of $\mathbb{N}$ | uncountable | $\{A, \mathbb{N} \setminus A\}$ for each $A$ gives uncountably many |

**AY22/23 Q17.** Countable among (i) relations on $\mathbb{Z}$, (ii) rationals in $(0,1)$, (iii) $\mathbb{Z}^*$: only (ii) and (iii), option D.
**AY25 Q17.** $A$ countably infinite, $B$ uncountable: $A \cup B$ is uncountable (it contains $B$) and $A \cap B$ is countable (a subset of $A$). Both true.
**AY25 Q13.** $(\mathbb{Z} \times \mathbb{Z}^+)/\sim$ with $(a,b) \sim (c,d) \leftrightarrow ad = bc$ is in bijection with $\mathbb{Q}$, hence with $\mathbb{Z}$ and with $\mathbb{Z} \times \mathbb{Z}^+$: all countably infinite.

# Part IX · Counting and Probability

## 39 · Sample spaces, the multiplication rule, permutations

DEFINITION **Sample space, event (L10 s8) · Equally likely probability (L10 s9)**

A **sample space** is the set of all possible outcomes of a random process or experiment. An **event** is a subset of a sample space.

If $S$ is a **finite** sample space in which **all outcomes are equally likely** and $E$ is an event in $S$, then

$$P(E) = \frac{|E|}{|S|} = \text{the number of outcomes in } \frac{E}{\text{the total number of outcomes in}} S.$$

THEOREM **Theorem 9.1.1 · Number of elements in a list (L10 s12)** **Theorem 9.2.1 · Multiplication rule (L10 s21)**

- If $m \le n$ are integers, there are $n - m + 1$ integers from $m$ to $n$ inclusive.
- If an operation consists of $k$ steps, where step $i$ can be performed in $n_i$ ways **regardless of how the preceding steps were performed**, then the whole operation can be performed in $n_1 n_2 \cdots n_k$ ways.
- **Theorem 5.2.4 (L10 s24).** A finite set $A$ has $|\mathcal{P}(A)| = 2^{|A|}$ subsets: each element is picked or dropped.

WORKED EXAMPLE **L10 s13, Ex. 1–5 · Lists, possibility trees, the multiplication rule**

- **Multiples of 5 from 100 to 999:** $5 \cdot 20, ..., 5 \cdot 199$, so $199 - 20 + 1 = 180$; probability $180/900 = 1/5$.
- **Tournament** (play until one team wins two in a row or three in total): the possibility tree has **10** leaves; 4 of them need five games, so $P = 4/10 = 2/5$ if all are equally likely.
- **PINs** of 4 symbols from 36: $36^4 = 1\,679\,616$ with repetition; $36 \cdot 35 \cdot 34 \cdot 33 = 1\,413\,720$ without; $P(\text{no repeat}) \approx 0.8417$.
- **Officers** (president, treasurer, secretary from Ann, Bob, Cyd, Dan; Ann is not president; Cyd or Dan is secretary). $3 \cdot 3 \cdot 2 = 18$ is **wrong**: the number of choices for secretary depends on earlier choices. Reorder the steps: secretary (2 ways), then president (2), then treasurer (2): $2 \cdot 2 \cdot 2 = 8$.
- **ICANDOIT** from 26 tiles: without replacement the probability is 0 (the letter I is needed twice); with replacement it is $(1/26)^8$.

DEFINITION **Permutation (L10 s32) · $r$-permutation, $P(n, r)$ (L10 s36)**

A **permutation** of a set of objects is an ordering of the objects in a row. An ***r*-permutation** of a set of $n$ elements is an ordered selection of $r$ of its elements; their number is $P(n, r)$.

THEOREM **Theorem 9.2.2 · Theorem 9.2.3 (L10 s34, s37) · Circular permutations (T9 Q5)**

- A set with $n \ge 1$ elements has $n!$ permutations; $0! = 1$.
- For $1 \le r \le n$: $P(n, r) = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}$. $P(5, 2) = 20$, $P(7, 4) = 840$, $P(5, 5) = 120$.
- $n$ distinct objects can be arranged around a circle in $(n-1)!$ ways (rotations are the same arrangement).

WORKED EXAMPLE **L10 Ex. 6 · T9 Q1, Q2, Q5**

- **COMPUTER:** $8! = 40\,320$ orderings; with CO kept together in order, treat CO as one unit: $7! = 5040$; probability $7!/8! = 1/8$.
- **T9 Q1.** 5-card sequences with at least one picture card (12 picture cards, 40 others): with replacement $52^5 - 40^5 = 277\,804\,032$; without replacement $P(52, 5) - P(40, 5) = 232\,914\,240$.
- **T9 Q2 (AY23/24 exam).** Five-digit cheque numbers readable upside down use digits $0, 1, 6, 8, 9$: $5^5$. Those that read the same after rotation (the middle digit is 0, 1 or 8; the last two digits are forced): $5 \cdot 5 \cdot 3 = 75$. Confusable numbers: $5^5 - 75 = 3050$.
- **T9 Q5.** (a) 8 boys and 4 girls at a round table, no two girls adjacent: seat the boys ($7!$), then put the girls into 4 of the 8 gaps ($P(8, 4)$): $7! \cdot 1680 = 8\,467\,200$. (b) 6 people, Eric not next to Freddy: $5! - 2 \cdot 4! = 72$. (c) $n - 1$ people at $n$ chairs: treat the empty chair as a person: $(n-1)!$.

METHOD **Arrangements in a row and round a table (L10 Ex. 6 · Week 10 s16, s74–76 · T9 Q5)**

- **Kept together (glue).** Treat the block as one object, arrange the objects, then order the inside of each block. 12 distinct cards (4 suits, 3 ranks each) in a row with each suit together: $4! \cdot (3!)^4 = 31\,104$. Identical letters need no inner ordering: MISSISSIPPI with the four S's consecutive is the block (SSSS) with M, I, I, I, I, P, P: $8!/(4!\,2!) = 840$.
- **Not adjacent.** Either total minus glued, or the **gap method**: arrange the other objects first, then drop the separated ones into distinct gaps. 5 people in a row with X and Y apart: $5! - 2 \cdot 4! = 72 = 3! \cdot P(4, 2)$. A row of $m$ objects has $m + 1$ gaps; a circle of $m$ objects has $m$ gaps (T9 Q5(a)).
- **Alternating in a row.** 3 red and 3 black cards, all distinct: the pattern is RBRBRB or BRBRBR, each filled in $3! \cdot 3!$ ways: $2 \cdot 36 = 72$. Round a table the two patterns coincide: $(n-1)!\,n!$.
- **Relative order.** 6 distinct cards in a row with A somewhere to the left of B: swapping A and B pairs the arrangements off, so exactly half qualify: $6!/2 = 360$.
- **Round a table.** Fix one person's seat to remove the rotations, then count the rest as a row.

WORKED EXAMPLE **Case split when a step's count depends on earlier choices (Week 10 s77)**

**4-digit even numbers with no repeated digit.** Choose the last digit first; the number of choices for the first digit then depends on whether 0 has been used, so split.

- **Last digit 0:** first digit 9 ways, then 8, then 7: $9 \cdot 8 \cdot 7 = 504$.
- **Last digit 2, 4, 6 or 8:** first digit 8 ways (not 0, not the last digit), second 8, third 7: $4 \cdot 8 \cdot 8 \cdot 7 = 1792$.

Total $504 + 1792 = 2296$ *(addition rule on disjoint cases)*.

PAST PAPER **AY20/21 Q20(d) · AY23/24 Q22, Q25(f)**

**AY20/21 Q20(d).** Directed graphs on vertices $a, b, c$ without parallel edges: each of the $3 \cdot 3 = 9$ ordered pairs (loops included) is an edge or not: $2^9 = 512$. With at least one loop: subtract the graphs with no loop, $2^9 - 2^6 = 448$.

**AY23/24 Q22.** 8 people round a table, the three roommates not **all three** together: all seatings 7!; glue the three into one block (6 units round the table: 5!, inner orders 3!): $7! - 5! \cdot 3! = 4320$ (option B).

**AY23/24 Q25(f).** Five-digit cheque numbers that can be read both ways up: each digit is one of $0, 1, 6, 8, 9$: $5^5 = 3125$. Excluding the 75 that read the same after rotation (the T9 Q2 version) leaves 3050.

## 40 · Addition, difference, inclusion and exclusion

THEOREM **Theorems 9.3.1–9.3.3 (L10 s40, s43, s50) · Complement (L10 s48)**

- **Addition rule.** If a finite set $A$ is the union of mutually disjoint subsets $A_1, ..., A_k$, then $|A| = |A_1| + |A_2| + \cdots + |A_k|$.
- **Difference rule.** If $A$ is finite and $B \subseteq A$, then $|A \setminus B| = |A| - |B|$. (Proof: $B$ and $A \setminus B$ are disjoint with union $A$; apply the addition rule.)
- **Inclusion/exclusion.** For finite $A, B, C$: $|A \cup B| = |A| + |B| - |A \cap B|$ and
$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$
- **Complement.** For an event $A$ in a finite sample space $S$: $P(\overline{A}) = 1 - P(A)$.

WORKED EXAMPLE **L10 Ex. 7–9 · T9 Q3**

- **Passwords** of 1 to 3 letters: $26 + 26^2 + 26^3 = 18\,278$ (addition rule over the lengths).
- **PINs with a repeated symbol:** $36^4 - 36 \cdot 35 \cdot 34 \cdot 33 = 265\,896$; probability $\approx 0.158 = 1 - 0.842$.
- **Multiples of 3 or 5 from 1 to 1000:** $333 + 200 - 66 = 467$ (multiples of both are the multiples of 15). **Neither:** $1000 - 467 = 533$, using De Morgan: $\overline{A} \cap \overline{B} = \overline{A \cup B}$.
- **T9 Q3.** Permutations of $1, ..., n$ ($n \geq 3$) with 1, 2 or 3 in its correct position: $|P_k| = (n-1)!$, $|P_i \cap P_j| = (n-2)!$, $|P_1 \cap P_2 \cap P_3| = (n-3)!$, so the count is $3(n-1)! - 3(n-2)! + (n-3)! = (3n^2 - 12n + 13)(n-3)!$.

METHOD **"At least one" and "at most"**

"At least one" is usually shortest by the difference rule: total minus "none". "At most one" is "none" plus "exactly one" by the addition rule. Overlapping conditions joined by "or" need inclusion/exclusion. Before multiplying, check that each step's count is independent of earlier choices; if not, reorder the steps or split into cases.

METHOD **Each at least once: complement, then inclusion/exclusion (T10 D3 · Week 10 s13–14 · AY23/24 Q25(g) · AY24/25 Q24(a))**

"Every value is used" is the complement of "some value is missed". Let $A_i$ be the set of outcomes that miss value $i$ and subtract $|A_1 \cup \cdots \cup A_m|$ from the total.

- **Two values:** $2^n - 2$ (remove the two constant strings). **Three values:** $3^n - 3 \cdot 2^n + 3$. These are the surjections from an $n$-set onto a 2-set and a 3-set (AY23/24 Q25(g)); the general formula is in §43.
- **Week 10 Ex. 3.** 7 balls in a row, balls of one colour identical, exactly 3 red and at least one blue and one green: place the reds, then fill 4 places with blue or green but not all one colour: $\binom{7}{3}(2^4 - 2) = 35 \cdot 14 = 490$.
- **Week 10 Ex. 2.** 7 books from 6 maths, 5 CS, 4 physics (all distinct), at least 2 per subject: split by distribution $(3,2,2)$, $(2,3,2)$, $(2,2,3)$: $\binom{6}{3}\binom{5}{2}\binom{4}{2} + \binom{6}{2}\binom{5}{3}\binom{4}{2} + \binom{6}{2}\binom{5}{2}\binom{4}{3} = 1200 + 900 + 600 = 2700$.
- **AY24/25 Q24(a).** 5 students each pick one of 5 GPUs at random. $P(\text{some GPU has more than one student}) = 1 - P(\text{all different}) = 1 - 5!/5^5 = 1 - 120/3125 = 0.9616$.

TRAP **Reserve first, then choose freely**

"Pick 2 books from each subject, then any 1 of the remaining 9" gives $15 \cdot 10 \cdot 6 \cdot 9 = 8100$, not 2700: a selection with 3 maths books is counted once for each choice of which maths book was the "extra". Reserving is safe only for **identical** objects (stars and bars); for distinct objects split into cases or use the complement.

## 41 · The pigeonhole principle in counting

THEOREM **Pigeonhole Principle · Generalized PHP and its contrapositive (L10 s55, s62, s65)**

- **PHP.** A function from one finite set to a smaller finite set cannot be one-to-one: at least two elements of the domain have the same image.
- **Generalized PHP.** For $f : X \to Y$ with $|X| = n$, $|Y| = m$ and a positive integer $k$: if $k < n/m$, then some $y \in Y$ is the image of at least $k+1$ distinct elements of $X$.
- **Contrapositive form.** If for each $y \in Y$ the set $f^{-1}(\{y\})$ has at most $k$ elements, then $n \le km$.

Equivalently: among $n$ pigeons in $m$ holes, some hole holds at least $\lceil n/m \rceil$ pigeons.

METHOD **Writing a pigeonhole argument**

State (1) what the pigeons are, (2) what the holes are, (3) the rule sending each pigeon to a hole, (4) the two counts, then conclude with the principle by name. For "how many must be drawn to guarantee ...", find the **largest** selection that avoids the target and add 1.

WORKED EXAMPLE **L10 Ex. 10–12 · Decimal expansions (L10 s59–61)**

- 13 people: two share a birth month ($13 > 12$). Singapore: two people have the same number of hairs (5.47 million people, at most 300 001 possible counts).
- **85 people, 26 initials:** $3 < 85/26$, so at least 4 share a last initial. By the contrapositive: if at most 3 shared each initial there would be at most $3 \cdot 26 = 78$ people.
- **Rational numbers have terminating or repeating decimals.** In the long division of $a$ by $b$ every remainder lies in $\{0, ..., b-1\}$ *(Theorem 4.4.1)*. If no remainder is 0, then among $b+1$ successive remainders two are equal, $r_j = r_k$ with $j < k$ *(PHP)*, and the digits repeat from there. $3/14 = 0.2\overline{142857}$: remainders $3, 2, 6, 4, 12, 8, 10, 2, ...$
- **42 students, 12 computers, at most 6 per computer: at least 5 computers are used by 3 or more students.** Let $k$ be the number of such computers. They serve at most $6k$ students; the other $12-k$ serve at most $2(12-k)$. So $42 \le 6k + 24 - 2k = 4k + 24$, giving $k \ge 4.5$, hence $k \ge 5$.

PROOF **T9 Q8 (AY21/22 exam) · Among any 5 distinct non-negative integers, two differ by a multiple of 4 T9 Q9 · Chess master**

**Q8.**

1. Let the pigeonholes be the classes $[0], [1], [2], [3]$ of congruence mod 4, and send each of the 5 integers to its class.
2. Two of the integers, $x$ and $y$, lie in the same class $[i]$ *(PHP: $5 > 4$)*. So $x = 4k + i$ and $y = 4l + i$ for some $k, l \in \mathbb{Z}$.
3. Then $x - y = 4(k-l)$ with $k - l \in \mathbb{Z}$ *(closure)*, so $4 \mid (x - y)$. ■

**Q9.** 77 days, at least one game a day, at most 12 a week. Let $P_i$ be the number of games played up to day $i$: $1 \le P_1 < P_2 < \cdots < P_{77} \le 132$. Let $Q_i = P_i + 21 \le 153$. The 154 numbers $P_1, ..., P_{77}, Q_1, ..., Q_{77}$ take values in $\{1, ..., 153\}$, so two are equal *(PHP)*. The $P_i$ are distinct and so are the $Q_i$, hence $P_j = Q_i = P_i + 21$ for some $i, j$: exactly 21 games were played on days $i+1$ to $j$. ■

PAST PAPER **T10 Q10 · T10 Q12 · AY25 final Q24(f)**

**T10 Q10.** At a party of $n \ge 2$ people where the host shakes hands with everyone, two people shook the same number of hands: every degree lies in $\{1, ..., n-1\}$ (each person shook the host's hand), so $n$ people share $n-1$ possible values *(PHP)*.

**T10 Q12.** Any simple graph $G$ on 6 vertices, or its complement, contains a triangle. Colour the edges of $K_6$ black (in $G$) or red (in $\overline{G}$). Fix a vertex $v$: of its 5 edges, at least 3 have one colour *(generalized PHP, $2 < 5/2$)*, say red, to $u_1, u_2, u_3$. If some edge $\{u_x, u_y\}$ is red, then $v, u_x, u_y$ form a red triangle; otherwise $u_1, u_2, u_3$ form a black triangle.

**AY25 Q24(f).** Socks: 8 red, 7 blue, 6 green, 5 black, 4 white. To avoid both "4 of one colour" and "3 each of two colours", take 3 of one colour and 2 of each of the other four: 11 socks. So **12** guarantee one of the two.

PROOF **AY20/21 Q20(f) · 21 students, non-negative integer scores with sum 200: two students have the same score**

1. Suppose not: the 21 scores are distinct. *(for contradiction)*
2. List them in increasing order, $s_1 < s_2 < \cdots < s_{21}$. Then $s_1 \ge 0$ and $s_{i+1} \ge s_i + 1$, so $s_i \ge i - 1$ for every $i$ *(scores are non-negative integers)*.
3. So $s_1 + \cdots + s_{21} \ge 0 + 1 + \cdots + 20 = (20 \cdot 21)/2 = 210 > 200$, contradicting the sum 200.
4. Hence two students have the same score. ■

The pattern: if every hole holds at most one pigeon, the pigeons are forced into the cheapest distinct holes, and even that total is too large.

PAST PAPER **AY22/23 Q22(d) · AY24/25 Q21 · AY23/24 Q21 (divisibility chains as holes)**

**AY22/23 Q22(d).** 838 students, 12 birth months: some month has at least $\lceil 838/12 \rceil = 70$ students ($69 \cdot 12 = 828 < 838$).

**AY24/25 Q21.** 25 labelled simple graphs on 3 vertices. Holes are the isomorphism classes, fixed by the number of edges $0, 1, 2, 3$: 4 holes, so at least $\lceil 25/4 \rceil = 7$ are isomorphic to one another (option C).

**AY23/24 Q21.** $S = \{1, 3, 5, ..., 21\}$; smallest $n$ such that any $n$ numbers from $S$ contain two with one dividing the other. Holes are **chains** in which every two members are comparable under "divides":

$$\{1, 3, 9\}, \quad \{5, 15\}, \quad \{7, 21\}, \quad \{11\}, \quad \{13\}, \quad \{17\}, \quad \{19\}.$$

These 7 chains cover $S$, so any 8 numbers put two in one chain *(PHP)*. Seven do not suffice: $\{9, 11, 13, 15, 17, 19, 21\}$ has no dividing pair. So $n = 8$ (option C).

**Same idea.** Among any $n+1$ numbers from $\{1, ..., 2n\}$ one divides another: write each as $2^k m$ with $m$ odd; there are only $n$ odd parts, and two numbers with the same odd part form a dividing pair.

## 42 · Combinations and multisets

DEFINITION **$r$-combination, $\binom{n}{r}$ (L11 s4) · Multiset (L11 s18)**

For integers $0 \le r \le n$, an ***r*-combination** of a set of $n$ elements is a **subset** of $r$ of them. $\binom{n}{r}$, read "$n$ choose $r$", is the number of $r$-combinations (also written $C(n, r)$, ${}_nC_r$).

An ***r*-combination with repetition allowed**, or **multiset of size *r***, chosen from a set $X$ of $n$ elements is an unordered selection of $r$ elements of $X$ with repetition allowed, written $\left[x_{i_1}, x_{i_2}, ..., x_{i_r}\right]$.

THEOREM **Theorems 9.5.1, 9.5.2, 9.6.1 (L11 s10, s15, s21)**

- **9.5.1.** $\binom{n}{r} = \frac{P(n,r)}{r!} = \frac{n!}{r!(n-r)!}$, because an $r$-permutation is an $r$-subset followed by one of its $r!$ orderings: $P(n, r) = \binom{n}{r} \cdot r!$.
- **9.5.2 Permutations with indistinguishable objects.** If $n$ objects consist of $n_1$ of type 1, ..., $n_k$ of type $k$, with $n_1 + \cdots + n_k = n$, the number of distinguishable orderings is

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\cdots-n_{k-1}}{n_k} = \frac{n!}{n_1!\, n_2! \cdots n_k!}.$$

- **9.6.1 Multisets.** The number of multisets of size $r$ from $n$ elements, equivalently the number of ways to select $r$ objects from $n$ categories with repetition, is $\binom{r+n-1}{r}$. (Stars and bars: a string of $r$ crosses and $n-1$ bars.)

| **Choosing $k$ from $n$ (L11 s23)** | **Order matters** | **Order does not matter** |
|---|---|---|
| Repetition allowed | $n^k$ | $\binom{k+n-1}{k}$ |
| Repetition not allowed | $P(n, k) = n!/(n-k)!$ | $\binom{n}{k} = n!/(k!(n-k)!)$ |

METHOD **Stars and bars (L11 s18–21)**

Line up $r$ identical stars and cut them into $n$ groups with $n-1$ bars: $\star\star\,|\,|\,\star\,|\,\star\star$ is the selection $(2, 0, 1, 2)$. Every string of $r$ stars and $n-1$ bars is one selection, so the count is $\binom{r+n-1}{r} = \binom{r+n-1}{n-1}$. The same count answers three questions: multisets of size $r$ from $n$ types; ways to put $r$ **identical** objects into $n$ **distinct** boxes; non-negative integer solutions of $x_1 + \cdots + x_n = r$.

- **Every $x_i \ge 1$.** Hand one to each box first, or place the bars in the $r-1$ gaps between stars: $\binom{r-1}{n-1}$.
- **Lower bounds $x_i \ge a_i$.** Substitute $y_i = x_i - a_i$, then solve $y_1 + \cdots + y_n = r - \sum a_i$.
- **Inequality $x_1 + \cdots + x_n \le r$.** Add a slack variable $x_{n+1} \ge 0$ and solve with equality: $\binom{r+n}{n}$.
- **Upper bound $x_1 \le c$.** Count all solutions and subtract those with $x_1 \ge c+1$; with several upper bounds use inclusion and exclusion.
- **Parity or multiples.** If every $x_i$ must be even, substitute $x_i = 2y_i$ and solve $y_1 + \cdots + y_n = r/2$ (no solutions when $r$ is odd). For $x_i$ odd use $x_i = 2y_i + 1$.
- **Bounded both ways** (dice, $1 \le x_i \le 6$): shift to $y_i = x_i - 1 \in \{0, ..., 5\}$, then remove the solutions with some $y_i \ge 6$.
- **Does not apply** when the objects are distinct ($n^r$ ways to place them) or when order within the selection matters.

**Checks.** $x_1 + x_2 + x_3 = 10$ with $x_1 \le 4$: $\binom{12}{2} - \binom{7}{2} = 66 - 21 = 45$. 12 identical sweets to 4 children, each getting at least 2: $y_1 + \cdots + y_4 = 4$, so $\binom{7}{3} = 35$.

WORKED EXAMPLE **L11 Ex. 1–7**

- **Committees.** 3-person committees from 4 people: $\binom{4}{3} = 4$. $P(4, 2) = 12 = \binom{4}{2} \cdot 2!$, so $\binom{4}{2} = 6$.

- **Teams of 5 from 5 men and 7 women.** 3 men and 2 women: $\binom{5}{3}\binom{7}{2} = 10 \cdot 21 = 210$. At least one man: $\binom{12}{5} - \binom{7}{5} = 792 - 21 = 771$. At most one man: $\binom{7}{5} + \binom{5}{1}\binom{7}{4} = 21 + 175 = 196$.
- **MISSISSIPPI** (4 S, 4 I, 2 P, 1 M): $\binom{11}{4}\binom{7}{4}\binom{3}{2}\binom{1}{1} = \frac{11!}{4!\,4!\,2!\,1!} = 34\,650$.
- **Multisets of size 3 from $\{1,2,3,4\}$:** $\binom{3+4-1}{3} = \binom{6}{3} = 20$; e.g. $[1,3,4]$ is the string $\times\,|\,|\,\times\,|\,\times$.
- **Lunch for 20 guests from 6 stalls:** $\binom{20+6-1}{20} = \binom{25}{20} = 53\,130$.
- **$x_1 + x_2 + x_3 = 20$:** non-negative integer solutions $\binom{22}{20} = 231$. Positive solutions: put $y_i = x_i - 1$, so $y_1 + y_2 + y_3 = 17$: $\binom{19}{17} = 171$.

PAST PAPER **T9 Q4, Q6, Q7 · T10 Q2 · AY25 final Q24(a)–(e), (g)**

**T9 Q6.** 5 from 7 men and 6 women with at least 3 men: $\binom{7}{3}\binom{6}{2} + \binom{7}{4}\binom{6}{1} + \binom{7}{5} = 525 + 210 + 21 = 756$; probability $756/\binom{13}{5} = 756/1287 \approx 0.5874$.

**T9 Q7 (AY21/22).** \$50 000 over 4 properties in \$1000 units: $x_1 + \cdots + x_4 = 50$ has $\binom{53}{50} = 23\,426$ solutions. Not necessarily all invested: add a slack variable $x_5$: $\binom{54}{50} = 316\,251$.

**T9 Q4, T10 Q2.** $n$ boxes, white balls in a non-empty consecutive block: $\sum_{k=1}^{n}(n-k+1) = n(n+1)/2$; or choose the two ends of the block among $n+1$ gaps: $\binom{n+1}{2}$.

**AY25 Q24.** (a) 3 red, 4 blue, 5 green balls in a row: $12!/(3!\,4!\,5!) = 27\,720$. (b) Committee of 5 from 6 men and 5 women with at least 3 women: $\binom{5}{3}\binom{6}{2} + \binom{5}{4}\binom{6}{1} + \binom{5}{5} = 150 + 30 + 1 = 181$. (c) 6 distinct letters with exactly 2 vowels: choose the 2 vowel positions, fill them, fill the rest: $\binom{6}{2} \cdot P(5,2) \cdot P(21,4) = 15 \cdot 20 \cdot 143\,640 = 43\,092\,000$. (d) $n$ couples at a round table, men and women alternating: $(n-1)!\,n!$. (e) Every woman next to her husband: seat the $n$ couples as units, then order each couple: $(n-1)!\,2^n$. (g) Three distinguishable dice with sum 10: $x_1 + x_2 + x_3 = 10$ with $1 \le x_i \le 6$. Put $y_i = x_i - 1$: $y_1 + y_2 + y_3 = 7$ with $0 \le y_i \le 5$. Unrestricted: $\binom{9}{2} = 36$. Bad solutions with a fixed $y_i \ge 6$: give that die 6 first, leaving sum 1 over three variables, $\binom{3}{2} = 3$; two dice cannot both be bad ($12 > 7$). So $36 - 3 \cdot 3 = 27$.

PAST PAPER **Stars and bars in the finals · AY20/21 Q20(a) · AY22/23 Q22(a) · AY23/24 Q25(d) · AY24/25 Q24(c)**

**AY20/21 Q20(a).** $x + y + z = 88$ with $x, y, z \ge 3$: subtract 3 from each, sum 79 over 3 variables: $\binom{81}{2} = 3240$.

**AY22/23 Q22(a).** $x_1 + x_2 + x_3 = 10$ with $x_i \ge i$: put $y_i = x_i - i$, sum $10 - 6 = 4$: $\binom{6}{2} = 15$.

**AY23/24 Q25(d).** 16 identical tasks to 4 distinct processors, each getting an even number: $x_i = 2y_i$, $y_1 + \cdots + y_4 = 8$: $\binom{11}{3} = 165$.

**AY24/25 Q24(c).** $x_1 + x_2 + x_3 + x_4 = 12$, $x_i \ge 0$: (i) $\binom{15}{3} = 455$. (ii) With $x_4 \le 3$: remove the solutions with $x_4 \ge 4$ (sum 8 after giving $x_4$ four): $455 - \binom{11}{3} = 455 - 165 = 290$.

PAST PAPER **Selections and arrangements · Week 10 Ex. 1, 4 · AY20/21 Q20(b) · AY23/24 Q25(c) · AY24/25 Q24(b)**

**Week 10 Ex. 1.** 5 distinct integers from $\{1, \ldots, 12\}$: exactly 2 even, $\binom{6}{2}\binom{6}{3} = 300$; at least one multiple of 3, $\binom{12}{5} - \binom{8}{5} = 792 - 56 = 736$.

**Week 10 Ex. 4.** BANANA with A first and last: arrange the remaining A, B, N, N: $4!/2! = 12$.

**AY20/21 Q20(b)** (a repeated letter). 4 tiles from I, C, A, N, D, O, I, T, order ignored. Split on the repeated letter: with a duplicate, take both I's and 2 of the other 6 letters, $\binom{6}{2} = 15$; with no duplicate, take 4 of the 7 distinct letters, $\binom{7}{4} = 35$. (This counts selections of letters; do not use $\binom{8}{4} = 70$, which treats the two I tiles as different.)

**AY23/24 Q25(c).** Morse messages with five dashes and three dots: choose the positions of the dots, $\binom{8}{3} = 8!/(5!\,3!) = 56$.

**AY24/25 Q24(b).** $\binom{n}{3} = k$ and $\binom{n-1}{2} = k/2$. Since $\binom{n}{3} = \frac{n}{3}\binom{n-1}{2}$, we need $n/3 = 2$: $n = 6$ (check: 20 and 10).

## 43 · Pascal's formula, the binomial theorem, combinatorial proofs

THEOREM **Theorem 9.7.1 Pascal's formula (L11 s26) · Theorem 9.7.2 Binomial theorem (L11 s34) · Identities (L11 s29–33)**

- **Pascal.** For positive integers $r \le n$: $\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$.
- **Binomial theorem.** For real $a, b$ and integer $n \ge 0$: $(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$. $\binom{n}{r}$ is a **binomial coefficient**.
- $\binom{n}{r} = \binom{n}{n-r}$; $k\binom{n}{k} = n\binom{n-1}{k-1}$; $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$.

DEFINITION **Combinatorial proof (L11 s27)**

A **combinatorial proof** uses counting as the basis of the proof. Two kinds: a **bijective proof** (a bijection between two sets shows they have the same size) and a **proof by double counting** (count one set in two ways; the two expressions are equal).

PROOF **Pascal's formula, algebraic and combinatorial (L11 s28) · Three double-counting proofs (L11 s30–33)**

**Algebraic.** $\binom{n}{r-1}+\binom{n}{r}=\frac{n!}{(n-r+1)!(r-1)!}+\frac{n!}{(n-r)!\,r!}=\frac{n!\,r+n!(n-r+1)}{(n-r+1)!\,r!}=\frac{(n+1)!}{(n+1-r)!\,r!}=\binom{n+1}{r}$.

**Combinatorial.**

1. $\binom{n+1}{r}$ counts the $r$-subsets of a set $A$ with $n+1$ elements. Fix $x\in A$.
2. The $r$-subsets containing $x$: choose the other $r-1$ elements from the remaining $n$: $\binom{n}{r-1}$.
3. The $r$-subsets not containing $x$: choose all $r$ from the remaining $n$: $\binom{n}{r}$.
4. The two cases are disjoint and exhaustive, so $\binom{n+1}{r}=\binom{n}{r-1}+\binom{n}{r}$ *(addition rule)*. ■

**$\binom{n}{k}=\binom{n}{n-k}$.** Choosing the $k$ pupils who go is the same as choosing the $n-k$ who stay.

**$k\binom{n}{k}=n\binom{n-1}{k-1}$.** Count committees of $k$ from $n$ with a chairperson: choose the committee then the chair, or choose the chair then the other $k-1$ members.

**$\sum_{k=0}^{n}\binom{n}{k}=2^n$.** Both count the subsets of an $n$-element set: by size on the left, by Theorem 6.3.1 on the right.

WORKED EXAMPLE **L11 Ex. 12, 13 · T10 Q1**

- $(a+b)^5=a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$.
- $(x-4y)^4=x^4-16x^3y+96x^2y^2-256xy^3+256y^4$ (substitute $b=-4y$ and keep the sign inside the power).
- $\sum_{k=0}^{n}\binom{n}{k}9^k=\sum_{k=0}^{n}\binom{n}{k}1^{n-k}9^k=(1+9)^n=10^n$.
- **T10 Q1.** Term independent of $x$ in $(2x^2+1/x)^9$: the general term is $\binom{9}{r}(2x^2)^{9-r}x^{-r}=\binom{9}{r}2^{9-r}x^{18-3r}$; $18-3r=0$ gives $r=6$: $\binom{9}{6}\cdot 2^3=84\cdot 8=672$.

THEOREM **Odd and even binomial sums · The $k$-th term (AY23/24 Q25(b) · AY20/21 Q20(c))**

- Put $a=1$, $b=-1$ in the binomial theorem: for $n\geq 1$, $\sum_{k=0}^{n}(-1)^k\binom{n}{k}=(1-1)^n=0$, so the even-$k$ and odd-$k$ coefficients have equal sums. Together they sum to $2^n$, hence
$$\binom{n}{0}+\binom{n}{2}+\binom{n}{4}+\cdots=\binom{n}{1}+\binom{n}{3}+\binom{n}{5}+\cdots=2^{n-1}.$$
A set with $n\geq 1$ elements has as many subsets of even size as of odd size.
- **AY23/24 Q25(b).** A fair coin tossed $n$ times shows an odd number of heads with probability $2^{n-1}/2^n=1/2$ for every $n\geq 1$, so $k=2$.
- **The $k$-th term** of $(a+b)^n$ is the one with $r=k-1$: $\binom{n}{k-1}a^{n-k+1}b^{k-1}$. The first term is $r=0$.
- **AY20/21 Q20(c).** Fifth term of $(1/(2\sqrt{x})-1/2)^{10}$ equal to 105: $r=4$, so $\binom{10}{4}\left(\frac{1}{2\sqrt{x}}\right)^6\left(-\frac{1}{2}\right)^4=\frac{210}{64x^3\cdot 16}=105$, giving $x^3=1/512$ and $x=1/8$.

METHOD **Counting by recurrence (Week 12 s25–29, AY18/19 Q17)**

When a direct count is awkward, ask what the $n$-th object adds to the count for $n-1$ objects, write the recurrence with its base case, then solve it by iteration and prove the closed form by induction (Part VII). Lazy caterer: the $n$-th cut crosses the $n-1$ earlier cuts and adds $n$ pieces, so $P(n)=P(n-1)+n$ with $P(0)=1$, giving $P(n)=(n^2+n+2)/2$.

THEOREM **Counting functions (T9 D3 · T10 D2, D3)**

For finite sets with $|A|=m$ and $|B|=n$:

| **Functions $A\to B$** | **Number** | $m=3,n=2$ | $m=n=3$ |
|---|---|---|---|
| all functions | $n^m$ | 8 | 27 |
| injections ($m\leq n$) | $P(n,m)=n!/(n-m)!$ | 0 | 6 |
| surjections ($m\geq n$) | $\sum_{i=0}^{n}(-1)^i\binom{n}{i}(n-i)^m$ | 6 | 6 |
| bijections ($m=n$) | $n!$ | 0 | 6 |

Surjections from a 5-element set to a 3-element set: $3^5-3\cdot 2^5+3\cdot 1^5=150$ (inclusion/exclusion on the values missed). **AY25 Q10:** strictly increasing $f:\{0,1\}\to\{0,1,2,3\}$: $\binom{4}{2}=6$; injective $g:\{0,1\}\to\{0,1,2\}$: $P(3,2)=6$. **AY22/23 Q22(g):** injections with $|A|=2$, $|B|=3$: 6; with $|A|=k\leq n=|B|$: $n!/(n-k)!$. **AY23/24 Q25(g):** surjections from an $n$-set onto a 2-set: $2^n-2$; onto a 3-set ($n\geq 2$): $3^n-3\cdot 2^n+3$.

## 44 · Probability axioms and expected value

**DEFINITION** **Probability axioms (L11 s39) · Expected value (L11 s43)**

Let $S$ be a sample space. A **probability function** $P$ from the set of all events in $S$ to $\mathbb{R}$ satisfies, for all events $A$, $B$:

1. $0 \le P(A) \le 1$; 2. $P(\varnothing) = 0$ and $P(S) = 1$; 3. if $A \cap B = \varnothing$, then $P(A \cup B) = P(A) + P(B)$.

If the possible outcomes of an experiment are real numbers $a_1, ..., a_n$ occurring with probabilities $p_1, ..., p_n$, the **expected value** is

$$\sum_{k=1}^{n} a_k p_k = a_1 p_1 + a_2 p_2 + \cdots + a_n p_n.$$

**THEOREM** **Complement · General union (L11 s40–41) · Linearity of expectation (L11 s48)**

- $P(\overline{A}) = 1 - P(A)$. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ for **any** events $A$, $B$.
- For random variables $X$, $Y$, **dependent or not**: $E[X + Y] = E[X] + E[Y]$; in general $E\left[\sum_{i=1}^{n} c_i X_i\right] = \sum_{i=1}^{n} c_i E[X_i]$.

**WORKED EXAMPLE** **L11 s41, Ex. 14–16 · T10 Q3, Q4, D1 · AY25 final Q23**

- **Heart or 7:** $\frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$.
- **Fair die:** $(1 + 2 + \cdots + 6)/6 = 3.5$. **Sum of two dice:** $3.5 + 3.5 = 7$ by linearity.
- **Lottery:** 500 000 tickets at \$5; prizes \$1 000 000 (1), \$1000 (10), \$500 (1000), \$10 (10 000). Expected net gain $= (999\,995 + 10 \cdot 995 + 1000 \cdot 495 + 10\,000 \cdot 5 - 488\,989 \cdot 5)/500\,000 = -1.78$ dollars.
- **T10 Q3 (AY20/21 S2).** Loaded die: each big number (4, 5, 6) is twice as likely as each small one, so $P(1) = P(2) = P(3) = 1/9$ and $P(4) = P(5) = P(6) = 2/9$. Maximum of two rolls: $P(\max = 1, ..., 6) = \frac{1}{81}, \frac{3}{81}, \frac{5}{81}, \frac{16}{81}, \frac{24}{81}, \frac{32}{81}$; expected value $398/81$.
- **T10 Q4.** Two balls from $\{1, 2, 2, 8, 8\}$: one ball has expectation $(1 + 2 + 2 + 8 + 8)/5 = 4.2$, so the sum has expectation 8.4 by linearity (the draws are dependent; linearity still applies).
- **T10 D1.** 2 bulbs from 8 with 3 defective: $E[X] = 2 \cdot 3/8 = 0.75$.
- **AY25 Q23.** Sundaes: 20% of customers choose 3 of 10 flavours, 80% choose 2: $0.2\binom{10}{3} + 0.8\binom{10}{2} = 24 + 36 = 60$.

**METHOD** **Expected count by indicator variables (L11 s48)**

To find the expected **number** of objects with some property, do not find the distribution of the count. For each object $i$ let $X_i = 1$ if it has the property and $X_i = 0$ otherwise, so $E[X_i] = P(\text{object } i \text{ has the property})$. The count is $X = X_1 + \cdots + X_n$, and by linearity

$$E[X] = \sum_{i=1}^{n} P(\text{object } i \text{ has the property}),$$

whether or not the $X_i$ are independent. If all $n$ probabilities equal $p$, then $E[X] = np$.

**METHOD** **Symmetry (Week 10 s72, s74)**

If swapping two roles maps outcomes to equally likely outcomes, the two events have the same probability. For two rolls of a fair die, $P(\text{second} > \text{first}) = P(\text{first} > \text{second})$, and with $P(\text{tie}) = 6/36$ the three cases sum to 1:

$$P(\text{second} > \text{first}) = \frac{1 - P(\text{tie})}{2} = \frac{1}{2} \cdot \frac{5}{6} = \frac{5}{12}.$$

Direct check: $(5 + 4 + 3 + 2 + 1 + 0)/36 = 15/36$. The same argument gives "A to the left of B" in half of all arrangements (§39) and $P(\text{second draw is red}) = P(\text{first draw is red})$ without replacement (§45).

**PAST PAPER** **AY20/21 Q10, Q11 · AY22/23 Q22(c), (e) · AY24/25 Q24(d)**

**AY20/21 Q10.** Expected sum of three fair dice: $3 \cdot 3.5 = 10.5$ (option D).

**AY20/21 Q11.** $(\sim b \wedge c \wedge d) \vee ((\sim a \vee b \vee \sim e \vee f) \wedge c) \vee \sim(d \wedge e) \vee c$ with six variables set at random. The statement is false iff every disjunct is false: $c$ false (which also kills the first two disjuncts) and $d$, $e$ both true; $a$, $b$, $f$ are free: $2^3/2^6 = 1/8$ (option C).

**AY22/23 Q22(c).** Second roll higher than the first: 5/12 by symmetry.

**AY22/23 Q22(e).** Two cards without replacement, scores 1 to 13: each card has expected score $(1 + \cdots + 13)/13 = 7$, so the combined score has expectation 14 (linearity; the draws are dependent).

**AY24/25 Q24(d).** Ten couples; 4 of the 20 persons leave, every set of 4 equally likely. (i) A fixed couple stays: all 4 leavers come from the other 18 persons, $\binom{18}{4}/\binom{20}{4} = 3060/4845 = 12/19$ (or $\frac{16}{20} \cdot \frac{15}{19}$: both of the couple are among the 16 who stay). (ii) Let $X_i = 1$ if couple $i$ stays. $E[X] = 10 \cdot 12/19 = 120/19 \approx 6.32$.

## 45 · Conditional probability, Bayes' theorem, independence

**DEFINITION Conditional probability (L11 s53) · Independent events (L11 s73) · Pairwise and mutually independent (L11 s79–80)**

For events $A$, $B$ in $S$ with $P(A) \neq 0$, the **conditional probability of $B$ given $A$** is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}, \quad \text{so} \quad P(A \cap B) = P(B \mid A) \cdot P(A).$$

$A$ and $B$ are **independent** iff $P(A \cap B) = P(A) \cdot P(B)$.

$A$, $B$, $C$ are **pairwise independent** iff (1) $P(A \cap B) = P(A)P(B)$, (2) $P(A \cap C) = P(A)P(C)$, (3) $P(B \cap C) = P(B)P(C)$; they are **mutually independent** iff also (4) $P(A \cap B \cap C) = P(A)P(B)P(C)$. Events $A_1, ..., A_n$ are mutually independent iff the probability of the intersection of **any subset** of them is the product of their probabilities.

**THEOREM Law of total probability (L11 s59, s64) · Independence in conditional form (L11 s72)**

- **Total probability.** If $S$ is the union of mutually disjoint events $B_1, ..., B_n$ of non-zero probability, then for every event $A$

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \cdots + P(A \mid B_n)P(B_n),$$

  because $A$ is the disjoint union of the $A \cap B_i$ *(axiom 3)* and $P(A \cap B_i) = P(A \mid B_i)P(B_i)$. Two cases: $P(A) = P(A \mid B)P(B) + P\left(A \mid \overline{B}\right)P\left(\overline{B}\right)$. This is the sum over the branches of the tree that end in $A$.
- **Conditional form of independence.** If $P(B) \neq 0$: $A$ and $B$ are independent iff $P(A \mid B) = P(A)$. If $P(A) \neq 0$: iff $P(B \mid A) = P(B)$. Knowing one event does not change the probability of the other.
- **Union of independent events.** $P(A \cup B) = P(A) + P(B) - P(A)P(B)$.
- **Complements.** If $A$ and $B$ are independent, so are $A$ and $\overline{B}$, since $P\left(A \cap \overline{B}\right) = P(A) - P(A \cap B) = P(A)(1 - P(B))$; likewise $\overline{A}$, $B$ and $\overline{A}$, $\overline{B}$.

**THEOREM Theorem 9.9.1 · Bayes' theorem (L11 s66)**

Suppose $S$ is the union of mutually disjoint events $B_1, ..., B_n$, and $A$ and all $B_i$ have non-zero probability. Then for $1 \leq k \leq n$:

$$P(B_k \mid A) = \frac{P(A \mid B_k) \cdot P(B_k)}{P(A \mid B_1) \cdot P(B_1) + P(A \mid B_2) \cdot P(B_2) + \cdots + P(A \mid B_n) \cdot P(B_n)}.$$

The denominator is $P(A)$, by the law of total probability.

**TRAP Conditional probability and independence**

- $P(B \mid A)$ and $P(A \mid B)$ are different numbers; Bayes converts one into the other.
- "At least one is a boy" is not "the first is a boy": of $BB, BG, GB, GG$ the condition leaves three outcomes, so $P(\text{both boys} \mid \text{at least one boy}) = 1/3$.
- Independent is not disjoint: disjoint events with positive probabilities are **dependent**.
- Pairwise independence does not give (4), and (4) does not give pairwise independence. Two fair coins with $A$ = first is heads, $B$ = second is heads, $C$ = both the same: every pair is independent, but $P(A \cap B \cap C) = 1/4 \neq 1/8$.

**METHOD With and without replacement**

**Counting $k$ draws from $n$ distinct objects.** With replacement the object goes back, so each draw has $n$ choices: $n^k$ ordered outcomes. Without replacement: $P(n, k)$ ordered outcomes, $\binom{n}{k}$ unordered ones. Unordered with replacement gives $\binom{k+n-1}{k}$ multisets, but these are **not equally likely**, so never use them as a sample space.

**Probability, urn with $b$ blue and $g$ grey balls, $N = b + g$, $k$ draws.**

- **With replacement:** the draws are independent and each is blue with probability $b/N$. $P(\text{exactly } j \text{ blue}) = \binom{k}{j}\left(\frac{b}{N}\right)^j\left(\frac{g}{N}\right)^{k-j}$.
- **Without replacement:** the draws are dependent. Either count unordered hands, $P(\text{exactly } j \text{ blue}) = \binom{b}{j}\binom{g}{k-j}/\binom{N}{k}$, or multiply conditional probabilities along the sequence: $P(\text{blue then blue}) = \frac{b}{N} \cdot \frac{b-1}{N-1}$.
- In both cases $P(\text{the } i\text{-th draw is blue}) = b/N$ for every $i$, by symmetry.
- **At least one:** use the complement, $1 - P(\text{none})$.

**Check, 5 blue and 7 grey, 3 draws.** Without replacement: $P(\text{exactly 2 blue}) = \binom{5}{2}\binom{7}{1}/\binom{12}{3} = 70/220 = 7/22$ and $P(\text{at least one blue}) = 1 - \binom{7}{3}/\binom{12}{3} = 37/44$. With replacement: $P(\text{exactly 2 blue}) = 3 \cdot (5/12)^2(7/12) = 175/576$.

WORKED EXAMPLE **L11 Ex. 17 · Urn with 5 blue and 7 grey balls, two drawn without replacement**
$P(B_1)=\frac{5}{12}$, $P(B_2 \mid B_1)=\frac{4}{11}$, $P\left(B_2 \mid \overline{B_1}\right)=\frac{5}{11}$. Multiply along the branches of the tree:

$$P(B_1\cap B_2)=\frac{20}{132},\quad P\left(B_1\cap\overline{B_2}\right)=\frac{35}{132},\quad P\left(\overline{B_1}\cap B_2\right)=\frac{35}{132},\quad P\left(\overline{B_1}\cap\overline{B_2}\right)=\frac{42}{132}.$$

- $P(B_2)=\frac{20}{132}+\frac{35}{132}=\frac{55}{132}=\frac{5}{12}$ *(axiom 3 on the disjoint cases)*.
- $P(B_1\cup B_2)=\frac{5}{12}+\frac{5}{12}-\frac{20}{132}=\frac{90}{132}=\frac{15}{22}$; so $P(\text{no blue})=\frac{7}{22}$.
- Expected number of blue balls: $0\cdot\frac{42}{132}+1\cdot\frac{70}{132}+2\cdot\frac{20}{132}=\frac{110}{132}=\frac{5}{6}\approx 0.83$.

WORKED EXAMPLE **Bayes · L11 s63–65, Ex. 18 · T10 Q5 · AY25 final Q24(h)**

- **Two urns** (3 blue, 4 grey; 5 blue, 3 grey), one chosen at random: $P(B)=\frac{3}{7}\cdot\frac{1}{2}+\frac{5}{8}\cdot\frac{1}{2}=\frac{59}{112}$ and $P(U_1 \mid B)=(3/14)/(59/112)=\frac{24}{59}\approx 40.7\%$.
- **Medical test.** Disease in 5 per 1000; false positive rate 3%; false negative rate 1%.
  $P(\text{disease}\mid +)=\frac{0.99\cdot 0.005}{0.99\cdot 0.005+0.03\cdot 0.995}\approx 14.2\%$. $P(\text{no disease}\mid -)=\frac{0.97\cdot 0.995}{0.01\cdot 0.005+0.97\cdot 0.995}\approx 99.995\%$.
- **T10 Q5 (AY21/22 S2).** $P(D)=0.001$, $P(+\mid D)=0.85$, $P(+)=0.1$. $P(D\mid +)=(0.85\cdot 0.001)/0.1=0.0085$. False positive: $P\left(+\mid\overline{D}\right)=\frac{P\left(\overline{D}\mid +\right)\cdot P(+)}{P\left(\overline{D}\right)}=(0.9915\cdot 0.1)/0.999\approx 0.0992$.
- **AY25 Q24(h).** $P(S)=0.2$, $P(O\mid S)=0.5$, $P(O\mid H)=0.08$: $P(S\mid O)=0.1/(0.1+0.064)\approx 0.61$. $P(U\mid S)=0.35$, $P(U\mid H)=0.04$: $P(S\mid U)=0.07/(0.07+0.032)\approx 0.69$. "Urgent" is the stronger signal.

WORKED EXAMPLE **Independence · L11 Ex. 19, 20 · T9 D1 · T10 Q6**

- **Loaded coin, $P(H)=0.6$, two tosses:** $P(\text{two heads})=0.36$; $P(\text{one head})=0.6\cdot 0.4+0.4\cdot 0.6=0.48$; $P(\text{no heads})=0.16$; $P(\text{at least one})=1-0.16=0.84$.
- **Ten tosses:** $P(\text{eight heads})=\binom{10}{8}(0.6)^8(0.4)^2\approx 0.121$; $P(\text{at least eight})=\binom{10}{8}(0.6)^8(0.4)^2+\binom{10}{9}(0.6)^9(0.4)+(0.6)^{10}\approx 0.167$. Probabilities of the form $\binom{n}{k}p^k(1-p)^{n-k}$ are **binomial probabilities**.
- **T9 D1.** 3 blue and 7 white balls, two draws with replacement: $P(\text{blue then white})=0.21$; $P(\text{both white})=0.49$; $P(\text{second blue})=0.3$.
- **T10 Q6 (AY15/16).** A random relation on a 4-element set is reflexive with probability $2^{12}/2^{16}=1/16$ and symmetric with probability $2^{10}/2^{16}=1/64$. For $|A|=n$: $1/2^n$ and $2^{(n^2+n)/2}/2^{n^2}=1/2^{(n^2-n)/2}$.
- **Birthday problem (T9 D4):** $P(\text{some shared birthday among } n)=1-(365\cdot 364\cdots(365-n+1))/365^n$, which first reaches 50% at $n=23$.

PAST PAPER **Total probability and Bayes in the finals · AY22/23 Q22(f) · AY23/24 Q25(e) · AY20/21 Q20(e)**

**AY22/23 Q22(f).** $P(\text{spam})=0.25$, $P(F\mid\text{spam})=0.6$, $P(F\mid\text{not spam})=0.3$, where $F=$ "free gifts" appears. (i) $P(F)=0.6\cdot 0.25+0.3\cdot 0.75=0.375$. (ii) $P(\text{spam}\mid F)=0.15/0.375=0.4$.

**AY23/24 Q25(e).** $P(G)=0.85$, $P(\text{bonus}\mid G)=0.8$, $P(\text{bonus}\mid\text{Apple})=0.95$: $P(G\mid\text{bonus})=0.68/(0.68+0.1425)=0.68/0.8225\approx 0.8267$.

**AY20/21 Q20(e), joint hypotheses.** One M&M from each of two bags (one 1994, one 1996); one is yellow, one is green. A hypothesis must describe the **whole** set-up: $A=$ "bag 1 is the 1994 bag and bag 2 the 1996 bag", $B=$ the reverse, $P(A)=P(B)=1/2$. Evidence $E=$ "yellow from bag 1 and green from bag 2". Multiply the two colour probabilities under each hypothesis:

$$P(E\mid A)=0.20\cdot 0.16=0.032,\quad P(E\mid B)=0.14\cdot 0.10=0.014,\quad P(A\mid E)=\frac{0.032}{0.032+0.014}=\frac{16}{23}\approx 0.696.$$

Using the yellow alone ($0.20/0.34=10/17$) ignores half of the evidence.

PAST PAPER **Independence in the finals · AY22/23 Q13, Q22(b) · AY23/24 Q25(a) · AY24/25 Q23, Q24(e)**

**AY22/23 Q22(b).** $A$, $B$ independent, $P(A)=0.2$, $P(A\cup B)=0.6$: $0.6=0.2+P(B)-0.2P(B)$, so $P(B)=0.5$.

**AY22/23 Q13, restricted pattern.** Ten tosses, $P(H)=p$; exactly 3 heads, all consecutive. Every sequence with 3 heads has probability $p^3(1-p)^7$; count only the sequences allowed: the block HHH starts at toss $1,\dots,8$. Answer $8p^3(1-p)^7$ (option D), not $\binom{10}{3}p^3(1-p)^7$.

**AY23/24 Q25(a).** $P(H)=0.6$, five tosses, three heads: $\binom{5}{3}(0.6)^3(0.4)^2=0.3456$.

**AY24/25 Q23.** One ball from $\{1,2,3,4\}$; $A=\{1,2\}$, $B=\{1,3\}$, $C=\{1,4\}$. Each has probability $1/2$ and each pairwise intersection is $\{1\}$ with probability $1/4=\frac{1}{2}\cdot\frac{1}{2}$, but $P(A\cap B\cap C)=1/4\neq 1/8$: pairwise independent, not mutually independent (option C).

**AY24/25 Q24(e).** $b$ blue and $r$ red markers, two drawn without replacement. (i) $P(R_1)=r/(b+r)$. (ii) True: $P(R_2)=\frac{r}{b+r}\cdot\frac{r-1}{b+r-1}+\frac{b}{b+r}\cdot\frac{r}{b+r-1}=\frac{r}{b+r}$ *(total probability)*. (iii) $P(\text{different colours})=\frac{2br}{(b+r)(b+r-1)}$. (iv) $b+r=25$ and $P(\text{same})=P(\text{different})=1/2$: $2br=\frac{1}{2}\cdot 25\cdot 24=300$, so $br=150$ with $r>b$: $r=15$, $b=10$.

# Part X · Graphs and Trees

## 46 · Graphs, special graphs, degree

DEFINITION **Undirected graph (L12 s7) · Directed graph (L12 s8)**

An **undirected graph** $G=(V,E)$ consists of two finite sets: a non-empty set $V$ of **vertices** and a set $E$ of **edges**, where each edge is associated with a set of one or two vertices called its **endpoints**. Write $e=\{v,w\}$.

An edge **connects** its endpoints and is **incident on** each of them; vertices joined by an edge are **adjacent**; an edge with a single endpoint is a **loop** (its vertex is adjacent to itself); edges with the same endpoints are **parallel**; edges sharing an endpoint are **adjacent edges**.

A **directed graph** (digraph) has directed edges, each associated with an **ordered pair** of vertices: $e=(v,w)$ goes from $v$ to $w$.

DEFINITION **Simple graph (L12 s19) · Complete graph $K_n$ (L12 s20) · Bipartite, $K_{m,n}$ (L12 s21) · Subgraph (L12 s22)**

- A **simple graph** is an undirected graph with no loops and no parallel edges.
- The **complete graph** $K_n$ ($n>0$) is the simple graph on $n$ vertices with exactly one edge between each pair of distinct vertices. It has $\binom{n}{2}=n(n-1)/2$ edges.
- A **bipartite graph** is a simple graph whose vertices can be divided into two disjoint sets $U$ and $V$ such that every edge connects a vertex in $U$ to one in $V$. In the **complete bipartite graph** $K_{m,n}$, with $|U|=m$ and $|V|=n$, every vertex of $U$ is joined to every vertex of $V$; it has $mn$ edges.
- $H$ is a **subgraph** of $G$ iff every vertex of $H$ is a vertex of $G$, every edge of $H$ is an edge of $G$, and every edge of $H$ has the same endpoints as in $G$.

DEFINITION **Degree, total degree (L12 s23) · Indegree, outdegree (L12 s26) · Complement, self-complementary, triangle (T10)**

The **degree** $\deg(v)$ is the number of edges incident on $v$, **with a loop counted twice**. The **total degree** of $G$ is the sum of the degrees of all its vertices.

In a digraph, the **indegree** $\deg^-(v)$ is the number of edges ending at $v$ and the **outdegree** $\deg^+(v)$ the number starting at $v$; $\sum_{v\in V}\deg^-(v)=\sum_{v\in V}\deg^+(v)=|E|$.

The **complement** $\overline{G}$ of a simple graph $G$ has the same vertices; distinct $v,w$ are adjacent in $\overline{G}$ iff they are not adjacent in $G$. $G$ is **self-complementary** iff $G\cong\overline{G}$. A **triangle** is a simple circuit of length 3.

A **vertex colouring** assigns colours to vertices so that adjacent vertices get different colours (L12 s13).

THEOREM **Theorem 10.1.1 Handshake theorem · Corollary 10.1.2 · Proposition 10.1.3 (L12 s25)**

- For any graph $G$ with vertices $v_1, ..., v_n$: $\deg(v_1) + \deg(v_2) + \cdots + \deg(v_n) = 2 \cdot |E|$.
- The total degree of a graph is even.
- In any graph there is an even number of vertices of odd degree.

PROOF **Handshake theorem · Proposition 10.1.3**

1. Let $e$ be an edge of $G$ with endpoints $v_i$ and $v_j$. If $i \neq j$, then $e$ contributes 1 to $\deg(v_i)$ and 1 to $\deg(v_j)$; if $i = j$ (a loop), it contributes 2 to $\deg(v_i)$ *(definition of degree)*.
2. So every edge contributes exactly 2 to the total degree, which is therefore $2\,|E|$. ■

**Proposition 10.1.3.** The total degree $T$ is even. Let $E$ be the sum of the even degrees and $O$ the sum of the odd degrees, so $O = T - E$ is even *(difference of even integers)*. A sum of odd integers is even only if the number of terms is even. ■

WORKED EXAMPLE **Degree arguments · T11 D1, D2 · T10 Q8**

- **No graph has degrees 1, 1, 2, 3:** the total would be 7, which is odd.
- **No simple graph has degrees 1, 1, 3, 3 on 4 vertices:** each vertex of degree 3 is adjacent to all three others, so every other vertex has degree at least 2.
- **T11 D1.** A connected simple graph on $n$ vertices has between $n-1$ edges (a tree) and $n(n-1)/2$ edges ($K_n$).
- **T11 D2.** Simple graphs on $n$ labelled vertices: each of the $\binom{n}{2}$ possible edges is present or absent, so $2^{n(n-1)/2}$; for $n = 3$ there are 8 (in 4 isomorphism classes).
- **T10 Q8.** Non-isomorphic undirected graphs with two vertices and two edges: **4** (two parallel edges; one edge and one loop; two loops at one vertex; one loop at each vertex). The question does not say "simple".
- **$K_n$ and $K_{m,n}$:** every vertex of $K_n$ has degree $n-1$; in $K_{m,n}$ the degrees are $n$ (in $U$) and $m$ (in $V$). Check: $n(n-1) = 2 \cdot \binom{n}{2}$ and $mn + nm = 2mn$.

**Colouring: chromatic number and chromatic index**

DEFINITION **Chromatic number $\chi(G)$ · Chromatic index $\chi'(G)$ (AY24/25 Q25 · W12 s52)**

The **chromatic number** $\chi(G)$ is the minimum number of colours needed to colour the **vertices** of $G$ so that no two adjacent vertices have the same colour.

The **chromatic index** $\chi'(G)$ is the minimum number of colours needed to colour the **edges** of $G$ so that no two adjacent edges (edges sharing an endpoint) have the same colour.

$\Delta(G)$ denotes the maximum degree of $G$.

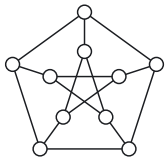
THEOREM **Values and bounds for $\chi$ and $\chi'$ (W12 s53–54, s58)**

| Graph | $\chi$ | $\chi'$ | Reason |
|---|---|---|---|
| $K_n$ | $n$ | $n-1$ ($n$ even), $n$ ($n$ odd) | all vertices mutually adjacent; for odd $n$ a colour covers at most $(n-1)/2$ of the $n(n-1)/2$ edges |
| $C_n$ | 2 ($n$ even), 3 ($n$ odd) | 2 ($n$ even), 3 ($n$ odd) | alternate two colours round the cycle; an odd cycle needs a third |
| bipartite with an edge: $K_{m,n}$, trees | 2 | $\Delta$ ($\max(m,n)$ for $K_{m,n}$) | colour by side; König's theorem for $\chi'$ |
| Petersen graph | 3 | 4 | has odd cycles; $\chi' = \Delta + 1$ |
| friendship graph $F_n$, $n \geq 2$ | 3 | $2n$ | triangles; the centre has degree $2n$ |

- **Lower bounds:** if $G$ contains $K_k$ as a subgraph then $\chi(G) \geq k$ (a triangle, or any odd cycle, forces 3). $\chi'(G) \geq \Delta(G)$, since the edges at one vertex all need different colours.
- **Upper bounds:** $\chi(G) \leq \Delta(G) + 1$ (colour the vertices one at a time; at most $\Delta$ colours are blocked). For a simple graph $\chi'(G)$ is $\Delta$ or $\Delta + 1$.
- $\chi(G) = 1$ iff $G$ has no edges; $\chi(G) \leq 2$ iff $G$ is bipartite (§47).

**DEFINITION** **Petersen graph (AY24/25 Q25 · W12 s53–55)**

An outer 5-cycle, an inner 5-cycle drawn as a pentagram, and five spokes joining them.



- **10 vertices, 15 edges**, every vertex of degree 3; the shortest cycle has length 5 (no triangles, not bipartite).
- $\chi = 3$ and $\chi' = 4$, although $\Delta = 3$.
- **Not Eulerian** (every degree is odd). **Not Hamiltonian**, although a path through all 10 vertices exists.
- **Not planar:** every face would have at least 5 edges, so $5f \leq 2e = 30$ and $f \leq 6$; but Euler's formula (§49) needs $f = 15 - 10 + 2 = 7$.

**METHOD** **Modelling a conflict or scheduling problem as vertex colouring (L12 s13–18)**

1. **Vertices** are the objects to be assigned: items, guests, classes, radio stations, traffic signals.
2. **Edges** join two objects that **conflict** (cannot share a container, table, time slot, frequency).
3. **Colours** are the resources. Vertices of the same colour are objects that may share one resource.
4. The minimum number of resources is $\chi(G)$. **Lower bound:** find a triangle or a larger $K_k$. **Upper bound:** exhibit a colouring, taking vertices of high degree first. When the bounds meet, the answer is proved.

**Wedding planner (L12 s16–17):** guests $B, C, D$ are in mutual conflict, so 3 tables are needed; $\{A, B, E\}$, $\{C, F\}$, $\{D, G, H\}$ works. **3 tables.**

**PAST PAPER** **AY22/23 Q23(f)–(i) · Volatile item packing as graph colouring**

Items $A, ..., K$ (no $I$); conflicts A: B, C; B: A, C, D; C: A, B, D; D: B, C, E, F; E: D, F, G, H; F: D, E, J, K; G: E, H; H: E, G; J: F, K; K: F, J.

**(f)** Vertices are items; two vertices are adjacent iff the items conflict; vertices of the same colour go into the same container.

**(g)** 14 edges: $AB, AC, BC, BD, CD, DE, DF, EF, EG, EH, GH, FJ, FK, JK$.

**(h)** The triangle $ABC$ forces 3 colours, and 3 suffice: colour 1: $A, D, G, J$; colour 2: $B, E, K$; colour 3: $C, F, H$.

**(i) 3 containers:** $\{A, D, G, J\}$, $\{B, E, K\}$, $\{C, F, H\}$.

**PAST PAPER** **AY24/25 Q25(a)–(c), (f) · Chromatic number and index**

**(a)** $\chi(K_{1231}) = 1231$; $\chi(K_{1231,1231}) = 2$; $\chi(C_{1231}) = 3$ (odd cycle); $\chi(\text{Petersen}) = 3$.

**(b)** $\chi'(K_{100}) = 99$ ($n$ even); $\chi'(K_{1231,1231}) = 1231$ (bipartite, $\Delta = 1231$); $\chi'(\text{Petersen}) = 4$.

**(c)** The Petersen graph is not Eulerian and not Hamiltonian.

**(f)** The graph in (e) is the **friendship graph** (T10 Q7), here $F_6$. With $2n + 1$ vertices and $n > 1$: $\chi' = 2n$. The $2n$ edges at the centre need $2n$ colours; each rim edge then takes the colour of a spoke of another triangle. ($F_1 = K_3$ needs 3, hence $n > 1$.)

**Matchings and dominating sets**

**DEFINITION** **Matching · Maximal, maximum, perfect matching · Neighbours $N(S)$ (AY25/26 Q25)**

A **matching** in $G = (V, E)$ is a subset $F \subseteq E$ in which no two edges share a vertex.

- **Maximal matching:** cannot be extended by adding another edge (every other edge meets an edge of $F$).
- **Maximum matching:** contains the largest possible number of edges.
- **Perfect matching:** covers every vertex of the graph.

For a set $S$ of vertices, $N(S)$ is the set of vertices adjacent to at least one vertex of $S$.

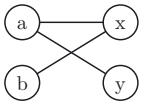
Every maximum matching is maximal, **not conversely**. Every perfect matching is maximum.

**THEOREM** **Hall's Marriage Theorem (AY25/26 Q25)**

A bipartite graph $G = (U, V, E)$ has a matching that covers all vertices in $U$ **iff** for every subset $S \subseteq U$: $|N(S)| \geq |S|$.

METHOD **Using Hall's theorem in both directions**
**To show a matching covering $U$ exists:** write one down. Start with the vertices of degree 1 (their edge is forced), delete the matched vertices, repeat. Checking Hall's condition directly means all $2^{|U|}$ subsets.
**To show none exists:** give one set $S \subseteq U$ with $|N(S)| < |S|$. Look for several vertices of $U$ whose neighbours all lie in a small part of $V$; a vertex of degree 0 and the case $|U| > |V|$ ($S = U$) are instant.
**Perfect matching:** a matching covering $U$ has $|U|$ edges and meets $|U|$ vertices of $V$, so it is perfect iff also $|U| = |V|$.

PAST PAPER **AY25/26 Q25(e)–(h) · Matchings**
**(e)** The additional condition is $|U| = |V|$.
**(f)** $U = \{a, b\}$, $V = \{x, y\}$, $E = \{ax, ay, bx\}$ (drawn on the right). $\{ax\}$ is **maximal** ($ay$ shares $a$, $bx$ shares $x$) but has 1 edge; $\{ay, bx\}$ is **maximum** with 2 edges.
**(g)** $G_1$: $a$ has only $x$ and $c$ has only $y$, so $ax$ and $cy$ are forced; then $bw$ and $dz$. **Yes:** $\{(a, x), (b, w), (c, y), (d, z)\}$.
**(h)** $G_2$: take $S = \{a, b, c\}$. $N(S) = \{w, x\}$, so $|N(S)| = 2 < 3 = |S|$. **No perfect matching.**




PAST PAPER **T10 Q11 (AY21/22 exam) · Dominating sets**
$D \subseteq V$ is a **dominating set** of a simple graph iff every vertex not in $D$ is adjacent to at least one vertex in $D$; it is **minimal** iff no proper subset of $D$ is dominating.
**(a) Size one** (one vertex adjacent to all others), on four vertices: $K_4$ with any vertex; the star $K_{1,3}$ with its centre.
**(b) Size three**, on four vertices: one edge plus two isolated vertices (the two isolated vertices and one end of the edge); the star $K_{1,3}$ with its three leaves.
Minimal is not minimum: the star has minimal dominating sets of sizes 1 and 3.

## 47 · Walks, connectedness, Euler and Hamiltonian circuits

DEFINITION **Walk, trail, path, closed walk, circuit, simple circuit (L12 s32–33)**
A **walk** from $v$ to $w$ is a finite alternating sequence $v_0 e_1 v_1 e_2 \cdots v_{n-1} e_n v_n$ of adjacent vertices and edges with $v_0 = v$, $v_n = w$, and $v_{i-1}, v_i$ the endpoints of $e_i$. Its **length** is $n$, the number of edges. The **trivial walk** from $v$ to $v$ is the single vertex $v$.

| | Repeated edge? | Repeated vertex? | Closed? | Notes |
|---|---|---|---|---|
| Walk | allowed | allowed | allowed | |
| Trail | no | allowed | allowed | a walk with no repeated edge |
| Path | no | no | no | a trail with no repeated vertex |
| Closed walk | allowed | allowed | yes | |
| Circuit (cycle) | no | allowed | yes | closed walk of length at least 3, no repeated edge |
| Simple circuit | no | first and last only | yes | |

An undirected graph is **cyclic** if it contains a loop or a cycle; otherwise it is **acyclic**. (Other books say "path" for trail and "cycle" for simple circuit; the course follows Epp.)

DEFINITION **Connected (L12 s35) · Connected component (L12 s38)**
Vertices $v$ and $w$ are **connected** iff there is a walk from $v$ to $w$. $G$ is **connected** iff $\forall v, w \in V$ there is a walk from $v$ to $w$.
$H$ is a **connected component** of $G$ iff (1) $H$ is a subgraph of $G$, (2) $H$ is connected, and (3) no connected subgraph of $G$ has $H$ as a subgraph and contains vertices or edges not in $H$. A connected component is a connected subgraph of largest possible size.

THEOREM **Lemma 10.2.1 (L12 s37)**
Let $G$ be a graph.
- (a) If $G$ is connected, then any two distinct vertices of $G$ can be connected by a **path**.
- (b) If $v$ and $w$ are part of a circuit and one edge is removed from the circuit, there is still a trail from $v$ to $w$.
- (c) If $G$ is connected and contains a circuit, then an edge of the circuit can be removed without disconnecting $G$.

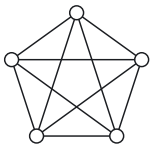
DEFINITION **Euler circuit, Eulerian graph (L12 s41) · Euler trail (L12 s43) · Hamiltonian circuit, Hamiltonian graph (L12 s49)**
An **Euler circuit** for $G$ is a circuit that contains every vertex and traverses **every edge exactly once**; a graph with one is **Eulerian**.
An **Euler trail** from $v$ to $w$ ($v \neq w$) starts at $v$, ends at $w$, passes through every vertex at least once and traverses every edge exactly once.
A **Hamiltonian circuit** for $G$ is a simple circuit that includes **every vertex** (each exactly once, except that the first and last are the same); a graph with one is **Hamiltonian**.

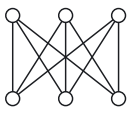
THEOREM **Theorems 10.2.2–10.2.4 · Corollary 10.2.5 · Proposition 10.2.6 (L12 s41–43, s52)**
- **10.2.2.** If a graph has an Euler circuit, then every vertex has positive even degree. (Contrapositive: an odd-degree vertex rules out an Euler circuit.)
- **10.2.3.** If $G$ is connected and every vertex has positive even degree, then $G$ has an Euler circuit.
- **10.2.4.** $G$ has an Euler circuit **iff** $G$ is connected and every vertex has positive even degree.
- **10.2.5.** There is an Euler trail from $v$ to $w$ iff $G$ is connected, $v$ and $w$ have odd degree, and every other vertex has positive even degree.
- **10.2.6.** If $G$ has a Hamiltonian circuit, then $G$ has a subgraph $H$ that (1) contains every vertex of $G$, (2) is connected, (3) has as many edges as vertices, and (4) has every vertex of degree 2.

There is no simple criterion for Hamiltonian circuits. To show none exists, use the contrapositive of 10.2.6.

PROOF **Theorem 10.2.2 · An Euler circuit forces every degree to be positive and even**
1. Let $C$ be an Euler circuit of $G$ and $v$ any vertex. $C$ contains $v$, so $\deg(v) > 0$.
2. Follow $C$ once around. Each time it passes through $v$ it arrives on one edge and leaves on another, using two edge-ends at $v$; if $v$ is the starting vertex, the first departure pairs with the final arrival.
3. $C$ uses every edge exactly once, so the edge-ends at $v$ are split into pairs. Hence $\deg(v)$ is even. ■

WORKED EXAMPLE **Euler and Hamilton · Königsberg · T10 Q7 · AY25 final Q25(b)–(d)**
- **Königsberg** (4 land masses, 7 bridges): degrees $5, 3, 3, 3$. No Euler circuit; no Euler trail either, since four vertices are odd.
- "Every vertex has positive even degree" alone is not enough: two disjoint triangles are not connected.
- $\boldsymbol{K_n}$ ($n \geq 3$) is Hamiltonian; it is Eulerian iff $n$ is odd.
- $\boldsymbol{K_{m,n}}$ ($m, n \geq 2$) is Eulerian iff $m$ and $n$ are both even; it is Hamiltonian iff $m = n$ (a circuit alternates between the two sides).
- **Cycle graph $\boldsymbol{C_n}$** is bipartite iff $n$ is even: $C_{12}$ and $C_{8888}$ are; $C_{57}$, $C_{123}$, $C_{1231}$, $C_{12345}$ are not.
- **Friendship graph $\boldsymbol{F_n}$** ($n$ triangles sharing a centre, T10 Q7): Eulerian, since the degrees are 2 and $2n$. Not Hamiltonian for $n \geq 2$: passing from one triangle to another repeats the centre.


$K_5$


$K_{3,3}$

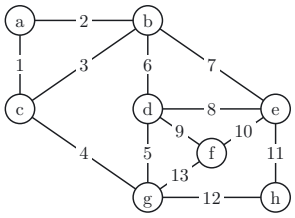
PAST PAPER **AY25 final Q9 · Relations from walks**
On the vertex set of a graph let $xRy$ iff there is a walk from $x$ to $y$.
**Undirected:** $R$ is an equivalence relation (reflexive by the trivial walk; symmetric by reversing a walk; transitive by joining walks). Its classes are the vertex sets of the connected components.
**Directed:** $R$ need not be a partial order. Even with no pair of opposite edges, the directed cycle $a \to b \to c \to a$ gives $aRb$ and $bRa$ with $a \neq b$, so antisymmetry fails. Only (i) is true.

PAST PAPER **AY22/23 Q23(a), (b) · AY23/24 Q24(c)(i) · Where an Euler trail starts and ends**

By Corollary 10.2.5 an Euler trail exists iff the graph is connected with **exactly two** odd vertices, and every Euler trail **starts at one of them and ends at the other**. No odd vertex: an Euler circuit, which may start anywhere. More than two: neither.

**AY22/23 Q23.** Degrees: $a$ 2, $b$ 4, $c$ 3, $d$ 4, $e$ 4, $f$ 3, $g$ 4, $h$ 2. **(a)** One trail: $c, a, b, c, g, d, b, e, d, f, e, h, g, f$ (the edge labels on the right give the order). **(b)** It may start at $\boldsymbol{c}$ **or** $\boldsymbol{f}$ only.

**AY23/24 Q24(c)(i).** Remove an edge $\{u, v\}$, $u \neq v$, from a graph with an Euler circuit: the edge lies on a circuit, so $G'$ stays connected (Lemma 10.2.1(c)), and exactly $u$ and $v$ become odd. $G'$ has an Euler trail from $u$ to $v$: **true**. (If the removed edge were a loop, $G'$ would keep an Euler circuit instead.)




**Bipartite graphs and odd cycles**

THEOREM **A graph is bipartite iff it has no cycle of odd length (AY24/25 Q22 · AY22/23 Q19)**

Equivalently: $G$ is bipartite iff its vertices can be coloured with at most 2 colours ($\chi(G) \leq 2$).

So $K_{m,n}$, trees, forests and even cycles are bipartite; a graph containing a triangle, or any odd cycle, is not.

PROOF **Bipartite ⇔ no odd cycle**

1. ($\Rightarrow$) Let $G$ be bipartite with sides $U$, $V$, and let $v_0 v_1 \cdots v_k$ with $v_k = v_0$ be a cycle. Every edge joins $U$ to $V$, so the vertices alternate sides *(definition of bipartite)*.
2. After $k$ steps the cycle is back on the side of $v_0$, so $k$ is even.
3. ($\Leftarrow$) Suppose $G$ has no odd cycle. Treat each connected component separately; pick a vertex $r$ in it, and let $d(x)$ be the length of a shortest path from $r$ to $x$ *(Lemma 10.2.1(a))*.
4. Put $x$ in $U$ if $d(x)$ is even and in $V$ if $d(x)$ is odd.
5. Suppose an edge $\{x, y\}$ had both ends on the same side. The shortest paths from $r$ to $x$ and from $r$ to $y$, with the edge $\{x, y\}$, form a closed walk of length $d(x) + d(y) + 1$, which is odd.
6. A closed walk of odd length contains a cycle of odd length: contradiction. So every edge joins $U$ to $V$. ■

PAST PAPER **AY24/25 Q22 · AY22/23 Q19 · True or false**

**AY24/25 Q22.** (i) Bipartite iff no odd cycle: **true**. (ii) Bipartite iff the sum of all degrees is even: **false**, since that sum is even for every graph (handshake theorem), for example $K_3$. (iii) $K_{2,n}$ is planar for $n > 3$: **true** (§49). Answer: only (i) and (iii).

**AY22/23 Q19.** (i) A simple graph with 6 vertices, 5 edges and no cycles cannot have exactly one vertex of degree 1: **true**. A forest with $c$ components has $v - c$ edges, so $c = 1$: it is a tree, with at least two leaves. (ii) Not connected $\Rightarrow e < n - 1$: **false** ($K_4$ plus an isolated vertex: $n = 5$, $e = 6$). (iii) All cycles in $K_{3,4}$ have an even number of edges: **true** (bipartite). Answer: only (i) and (iii).

**Cut-points, bridges, vertex cut sets**

DEFINITION **$G - v$, cut-point · $G - e$, bridge (AY22/23 Q23) · Vertex cut set (AY24/25 Q25)**

Let $G = (V, E)$ be a connected undirected graph.

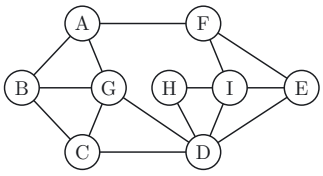
$G - v$ is $G$ with the vertex $v$ and all edges incident to $v$ deleted; $v$ is a **cut-point** (cut vertex) iff $G - v$ is disconnected.

$G - e$ is $G$ with the edge $e$ deleted (its endpoints stay); $e$ is a **bridge** iff $G - e$ is disconnected.

A **vertex cut set** is a set $S \subseteq V$ whose removal, with all incident edges, disconnects $G$. A **minimum vertex cut set** is one of the smallest possible size.

THEOREM **Bridges and cut-points · AY22/23 Q23(c), (d)**

- An edge is a bridge **iff it lies on no circuit**. (On a circuit: removing it keeps $G$ connected, Lemma 10.5.3. On no circuit: if $G-e$ were connected, a path between the endpoints of $e$ together with $e$ would be a circuit.)
- **In a tree with 2 or more vertices every edge is a bridge** (a tree has no circuits): Q23(d) **true**.
- **In a tree every vertex of degree greater than 1 is a cut-point**: Q23(c) **true**. Two neighbours of $v$ are joined by the unique path through $v$ (§50), so they are separated in $T-v$. A leaf is never a cut-point.
- $K_n$ has no vertex cut set. In $C_n$ ($n \geq 4$) no single vertex is a cut-point; two non-adjacent vertices form a minimum vertex cut set.
- The neighbours of a vertex $v$ form a vertex cut set whenever other vertices remain, so a minimum vertex cut set has at most (minimum degree) vertices. Search sizes $1, 2, \ldots$ in turn.




PAST PAPER **AY24/25 Q25(d) · Minimum vertex cut set**

No single vertex disconnects the graph, so the minimum size is 2. The left part $\{A, B, C, G\}$ is joined to the rest only by $AF$, $CD$ and $GD$.
$\boldsymbol{\{A, D\}}$ separates $\{B, C, G\}$ from $\{E, F, H, I\}$; $\boldsymbol{\{D, F\}}$ separates $\{A, B, C, G\}$ from $\{E, H, I\}$; $\boldsymbol{\{D, I\}}$ isolates $H$, whose only neighbours are $D$ and $I$. Any one of these three is a correct answer, and there are no others of size 2.

PAST PAPER **Edge counts and components · AY23/24 Q19, Q20, Q24(c)(iii), Q24(d)**

**Extremes.** A simple graph with $n$ vertices and $k$ connected components has between $n-k$ edges (a forest) and $\binom{n-k+1}{2}$ edges ($K_{n-k+1}$ with $k-1$ isolated vertices). So a disconnected simple graph has at most $\binom{n-1}{2}$ edges, and $\binom{n-1}{2}+1$ edges force connectedness.
**Q24(d).** $G$ is connected on $n$ vertices ($n$ even); removing one edge leaves two complete graphs $K_a$ and $K_{n-a}$. $|E| = \binom{a}{2} + \binom{n-a}{2} + 1$, least when $a = n/2$: $\boldsymbol{n(n-2)/4+1}$ (check $n=4$: 3, the path).
**Q24(c)(iii).** "Every vertex of degree $n-2$ is possible for all $n > 3$": **false**. The total degree $n(n-2)$ must be even, so $n$ must be even (then $K_n$ minus a perfect matching works).
**Q19, Q20.** $v_i$ and $v_j$ are joined iff $5 \mid (i+j)$. Indices with residues 1 and 4 (mod 5) form one component, residues 2 and 3 another, residue 0 a third. At most 3 components; the residue-0 component has $\lfloor n/5 \rfloor$ vertices; the other two differ in size by at most 1: **all of (i), (ii), (iii)**. If $5 \mid n$ the sizes are $2n/5$, $2n/5$, $n/5$: **two components with $2n/5$ vertices each**.

## 48 · Adjacency matrices and counting walks

DEFINITION **Adjacency matrix (L12 s61–62) · Matrix product (L12 s64) · Identity, powers (L12 s69–70)**

For a graph $G$ with ordered vertices $v_1, \ldots, v_n$, the **adjacency matrix** is the $n \times n$ matrix $A = (a_{ij})$ over the non-negative integers with

- **directed:** $a_{ij}$ = the number of arrows from $v_i$ to $v_j$;
- **undirected:** $a_{ij}$ = the number of edges connecting $v_i$ and $v_j$.

The matrix of an undirected graph is **symmetric**: $a_{ij} = a_{ji}$.
For an $m \times k$ matrix $A$ and a $k \times n$ matrix $B$, the **product** $AB = (c_{ij})$ has $c_{ij} = \sum_{r=1}^{k} a_{ir} b_{rj}$ (row $i$ of $A$ times column $j$ of $B$). Matrix multiplication is associative but **not commutative**.
$I_n$ is the $n \times n$ **identity matrix** (1 on the main diagonal, 0 elsewhere). **Powers:** $A^0 = I$ and $A^n = AA^{n-1}$ for $n \geq 1$.

THEOREM **Theorem 10.3.2 · Counting walks (L12 s74)**

If $A$ is the adjacency matrix of a graph with vertices $v_1, \ldots, v_m$, then for every positive integer $n$ and all $i, j$:

the $ij$-th entry of $A^n$ = the number of walks of length $n$ from $v_i$ to $v_j$.

WORKED EXAMPLE **L12 s72–73 · Walks of length 2**

$G$ has vertices $v_1, v_2, v_3$ and edges $e_1 = \{v_1, v_2\}$, a loop $e_2$ at $v_2$, and parallel edges $e_3, e_4 = \{v_2, v_3\}$.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 6 & 2 \\ 2 & 2 & 4 \end{pmatrix}.$$

Entry $(2,2)$ of $A^2$ is $1 \cdot 1 + 1 \cdot 1 + 2 \cdot 2 = 6$: one walk via $v_1$ ($v_2 e_1 v_1 e_1 v_2$), one via the loop ($v_2 e_2 v_2 e_2 v_2$), and four via $v_3$ (choose $e_3$ or $e_4$ each way). A walk may repeat edges, and different parallel edges give different walks.

**Reading a matrix:** a non-zero diagonal entry is a loop; an entry greater than 1 means parallel edges; in a simple graph the row sum of $v_i$ is $\deg(v_i)$, and the diagonal of $A^2$ lists the degrees.

METHOD **Walks of length 3 or 4 without computing every entry**
Rows are "from", columns are "to"; a digraph's matrix need not be symmetric, and a loop puts 1 on the diagonal. Compute $A^2$ in full, then only the entry wanted:

$$(A^3)_{ij} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } A^2), \qquad (A^4)_{ij} = (\text{row } i \text{ of } A^2) \cdot (\text{column } j \text{ of } A^2).$$

PAST PAPER **AY25/26 Q20 · AY24/25 Q20 · AY23/24 Q24(b) · AY20/21 Q12 · Counting walks in a digraph**
**AY25/26 Q20.** Arrows $a \to c$, $b \to a$, $b \to b$, $c \to a$, $c \to b$ (rows and columns in the order $a, b, c$):

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad A^4 = (A^2)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 3 & 3 & 2 \\ 3 & 3 & 2 \end{pmatrix}.$$

Walks of length 4: $a$ to $c$: 1; $b$ to $b$: 3; $c$ to $a$: 3; $c$ to $b$: 3. The least of all nine entries is ***a* to *c*** (option A).
**AY24/25 Q20.** Arrows $a \to a$, $a \to b$, $a \to c$, $b \to a$, $b \to c$, $c \to a$:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad A^4 = \begin{pmatrix} 13 & 6 & 9 \\ 9 & 4 & 6 \\ 6 & 3 & 4 \end{pmatrix}.$$

$a$ to $b$: 6; $b$ to $b$: 4; $c$ to $a$: 6; $c$ to $b$: 3, the least entry: ***c* to *b*** (option D).
**AY23/24 Q24(b).** $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$, $A^2 = \begin{pmatrix} 5 & 4 & 4 \\ 4 & 7 & 6 \\ 2 & 3 & 6 \end{pmatrix}$. Walks of length 4 from $v_1$ to $v_3$: $(5,4,4) \cdot (4,6,6)$ $= 20 + 24 + 24 =$ **68**.
**AY20/21 Q12.** Rows of $A$: $(1,1,0,1)$, $(0,0,1,1)$, $(0,0,1,1)$, $(0,1,0,1)$. Column 4 of $A^2$ is $(3,2,2,2)$, so walks of length 3 from vertex 1 to vertex 4: $(1,1,0,1) \cdot (3,2,2,2) =$ **7**.

## 49 · Isomorphism and planar graphs

DEFINITION **Isomorphic graphs (L12 s79)**
$G = (V_G, E_G)$ is **isomorphic** to $G' = (V_{G'}, E_{G'})$, written $G \cong G'$, iff there are bijections $g : V_G \to V_{G'}$ and $h : E_G \to E_{G'}$ that preserve the edge-endpoint functions: for all $v \in V_G$ and $e \in E_G$,

$$v \text{ is an endpoint of } e \Leftrightarrow g(v) \text{ is an endpoint of } h(e).$$

**For simple graphs:** $G \cong G'$ iff there is a bijection $\pi : V_G \to V_{G'}$ such that $\{u, v\} \in E_G$ $\Leftrightarrow \{\pi(u), \pi(v)\} \in E_{G'}$.

THEOREM **Theorem 10.4.1 (L12 s81) · Invariants**
Graph isomorphism is an equivalence relation on any set of graphs. (Reflexive: identity maps. Symmetric: inverse bijections. Transitive: compose the bijections.)
Isomorphic graphs agree on every structural property: number of vertices, number of edges, the list of degrees, number of connected components, number of circuits of each length, being bipartite, being Eulerian or Hamiltonian.

METHOD **Isomorphic or not?**
**To prove isomorphic:** write the bijection on vertices explicitly and check that every edge maps to an edge (and the edge counts agree).
**To prove not isomorphic:** name one invariant that differs. Compare in this order: vertices, edges, sorted degree list, then circuits of length 3 or 4, then what the vertices of a given degree are adjacent to. Equal degree lists do **not** prove isomorphism.

DEFINITION **Planar graph (L12 s82) · Subdivision (L12 s84)**
A **planar graph** is a graph that **can be** drawn on a plane without edges crossing. One drawing with crossings does not show a graph is non-planar. A planar drawing divides the plane into **faces** (regions), including the outer one.
A **subdivision** of $G$ is obtained by inserting new vertices into edges of $G$ (replace an edge by a path through a new vertex).

THEOREM **Kuratowski's theorem (L12 s83) · Euler's formula (L12 s86)**
- $K_5$ and $K_{3,3}$ are not planar. A finite graph is planar **iff** it has no subgraph that is a subdivision of $K_5$ or of $K_{3,3}$.
- For a **connected planar** simple graph with $v$ vertices, $e$ edges and $f$ faces: $f = e - v + 2$.

Examples: $K_4$ is planar with $v = 4$, $e = 6$, $f = 4$. $K_{3,4}$ is not planar, since it contains $K_{3,3}$ (AY25 Q25(a)). A tree has $f = (v-1) - v + 2 = 1$ face.

THEOREM **Consequences of Euler's formula (AY22/23 Q18 · AY23/24 Q18 · AY24/25 Q22)**
Let $G$ be a **simple connected planar** graph with $v \geq 3$ vertices and $e$ edges.
- $e \leq 3v - 6$. If $G$ has no triangles (for example $G$ bipartite): $e \leq 2v - 4$.
- **Every simple planar graph has a vertex of degree at most 5.**
- $K_5$: $e = 10 > 9$. $K_{3,3}$: no triangles and $e = 9 > 8$. Both are non-planar. The inequalities are necessary only: the Petersen graph has $15 \leq 24$ edges and is not planar.
- **Planar:** $K_n$ iff $n \leq 4$; $K_{m,n}$ iff $\min(m,n) \leq 2$. $\boldsymbol{K_{2,n}}$ **is planar for every** $\boldsymbol{n}$**:** put the $n$ vertices in a column and the other two on its left and right. $\boldsymbol{K_5}$ **minus one edge is planar:** draw $K_4$ as a triangle with a centre and join a fifth vertex, placed outside, to the three corners; only the edge from centre to outside is absent.
- With $c$ connected components the formula becomes $f = e - v + c + 1$.

PROOF $e \leq 3v - 6$ · $e \leq 2v - 4$ · **A vertex of degree at most 5**
1. Every face is bounded by at least 3 edges *(simple, $v \geq 3$)* and every edge borders at most 2 faces, so $3f \leq 2e$.
2. Substitute $f = e - v + 2$ *(Euler's formula)*: $3(e - v + 2) \leq 2e$, that is $e \leq 3v - 6$.
3. With no triangles every face has at least 4 edges: $4f \leq 2e$, so $2(e - v + 2) \leq e$, that is $e \leq 2v - 4$.
4. Suppose every vertex of a simple planar graph had degree at least 6; pick a connected component (it has $v \geq 7$). Then $2e \geq 6v$ *(handshake theorem)*, so $e \geq 3v > 3v - 6$: contradiction with line 2. ■

PAST PAPER **AY22/23 Q18 · AY23/24 Q18 · Planarity true or false**
**AY22/23 Q18.** (i) $K_5 - \{e\}$ is planar: **true**. (ii) $K_5$ is Eulerian and Hamiltonian: **true** (connected, every degree 4; $K_n$ is Hamiltonian for $n \geq 3$). (iii) A simple planar graph with 6 vertices, 8 edges and 7 faces: **false**. $f = e - v + c + 1 = 3 + c$ needs $c = 4$ components, but 6 vertices in 4 components carry at most $\binom{3}{2} = 3$ edges. Answer: only (i) and (ii).
**AY23/24 Q18.** (i) Two spanning trees of a graph always share an edge: **false** (§52). (ii) Every simple planar graph has a vertex of degree at most 5: **true**.

PAST PAPER **T11 Q1 · Self-complementary graphs on $n = 3, 4, 5, 6$ vertices**
If $G \cong \overline{G}$ then $G$ and $\overline{G}$ have the same number of edges, so $\binom{n}{2}$ is even. For $n = 3$ and $n = 6$, $\binom{n}{2} = 3$ and 15: **none**.
$n = 4$ (3 edges): the path on 4 vertices, and only that. $n = 5$ (5 edges): the cycle $C_5$, and the "bull" (a triangle with a pendant edge at two of its vertices): **two**.

THEOREM **Self-complementary graphs need $n \equiv 0$ or $1 \pmod 4$ · AY20/21 Q21(b), (c)**
A self-complementary simple graph on $n$ vertices has exactly $n(n-1)/4$ edges. Hence $n \equiv 0$ or $1 \pmod 4$; in particular there is **no** self-complementary graph with $4k + 2$ (or $4k + 3$) vertices.

PROOF **Edge count of a self-complementary graph**
1. Let $G \cong \overline{G}$ with $n$ vertices. Isomorphic graphs have the same number of edges, so $|E| = |\overline{E}|$ *(Theorem 10.4.1, invariants)*.
2. Each of the $\binom{n}{2}$ pairs of vertices is an edge of exactly one of $G$, $\overline{G}$: $|E| + |\overline{E}| = n(n-1)/2$. So $|E| = n(n-1)/4$.
3. $|E|$ is an integer, so $4 \mid n(n-1)$. One of $n$, $n-1$ is odd, so 4 divides the other: $n \equiv 0$ or $1 \pmod 4$.
4. For $n = 4k + 2$: $n(n-1)/2 = (2k+1)(4k+1)$ is odd, so it cannot be split into two equal parts. No such graph exists. ■

WORKED EXAMPLE **Small enumerations up to isomorphism · AY22/23 Q20 · AY24/25 Q21 · AY20/21 Q20(d)**
- **Simple graphs on $\boldsymbol{n = 1, 2, 3, 4, 5}$ vertices:** 1, 2, 4, 11, 34 classes; connected ones: 1, 1, 2, 6, 21.
- **The 11 on 4 vertices, by number of edges** $0, 1, \ldots, 6$: $1, 1, 2, 3, 2, 1, 1$ (the list is symmetric under taking complements). With 3 edges: path, star, triangle plus isolated vertex.
- **AY22/23 Q20.** Disconnected simple graphs on 4 vertices: $11 - 6 = \mathbf{5}$ (no edge; one edge; two disjoint edges; two adjacent edges plus an isolated vertex; triangle plus an isolated vertex).

- **AY24/25 Q21.** 25 labelled simple graphs on 3 vertices fall into 4 isomorphism classes (0, 1, 2 or 3 edges), so at least $\lceil 25/4 \rceil = \mathbf{7}$ are isomorphic to one another (generalized pigeonhole, §41).
- **AY20/21 Q20(d).** Directed graphs on 3 labelled vertices, no parallel edges: each of the 9 ordered pairs (loops included) is present or absent, $2^9 = \mathbf{512}$. With at least one loop: $512 - 2^6 = \mathbf{448}$.

## 50 · Trees

DEFINITION **Circuit-free, tree, trivial tree, forest (L13 s4) · Leaf, internal vertex (L13 s9)**

A graph is **circuit-free** iff it has no circuits. A simple graph is a **tree** iff it is **circuit-free and connected**. A **trivial tree** is a single vertex. A simple graph is a **forest** iff it is circuit-free and not connected.

In a tree with one or two vertices each vertex is a **terminal vertex** (**leaf**). In a tree with at least three vertices, a vertex of degree 1 is a **leaf** and a vertex of degree greater than 1 is an **internal vertex**.

THEOREM **Lemma 10.5.1 · Theorem 10.5.2 · Lemma 10.5.3 · Theorem 10.5.4 · Lemma 10.5.5 (L13 s7–15, T11)**

- **10.5.1.** Any non-trivial tree has at least one vertex of degree 1 (in fact at least two).
- **10.5.2.** Any tree with $n$ vertices ($n > 0$) has $n - 1$ edges.
- **10.5.3.** If $G$ is connected and $C$ is a circuit in $G$, removing one edge of $C$ leaves $G$ connected.
- **10.5.4.** If $G$ is a **connected** graph with $n$ vertices and $n - 1$ edges, then $G$ is a tree.
- **10.5.5 (T11).** In a simple graph, if there are two distinct paths from $v$ to a different vertex $w$, then $G$ contains a cycle.

Connectedness is needed in 10.5.4: a triangle with a pendant edge, together with one isolated vertex, has 5 vertices and 4 edges but is not a tree (L13 s17).

PROOF **Lemma 10.5.1 (L13 s7) · Theorem 10.5.2 by induction (L13 s10–11) · At least two leaves (L13 s12)**

**10.5.1.** Start at any vertex $v$ and walk along edges, never reusing an edge. A vertex cannot be revisited *(T is circuit-free)*, and $T$ has finitely many vertices, so the walk stops at a vertex with no unused edge: a vertex of degree 1.

**10.5.2.** Let $P(n) \equiv$ "any tree with $n$ vertices has $n - 1$ edges".

1. $P(1)$: a tree with one vertex has no edges *(a simple graph has no loops)*.
2. Assume $P(k)$ for an arbitrary $k \geq 1$. Let $T$ be any tree with $k + 1$ vertices.
3. $T$ is non-trivial, so it has a vertex $v$ of degree 1 *(Lemma 10.5.1)* let $e$ be the edge at $v$.
4. Let $T'$ be $T$ with $v$ and $e$ removed. $T'$ has $k$ vertices, is circuit-free (a subgraph of $T$) and is connected (no path between other vertices passes through $v$, as $\deg(v) = 1$). So $T'$ is a tree.
5. $T'$ has $k - 1$ edges *(inductive hypothesis)*, so $T$ has $k = (k + 1) - 1$ edges. Thus $P(k + 1)$ is true. ■

**Two leaves.** Let $T$ be a tree with $n \geq 2$ vertices. Every degree is at least 1 *(connected)*. If at most one vertex had degree 1, the total degree would be at least $1 + 2(n - 1) = 2n - 1$; but it equals $2(n - 1) = 2n - 2$ *(Theorem 10.5.2 and the handshake theorem)*. Contradiction. ■

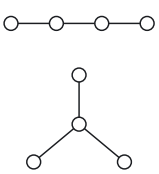
PROOF **Theorem 10.5.4 · A connected graph with $n$ vertices and $n - 1$ edges is a tree (L13 s15–16)**

1. Let $G$ be connected with $n$ vertices and $n - 1$ edges. It suffices to show that $G$ is circuit-free.
2. Suppose $G$ has a circuit. Remove an edge of it; the graph stays connected *(Lemma 10.5.3)*. Repeat while a circuit remains.
3. The result $G''$ is connected and circuit-free, so it is a tree with $n$ vertices and therefore $n - 1$ edges *(Theorem 10.5.2)*.
4. But at least one edge was removed, so $G''$ has at most $n - 2$ edges: contradiction.
5. Hence $G$ is circuit-free, and $G$ is a tree. ■

WORKED EXAMPLE **L13 s13 · T11 Q3 · Trees on few vertices**

A tree with 4 vertices has 3 edges and total degree 6, with at least two leaves: the degree lists are $1, 1, 2, 2$ (a path) and $1, 1, 1, 3$ (a star). So there are **two** non-isomorphic trees with 4 vertices.

**Labelled copies (T11 Q3):** $n = 1$: 1; $n = 2$: 1; $n = 3$: one tree (the path), 3 copies (choose the middle vertex); $n = 4$: the path has $4!/2 = 12$ copies and the star has 4 (choose the centre), 16 in all. In general there are $n^{n-2}$ labelled trees.



WORKED EXAMPLE **Counting trees · W12 s51 · AY20/21 Q21(e)**

- **Non-isomorphic trees on $n = 1, ..., 6$ vertices:** $1, 1, 1, 2, 3, 6$.
- **5 vertices (W12 s51):** 4 edges, total degree 8, at least two leaves; the other three degrees sum to 6: $\{1, 1, 4\}$ (star), $\{1, 2, 3\}$, $\{2, 2, 2\}$ (path). **3 trees.**
- **Rooted trees** (an isomorphism must send root to root) on $n = 1, ..., 5$ vertices: $1, 1, 2, 4, 9$. **AY20/21 Q21(e), 5 vertices:** delete the root and list the sizes of the subtrees hanging from it. Sizes $\{4\}$: 4 trees (one per rooted tree on 4 vertices); $\{3, 1\}$: 2; $\{2, 2\}$: 1; $\{2, 1, 1\}$: 1; $\{1, 1, 1, 1\}$: 1. Total **9**.
- **Forest:** with $c$ components, $e = v - c$. So "no cycles and $e = v - 1$" already forces a tree (AY22/23 Q19(i), §47).

PROOF **T11 Q4–Q6 · Edge bounds and the unique-path characterization**

Let $G = (V, E)$ be a simple graph.

1. **Connected $\Rightarrow |E| \geq |V| - 1$.** While $G$ has a circuit, remove one of its edges; $G$ stays connected *(Lemma 10.5.3)*. The process ends with a connected circuit-free graph on $V$, a tree with $|V| - 1$ edges *(Theorem 10.5.2)*, all of which are edges of $G$. **Converse false:** a triangle plus an isolated vertex has $3 \geq 4 - 1$ edges and is not connected.
2. **Acyclic $\Rightarrow |E| \leq |V| - 1$.** Each connected component is a tree. With $c \geq 1$ components of sizes $n_1, ..., n_c$, $|E| = \sum(n_i - 1) = |V| - c \leq |V| - 1$. **Converse false:** the same graph has $3 \leq 3$ edges and contains a cycle.
3. ***G* is a tree iff there is exactly one path between every pair of distinct vertices.**
   ($\Rightarrow$) $G$ is connected, so any two distinct vertices are joined by a path *(Lemma 10.2.1(a))*. Two distinct paths between them would give a cycle *(Lemma 10.5.5)*, contradicting circuit-freeness.
   ($\Leftarrow$) A path between every pair makes $G$ connected. If $G$ had a circuit, two vertices on it would be joined by two distinct paths (the two ways round), contradicting uniqueness. So $G$ is circuit-free and connected. ■

PAST PAPER **T11 Q7 (AY17/18 exam) · Splitting a pile of $n$ stones**

Splitting a pile of $k$ into $k_1$ and $k_2$ earns $k_1 k_2$ dollars. Model the stones as the vertices of $K_n$, where an edge means "still in the same pile". A split into $k_1$ and $k_2$ deletes exactly the $k_1 k_2$ edges between the two parts and earns one dollar per deleted edge. At the end every stone is alone, so all $\binom{n}{2}$ edges have been deleted exactly once. The total is $\boldsymbol{n(n-1)/2}$ **whatever the strategy**.

## 51 · Rooted trees and binary trees

DEFINITION **Rooted tree, level, height · Child, parent, sibling, ancestor, descendant (L13 s19)**

A **rooted tree** is a tree with one distinguished vertex, the **root**. The **level** of a vertex is the number of edges on the unique path between it and the root (the root has level 0). The **height** of a rooted tree is the maximum level of any vertex.

The **children** of $v$ are the vertices adjacent to $v$ and one level farther from the root; then $v$ is their **parent**; distinct vertices with the same parent are **siblings**. If $v$ lies on the unique path between $w$ and the root ($v \neq w$), then $v$ is an **ancestor** of $w$ and $w$ a **descendant** of $v$.

DEFINITION **Binary tree, full binary tree · Left and right subtree (L13 s22)**

A **binary tree** is a rooted tree in which every parent has at most two children, each designated a **left child** or a **right child** (not both), with at most one of each. A **full binary tree** is a binary tree in which each parent has **exactly two** children.

The **left subtree** of a parent $v$ is the binary tree whose root is the left child of $v$ and whose vertices are that child and all its descendants; the **right subtree** is defined analogously.

THEOREM **Theorem 10.6.1 Full binary tree theorem (L13 s27) · Theorem 10.6.2 (L13 s29)**

- **10.6.1.** A full binary tree with $k$ internal vertices has $2k + 1$ vertices in total and $k + 1$ leaves.
- **10.6.2.** A binary tree of height $h$ with $t$ leaves satisfies $t \leq 2^h$; equivalently $\log_2 t \leq h$.

Hence: no full binary tree has 10 internal vertices and 13 leaves (it would have 11 leaves); no binary tree of height 5 has 38 leaves ($38 > 32$). A binary tree of height $h$ has at most $2^{h+1} - 1$ vertices.

PROOF **Theorem 10.6.1 (L13 s27) · Theorem 10.6.2 by strong induction (L13 s30–34)**

**10.6.1.** Every vertex except the root has a parent. Each of the $k$ internal vertices has exactly two children, so $2k$ vertices have a parent. Total: $2k+1$. Leaves: $(2k+1)-k=k+1$. ■

**10.6.2.** Let $P(h) \equiv$ "every binary tree of height $h$ has at most $2^h$ leaves".

1. $P(0)$: the tree is a single vertex, a leaf: $t=1=2^0$.
2. Let $k \geq 0$ and assume $P(i)$ for $0 \leq i \leq k$. Let $T$ have height $k+1$, root $v$ and $t$ leaves. Since $k+1 \geq 1$, $v$ has a child.
3. Case 1 ($v$ has one child): its subtree $T_L$ has height $k$, so $t_L \leq 2^k$ *(inductive hypothesis)*. In the slides the root, having degree 1, is also counted, so $t = t_L + 1 \leq 2^k + 1 \leq 2^k + 2^k = 2^{k+1}$ *($2^k \geq 1$)*.
4. Case 2 ($v$ has two children): the subtrees $T_L$, $T_R$ have heights $h_L, h_R \leq k$, so $t = t_L + t_R \leq 2^{h_L} + 2^{h_R} \leq 2^k + 2^k = 2^{k+1}$ *(inductive hypothesis)*.
5. In both cases $P(k+1)$ is true. ■

DEFINITION **Tree traversal: BFS and DFS (L13 s36–39)**

**Breadth-first search:** visit the root, then the vertices level by level, left to right.

**Depth-first search**, three orders, each applied recursively:

- **Pre-order:** root, left subtree, right subtree.
- **In-order:** left subtree, root, right subtree.
- **Post-order:** left subtree, right subtree, root.

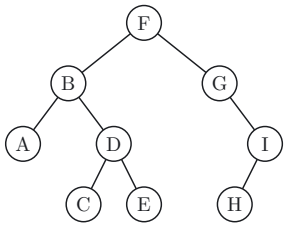
For the tree on the right (L13 s40):

BFS: $F, B, G, A, D, I, C, E, H$

pre-order: $F, B, A, D, C, E, G, I, H$

in-order: $A, B, C, D, E, F, G, H, I$

post-order: $A, C, E, D, B, H, I, G, F$




METHOD **Rebuilding a binary tree from two traversals**

**Pre-order + in-order** (or **post-order + in-order**) determine the tree. The root is the **first** item of the pre-order (the **last** of the post-order). Find it in the in-order: everything to its left is the left subtree, everything to its right is the right subtree. Split the other traversal into blocks of the same sizes and recurse.

**Pre-order + post-order** do not determine the tree: a vertex with exactly one child could have it on the left or the right, so there are $2^s$ trees, where $s$ is the number of one-child vertices.

**Expression trees (L13 s24–26):** operators at internal vertices, operands at leaves; $((a-b)\cdot c)+(d/e)$ has root $+$.

PAST PAPER **T11 Q8, Q9, D3 · AY25 final Q19, Q21, Q22**

**T11 Q8(a).** In-order E A C K F H D B G, pre-order F A E K C D H G B. Root $F$; left part E A C K, right part H D B G. Left subtree: root $A$, left child $E$, right child $K$ with left child $C$. Right subtree: root $D$, left child $H$, right child $G$ with left child $B$. **Post-order: E, C, K, A, H, B, G, D, F.**

**T11 Q8(b).** In-order D B F E A G C H K, post-order D F E B G K H C A. Root $A$; left subtree: root $B$, left child $D$, right child $E$ with left child $F$; right subtree: root $C$, left child $G$, right child $H$ with right child $K$. **Pre-order: A, B, D, E, F, C, G, H, K.**

**T11 Q9.** Binary trees with in-order X Y Z: **5**; with in-order A B C D: **14** (the Catalan numbers: choose the root, then the shapes of the two sides).

**AY25 Q21.** Pre-order G B E D F C A, in-order B G F D C E A: root $G$ with left child $B$; right subtree has root $E$, left child $D$ (children $F$ and $C$), right child $A$. Post-order: B, F, C, D, A, E, G.

**AY25 Q22.** Pre-order A B C D E, post-order C E D B A: $A$ has the single child $B$; $B$ has children $C$ and $D$; $D$ has the single child $E$. Two one-child vertices: $2^2 = 4$ trees.

**AY25 Q19.** Unlabelled rooted binary trees, left and right not distinguished: 2 with 3 vertices, 3 with 4 vertices, **6** with 5 vertices.

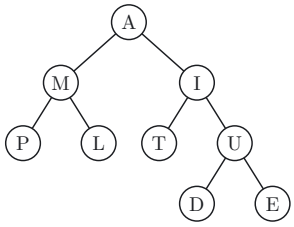
METHOD **Pre-order + post-order · When the tree is unique, and how to rebuild it**

1. The root is the **first** item of the pre-order and the **last** of the post-order.
2. If more items remain, the **second** item $c$ of the pre-order is the root of the first subtree. Find $c$ in the post-order: the items up to and including $c$ form that subtree (say $k$ items); the next $k$ items of the pre-order after the root are the same set.
3. The items left over (before the root in the post-order) form the second subtree. Recurse on each block.
4. **Ambiguity:** if $c$ is the second-last item of the post-order, nothing is left over: the vertex has **one** child, which may be a left or a right child. Each such vertex doubles the count ($2^s$ trees, AY25/26 Q22).

**Unique iff no vertex has exactly one child.** For a **full** binary tree (given, as in AY23/24 Q24(e), or found, as in AY24/25 Q25(g)) step 4 never occurs: the first block is the left subtree and the rest the right subtree.

PAST PAPER **AY24/25 Q25(g) · AY23/24 Q24(e) · In-order from pre-order and post-order**

**AY24/25 Q25(g).** Pre A M P L I T U D E, post P L M T D E U I A. Root $A$. Next is $M$; in the post-order $M$ ends the block P L M, so the left subtree is $\{M, P, L\}$ and the right is $\{I, T, U, D, E\}$.
Left (pre M P L, post P L M): $M$ with children $P$, $L$. Right (pre I T U D E, post T D E U I): $I$ with left child $T$ (block T) and right subtree pre U D E, post D E U: $U$ with children $D$, $E$. No one-child vertex, so the tree is unique. **In-order: P M L A T I D U E.**




**AY23/24 Q24(e)** (full binary tree). Pre S U R E C A N D O I T, post E C R A U O I D T N S. Root $S$; left block E C R A U (root $U$: left $R$ with children $E$, $C$; right $A$); right block O I D T N (root $N$: left $D$ with children $O$, $I$; right $T$). **In-order: E R C U A S O D I N T.**

PAST PAPER **AY22/23 Q23(e) · AY20/21 Q21(d) · Rebuilding with the in-order**

**AY22/23 Q23(e).** In-order E D B C G F A H K, pre-order C B D E A G F H K. Root $C$; left part E D B is the chain $B$, left child $D$, left child $E$. Right part G F A H K: root $A$; left $G$ with right child $F$; right $H$ with right child $K$. **Post-order: E D B F G K H A C.**

**AY20/21 Q21(d).** In-order U C A N D O I T Y E H, post-order C U N A T Y I H E O D. Root $D$. Left part U C A N: root $A$, left child $U$ (with right child $C$), right child $N$. Right part O I T Y E H: root $O$ with no left child and right child $E$; $E$ has left child $I$ and right child $H$; $I$ has right child $Y$; $Y$ has left child $T$. (Pre-order: D A U C N O E I Y T H.)

DEFINITION **Binary search tree (AY23/24 Q24(f)) · Height-balanced binary tree (W13 s7)**

A **binary search tree** (BST) is a binary tree with distinct values at its vertices such that, for every vertex with value $x$, all values in its left subtree are smaller than $x$ and all values in its right subtree are greater than $x$. Its in-order traversal lists the values in increasing order.

A **height-balanced** binary tree is a binary tree in which, at every vertex, the heights of the left and right subtrees differ by at most 1 (an absent subtree has height $-1$).

THEOREM **Catalan numbers count binary trees · Minimum size of a height-balanced tree**

- The number of BSTs on $n$ given values = the number of binary trees with a given in-order = the number of binary tree shapes with $n$ vertices = the **Catalan number**

$$C_n = \frac{(2n)!}{(n+1)!\,n!} = \frac{1}{n+1}\binom{2n}{n}: \quad 1, 2, 5, 14, 42, 132, \ldots \quad (n = 1, 2, 3, \ldots).$$

  Choosing the root leaves $i$ values on the left and $n-1-i$ on the right: $C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$, $C_0 = 1$.
- **AY23/24 Q24(f):** BSTs with 2, 3, 4 values: **2, 5, 14** (compare T11 Q9).
- **Height-balanced, minimum number of vertices $N(h)$ for height $h$:** a root, a subtree of height $h-1$ and one of height $h-2$: $N(h) = 1 + N(h-1) + N(h-2)$ with $N(-1) = 0$, $N(0) = 1$, giving $1, 2, 4, 7, 12, 20$. Height 4 needs **12** vertices (AY22/23 Sem 2 Q20, W13 s7).
- **AY23/24 Q24(g)(i):** "a binary tree has $k$ leaves and $k+1$ internal vertices" is **false** as a general claim; a full binary tree has it the other way round ($k$ internal, $k+1$ leaves).

## 52 · Spanning trees and minimum spanning trees

**DEFINITION Spanning tree (L13 s45) · Weighted graph, minimum spanning tree (L13 s48)**

A **spanning tree** for a graph $G$ is a subgraph of $G$ that contains every vertex of $G$ and is a tree.
A **weighted graph** is a graph in which each edge $e$ has a positive real **weight** $w(e)$; the **total weight** $w(G)$ is the sum of the weights of all edges. A **minimum spanning tree** (MST) for a connected weighted graph is a spanning tree with the least possible total weight.

**THEOREM Proposition 10.7.1 (L13 s45)**

(1) Every connected graph has a spanning tree. (2) Any two spanning trees for a graph have the same number of edges ($n-1$ for $n$ vertices).
**Counting spanning trees:** if $G$ has exactly one circuit, of length $k$, removing any one of its edges gives a spanning tree: $k$ spanning trees (L13 s46: 3). $K_n$ has $n^{n-2}$.

**THEOREM An edge lies in every spanning tree iff it is a bridge (AY23/24 Q24(g)(iii))**

Let $e$ be an edge of a connected graph $G$. Then $e$ is contained in every spanning tree of $G$ iff removing $e$ disconnects $G$.

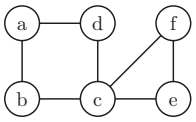
**PROOF Bridges and spanning trees**

1. ($\Leftarrow$) Let $e$ be a bridge and suppose a spanning tree $T$ does not contain $e$. Then $T$ is a connected subgraph of $G-e$ containing every vertex, so $G-e$ is connected: contradiction.
2. ($\Rightarrow$, contrapositive) If $e$ is not a bridge, $G-e$ is connected, so it has a spanning tree $T$ *(Proposition 10.7.1)*. $T$ contains every vertex of $G$, so it is a spanning tree of $G$ that does not contain $e$. ■

**METHOD Counting spanning trees when there is more than one circuit (T11 Q2 · L13 s46)**

A spanning tree keeps $n-1$ edges: delete $e-(n-1)$ edges so that every circuit is broken and the graph stays connected. Bridges are never deleted.

- **Circuits sharing no edge** (they meet only at vertices): delete one edge from each, independently, and **multiply** the circuit lengths.
  **T11 Q2** (right): the 4-circuit $adcb$ and the triangle $cfe$ share only the vertex $c$: $4 \times 3 =$ **12**.
- **Two circuits sharing an edge** (or a path): split into cases. Shared edge deleted: one large circuit is left, delete any one of its edges. Shared edge kept: delete one other edge from each circuit.
  A 4-circuit and a triangle sharing an edge: $5 + 3 \times 2 = 11$. $C_4$ plus a chord: $4 + 2 \times 2 = 8$.
- Check on a known value: $K_n$ has $n^{n-2}$ ($K_4$: 16).




**METHOD Algorithm 10.7.1 · Kruskal (L13 s50)**

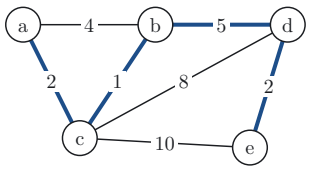
Input: a connected weighted graph $G$ with $n$ vertices.
1. Initialize $T$ with all vertices of $G$ and no edges.
2. Let $E$ be the set of all edges of $G$; let $m = 0$.
3. While $m < n-1$:
   3.1. find an edge $e$ in $E$ of least weight;
   3.2. delete $e$ from $E$;
   3.3. if adding $e$ to $T$ does not produce a circuit, add $e$ to $T$ and set $m = m+1$.

Output: $T$, a minimum spanning tree for $G$.

**METHOD Algorithm 10.7.2 · Prim (L13 s55)**

Input: a connected weighted graph $G$ with $n$ vertices.
1. Pick a vertex $v$; let $T$ be the graph with this vertex only.
2. Let $V$ be the set of all vertices of $G$ except $v$.
3. For $i = 1$ to $n-1$:
   3.1. find an edge $e$ of least weight among those connecting $T$ to a vertex in $V$; let $w$ be its endpoint in $V$;
   3.2. add $e$ and $w$ to $T$, and delete $w$ from $V$.

Output: $T$, a minimum spanning tree for $G$.




**WORKED EXAMPLE Kruskal and Prim on one graph**

Edges: $ab = 4$, $ac = 2$, $bc = 1$, $bd = 5$, $cd = 8$, $ce = 10$, $de = 2$.
**Kruskal:** $bc$ (1) added; $ac$ (2) added; $de$ (2) added; $ab$ (4) rejected, since it closes the circuit $abc$; $bd$ (5) added. Four edges: stop.
**Prim from $a$:** add $ac$ (2), then $cb$ (1), then $bd$ (5), then $de$ (2).

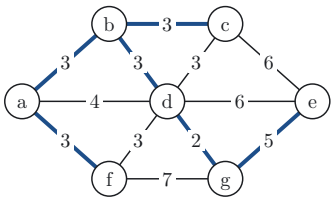
Both give the same tree, of weight $1+2+2+5=10$, with the edges added in different orders.

WORKED EXAMPLE **L13 s52, s57 · The airline graph (8 cities)**
**Kruskal:** Chi–Mil 74, Lou–Cin 83, Lou–Nas 151, Cin–Det 230, StL–Lou 242, StL–Chi 262 added; Chi–Lou 269, Lou–Det 306, Lou–Mil 348 **not added** (each would close a circuit); Min–Chi 355 added. Total weight **1397**.
**Prim from Minneapolis:** Min–Chi 355, Chi–Mil 74, Chi–StL 262, StL–Lou 242, Lou–Cin 83, Lou–Nas 151, Cin–Det 230. The same tree, total 1397.

TRAP **Minimum spanning trees**
- Kruskal may hold a **disconnected** forest midway; Prim's partial tree is always connected. Marking a Prim step that does not touch the current tree is an error.
- With repeated weights there may be several MSTs, but all have the same total weight; with all weights distinct the MST is unique. To count MSTs, look at each tie and see which choices are interchangeable.
- An MST is not a shortest-path tree. Dijkstra's algorithm (L13 s59 onward) is not examinable this semester.
- Stop after $n-1$ edges. If a question asks for the order of edges added, follow the named algorithm exactly.

METHOD **Counting minimum spanning trees when weights tie**
Run Kruskal one weight class at a time. For each weight, look at the edges of that weight that do not close a circuit with **lighter** edges, and count the ways to choose among them; ties in different parts of the graph are independent, so **multiply**. A tie where every tied edge is rejected, or every tied edge is accepted, gives no choice.
Equivalently: fix the forced edges, then count the spanning trees of what is left of the tied part.

PAST PAPER **AY24/25 Q25(e) · W12 s60 (AY20/21 Sem 2 Q21b) · Number of MSTs**
**AY24/25 Q25(e).** Friendship graph $F_6$; the triangles (spoke, spoke, rim) have weights $(3,3,3)$, $(9,9,3)$, $(10,10,7)$, $(6,3,3)$, $(9,3,6)$, $(9,3,6)$. The triangles share only the centre, so each one independently drops one heaviest edge and keeps the other two.
(i) Weight $6+12+17+6+9+9=$ **59**. (ii) Choices of the dropped edge: $3\times2\times2\times1\times1\times1=$ **12** MSTs.
**W12 s60.** Edges $ab=1$, $bc=1$, $ae=2$, $be=2$, $cd=2$, $bd=2$, a second $bd=3$, $de=3$. Both edges of weight 1 are taken. Then $e$ is reached by $ae$ or $be$ (not both: circuit $abe$) and $d$ by $cd$ or the lighter $bd$ (not both: circuit $bcd$): $2\times2=$ **4** MSTs, each of weight 6.




PAST PAPER **T11 Q10 (AY23/24 Sem 2 exam) · Weight and number of MSTs**
Vertices are labelled here for reference. **Kruskal:** $dg$ (2) is taken. Weight 3: the six edges $ab, bc, bd, cd, af, fd$ form a 4-circuit $abdf$ and a triangle $bcd$ sharing $bd$; any spanning tree of this part (4 edges) is taken. Then $ad$ (4) is rejected and $ge$ (5) completes the tree; the edges of weight 6 and 7 are never used.
**(a)** Weight $2+4\times3+5=$ **19**.
**(b)** The tied part has $5+3\times2=11$ spanning trees, so there are 11 MSTs on the labelled graph. Up to isomorphism there are **5**: degree lists $1,1,1,1,2,2,4$; $1,1,1,1,2,3,3$; $1,1,2,2,2,2,2$ (a path); and two different trees with $1,1,1,2,2,2,3$ (longest path of 4 edges, or of 5 edges). One MST is shown in bold.

PAST PAPER **AY20/21 Q21(a) · AY23/24 Q24(a), Q24(g)(ii), Q18(i) · More on spanning trees**
**AY20/21 Q21(a).** Edges $ab$ 62, $bc$ 80, $ad$ 61, $bd$ 29, $be$ 37, $bg$ 37, $ce$ 70, $df$ 45, $dg$ 32, $eg$ 9, $eh$ 13, $fg$ 38, $gh$ 12. Kruskal takes $\{e,g\}$ 9, $\{g,h\}$ 12, $\{b,d\}$ 29, $\{d,g\}$ 32, $\{f,g\}$ 38, $\{a,d\}$ 61, $\{c,e\}$ 70 and rejects 13, both 37s (each closes a circuit), 45, 62, 80. **Weight 251**; the MST is unique in spite of the tie.
**AY23/24 Q24(a).** From the weight matrix: $w_{12}=w_{14}=w_{24}=3$, $w_{23}=w_{34}=4$, every edge at vertex 5 weighs at least 5 ($w_{15}=w_{25}=w_{45}=5$), $w_{35}=6$, $w_{13}=8$. Kruskal: two edges of weight 3, one of weight 4, one of weight 5. **Weight 15.**
**AY23/24 Q24(g)(ii).** With all weights distinct the MST is unique, so Prim and Kruskal give the same tree: **true**.
**AY23/24 Q18(i).** "Two spanning trees of a graph always share an edge": **false**. In $K_4$ the paths $1,2,3,4$ and $2,4,1,3$ are spanning trees with no common edge.

# Part XI · Exam Playbook: additions

METHOD **Final exam format and approach (AY25/26 paper)**
**Format:** 2 hours, open book, calculators allowed, 100 marks. Part A: 23 MCQ at 2 marks each (46). Part B: three structured questions (54), recently Counting and Probability, Graphs and Trees, and Functions; many parts need the answer only, "as a single number".

- **Coverage is the whole module.** Roughly the first third of Part A is logic, sets and relations (§1–25); the rest is functions, induction and recursion, cardinality, counting, graphs and trees (§26–52).
- **New definitions in the question** (matchings, dominating sets, order of a function, left inverses) are routine: copy the definition onto scratch paper and test it on the smallest example.
- **Counting answers:** decide ordered or unordered, repetition or not (table in §56), then check the formula on a case small enough to list.
- **"Single number" answers** get no method marks: recompute by a second route (difference rule against direct cases).
- **Proof parts:** use the class format for induction (§32), the injective/surjective templates (§28), and name the theorem used at each step.

## 56 · Formula sheet for Lectures 7–13

| Counting | Value |
|---|---|
| integers from $m$ to $n$ | $n-m+1$ |
| ordered, repetition allowed | $n^k$ |
| ordered, no repetition | $P(n,k)=n!/(n-k)!$ |
| unordered, no repetition | $\binom{n}{k}=n!/(k!(n-k)!)$ |
| unordered, repetition allowed | $\binom{k+n-1}{k}$ |
| orderings with repeated types | $n!/(n_1!\cdots n_k!)$ |
| circular arrangements of $n$ | $(n-1)!$ |
| $x_1+\cdots+x_n=r$, $x_i\geq 0$ | $\binom{r+n-1}{r}$ |
| same, $x_i\geq 1$ | $\binom{r-1}{n-1}$ |
| functions $A\to B$, $\lvert A\rvert=m$, $\lvert B\rvert=n$ | $n^m$ |
| injections; bijections ($m=n$) | $P(n,m)$; $n!$ |
| subsets of an $n$-set | $2^n$ |
| pigeons: some hole has at least | $\lceil n/m\rceil$ |

| Sums | Closed form |
|---|---|
| $1+2+\cdots+n$ | $n(n+1)/2$ |
| $1^2+2^2+\cdots+n^2$ | $n(n+1)(2n+1)/6$ |
| $\sum_{i=0}^{n} r^i$ $(r\neq 1)$ | $(r^{n+1}-1)/(r-1)$ |
| $\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$ | $(a+b)^n$ |
| $\sum_{k=0}^{n}\binom{n}{k}$ | $2^n$ |

| Probability | Formula |
|---|---|
| equally likely outcomes | $P(E)=\lvert E\rvert\ /\ \lvert S\rvert$ |
| complement | $P(\overline{A})=1-P(A)$ |
| union | $P(A)+P(B)-P(A\cap B)$ |
| conditional | $P(B\mid A)=P(A\cap B)/P(A)$ |
| independent | $P(A\cap B)=P(A)P(B)$ |
| expected value | $\sum a_k p_k$ |
| linearity | $E[X+Y]=E[X]+E[Y]$ |
| binomial | $\binom{n}{k}p^k(1-p)^{n-k}$ |

| Graphs and trees | Fact |
|---|---|
| total degree | $2\lvert E\rvert$ |
| edges of $K_n$; of $K_{m,n}$ | $n(n-1)/2$; $mn$ |
| simple graphs on $n$ labelled vertices | $2^{n(n-1)/2}$ |
| Euler circuit | connected, all degrees even |
| Euler trail $v$ to $w$ | connected, only $v, w$ odd |
| walks of length $n$ from $v_i$ to $v_j$ | entry $ij$ of $A^n$ |
| connected planar | $f=e-v+2$ |
| tree on $n$ vertices | $n-1$ edges, at least 2 leaves |
| full binary tree, $k$ internal | $2k+1$ vertices, $k+1$ leaves |
| binary tree of height $h$ | at most $2^h$ leaves |
| labelled trees on $n$ vertices | $n^{n-2}$ |

**Function properties under composition (§29).** $f, g$ injective $\Rightarrow$ $g\circ f$ injective; $f, g$ surjective $\Rightarrow$ $g\circ f$ surjective; $g\circ f$ injective $\Rightarrow$ $f$ injective; $g\circ f$ surjective $\Rightarrow$ $g$ surjective; $(g\circ f)^{-1}=f^{-1}\circ g^{-1}$. **Finite sets (§35):** injection $A\to B$ gives $\lvert A\rvert\leq\lvert B\rvert$; surjection gives $\lvert A\rvert\geq\lvert B\rvert$; bijection gives $\lvert A\rvert=\lvert B\rvert$. For $\lvert A\rvert=\lvert B\rvert$ finite, injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective.

## 58 · Question locators for Tutorials 6–11 and the final

| Tutorial question | Topic | Solved in |
|---|---|---|
| T6 D1, D3, D4; Q1, Q2 | Is it a function; equality; a wrong proof; images and preimages | §26, §27, §29, §26 |
| T6 Q3, Q6 | Inverse of a composition; $g\circ f$ injective | §29 |
| T6 Q4, Q5 | Classify four functions; left and right inverses | §28 |
| T6 Q7, Q8; T8 Q10(c) | Order of a bijection and of an input | §29 |
| T6 Q9, Q10; T8 Q10(a), (b) | $f^{-1}(f(X))$; well-defined operations on a quotient | §26, §30 |

| | | |
|---|---|---|
| T7 Q1, D2, D3 | Mistakes in induction proofs | §32 |
| T7 Q2, Q3, Q4 | Sum of squares; Bernoulli; divisibility by $2^{n+2}$ | §32 |
| T7 Q5, Q6, Q7 | \$3 and \$5 notes; sums of powers of 2; three-term recurrence | §33 |
| T7 Q8, Q9; T8 Q9 | Fibonacci identities; Tower of Hanoi | §33 |
| T7 Q10, Q11 | Membership of recursively defined sets | §34 |
| T8 D1, D2, D3; Q1 | Countable or not; bijection $\mathbb{N} \to \mathbb{Z}$ | §37, §38 |
| T8 Q2, Q3, Q4, Q5 | Unions of countable sets; limits of induction | §37 |
| T8 Q6, Q7, Q8 | $X \cup Y \to X$; $\mathcal{P}(A)$ uncountable; $\lvert A\rvert \leq \lvert R\rvert$ | §38, §38, §35 |
| T9 D1, D4; Q1, Q2, Q5 | Replacement; birthdays; cards; cheque numbers; round tables | §45, §45, §39 |
| T9 Q3, Q4, Q6, Q7 | Inclusion/exclusion; consecutive boxes; committees; investments | §40, §42 |
| T9 Q8, Q9; T10 Q10, Q12 | Pigeonhole arguments | §41 |
| T10 D1, Q3, Q4; Q5, Q6 | Expected value; Bayes; random relations | §44, §45 |
| T10 D2, D3, Q1, Q2 | Counting functions; binomial term; choosing block ends | §43, §42 |
| T10 Q7, Q8; T11 D1, D2 | Friendship graph; small graphs; edge counts | §47, §46 |
| T11 Q1, Q3, Q4–Q7 | Self-complementary graphs; trees; edge bounds; stones | §49, §50 |
| T11 D3, Q8, Q9 | Traversals and rebuilding binary trees | §51 |
| T10 Q11; T11 Q2, Q10 | Dominating sets; counting spanning trees and MSTs | §46; §52 |

| **AY25/26 final question** | **Solved in** |
|---|---|
| Q9; Q10; Q11, Q12, Q26(b) | §47; §43; §29 |
| Q13, Q17, Q18; Q14–Q16 | §38; §34 |
| Q19, Q21, Q22; Q23 | §51; §44 |
| Q24(a)–(e), (g); Q24(f); Q24(h) | §42; §41; §45 |
| Q25(a); Q25(b)–(d) | §49; §47 |
| Q20; Q25(e)–(h) | §48; §46 |

The folder has no official answers for the AY25/26 final; those solutions are worked here.

| **Earlier final, graphs and trees** | **Topic** | **Solved in** |
|---|---|---|
| AY24/25 Q20; Q21; Q22 | Walks of length 4; isomorphism classes; bipartite and planar facts | §48; §49; §47 |
| AY24/25 Q25(a)–(c), (f); (d); (e); (g) | Chromatic number and index, Petersen graph; vertex cut set; number of MSTs; pre-order + post-order | §46; §47; §52; §51 |
| AY23/24 Q18; Q19, Q20 | Planar degree bound, disjoint spanning trees; components | §49, §52; §47 |
| AY23/24 Q24(a), (g)(ii), (g)(iii); (b) | MST from a weight matrix, bridges; walks of length 4 | §52; §48 |
| AY23/24 Q24(c)(i), (c)(iii), (d); (e), (f), (g)(i) | Euler trail after deleting an edge, edge counts; full binary tree, BSTs | §47; §51 |
| AY22/23 Q18; Q19; Q20 | Planarity; forests and bipartite cycles; disconnected graphs on 4 vertices | §49; §47; §49 |
| AY22/23 Q23(a), (b); (c), (d); (e); (f)–(i) | Euler trail; cut-points and bridges; traversals; packing as colouring | §47; §47; §51; §46 |
| AY20/21 Q12; Q20(d); Q21(a) | Walks of length 3; counting digraphs; MST | §48; §49; §52 |
| AY20/21 Q21(b), (c); (d); (e) | Self-complementary graphs; rebuilding a tree; rooted trees on 5 vertices | §49; §51; §50 |

AY24/25 Q25(a)–(f) follows the answers in the Week 12 recitation; the other solutions are worked here.

## 60 · Self-test for Lectures 7–13

Decide each statement before reading the reason.

| **Statement** | | **Reason** |
|---|---|---|
| $f^{-1}(B)$ is defined only when $f$ is a bijection | **F** | the setwise preimage exists for every function (§26) |
| $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 3x + 1$ is a bijection | **F** | injective, but 0 has no preimage |

| | | |
|---|---|---|
| $g \circ f$ injective $\Rightarrow$ $g$ injective | **F** | only $f$ must be injective (§29) |
| the composition of two bijections is a bijection | **T** | Theorems 7.3.3 and 7.3.4 |
| $f : \mathbb{Z} \to \mathbb{Z}$ and $g : \mathbb{Z} \to \mathbb{Q}$ with the same formula are equal | **F** | different co-domains (Theorem 7.1.1) |
| $X = f^{-1}(f(X))$ for every $X \subseteq A$ | **F** | only $\subseteq$; equality needs injectivity |
| $[x]_3 \mapsto [x]_2$ is a function $\mathbb{Z}_3 \to \mathbb{Z}_2$ | **F** | $[0]_3 = [3]_3$ but $[0]_2 \neq [3]_2$ |
| $2n + 1 < 2^n$ for all $n \geq 1$ | **F** | fails at $n = 1, 2$; true from $n = 3$ |
| strong induction proves more statements than weak induction | **F** | 1PI, 2PI and WOP are equivalent |
| every non-empty subset of $\mathbb{Z}$ has a least element | **F** | WOP is about $\mathbb{Z}_{\geq 0}$; take $\mathbb{Z}^-$ |
| an empty sum is 0 and an empty product is 1 | **T** | conventions (§31) |
| $\lvert\mathbb{Z}\rvert = \lvert 2\mathbb{Z}\rvert$ | **T** | $n \mapsto 2n$ is a bijection |
| every infinite subset of $\mathbb{R}$ is uncountable | **F** | $\mathbb{Z}$, $\mathbb{Q}$ |
| every subset of $\mathbb{Q}$ is countable | **T** | Theorem 7.4.3 |
| a countable union of finite sets can be infinite | **T** | $\bigcup\{i\} = \mathbb{Z}^+$; still countable |
| $\mathcal{P}(\mathbb{N})$ is countable | **F** | diagonal argument (§38) |
| $P(n, r) = \binom{n}{r} \cdot r!$ | **T** | choose a subset, then order it |
| the number of multisets of size 3 from 4 elements is $\binom{4}{3}$ | **F** | $\binom{6}{3} = 20$ |
| $\binom{n}{r} = \binom{n}{n-r}$ | **T** | choose which to leave out |
| disjoint events are independent | **F** | if both have positive probability they are dependent |
| $E[X + Y] = E[X] + E[Y]$ needs independence | **F** | linearity always holds |
| $P(A \mid B) = P(B \mid A)$ | **F** | equal only if $P(A) = P(B)$ or $P(A \cap B) = 0$ |
| a graph can have exactly three vertices of odd degree | **F** | Proposition 10.1.3 |
| all degrees even $\Rightarrow$ the graph has an Euler circuit | **F** | it must also be connected |
| every Eulerian graph is Hamiltonian | **F** | friendship graph $F_2$ |
| a graph with $n$ vertices and $n - 1$ edges is a tree | **F** | only if connected (Theorem 10.5.4) |
| $K_{3,3}$ is planar | **F** | Kuratowski; $K_5$ is not planar either |
| equal degree lists $\Rightarrow$ isomorphic | **F** | $C_6$ and two disjoint triangles |
| pre-order and post-order determine a binary tree | **F** | a one-child vertex is ambiguous (§51) |
| a full binary tree has an odd number of vertices | **T** | $2k + 1$ |
| all spanning trees of a graph have the same number of edges | **T** | Proposition 10.7.1 |

## 62 · Symbols for Lectures 7–13

| Symbol | Read as | Symbol | Read as | Symbol | Read as |
|---|---|---|---|---|---|
| $f : X \to Y$ | function from $X$ to $Y$ | $x \mapsto y$ | $x$ maps to $y$ | $f(A)$ | setwise image |
| $f^{-1}(B)$ | setwise preimage | $f^{-1}$ | inverse function | $g \circ f$ | $f$ then $g$ |
| $\mathrm{id}_X$ | identity on $X$ | $A^*$ | strings over $A$ | $\varepsilon$ | empty string |
| $\mathbb{Z}_n$ | $\mathbb{Z}$ mod $n$ (L7); $\{1, ..., n\}$ (L9) | $\lfloor x \rfloor, \lceil x \rceil$ | floor, ceiling | $n!$ | $n$ factorial |
| $\sum_{k=m}^{n} a_k$ | sum of $a_m, ..., a_n$ | $\prod_{k=m}^{n} a_k$ | product | $F_n$ | Fibonacci number |
| $\aleph_0$ | $\lvert\mathbb{Z}^+\rvert$, aleph-zero | $\lvert A\rvert = \lvert B\rvert$ | same cardinality | $P(n, r)$ | $r$-permutations |
| $\binom{n}{r}$ | $n$ choose $r$ | $P(E)$ | probability of $E$ | $P(B \mid A)$ | $B$ given $A$ |
| $E[X]$ | expected value | $G = (V, E)$ | graph | $\deg(v)$ | degree |
| $\deg^-(v), \deg^+(v)$ | indegree, outdegree | $K_n$ | complete graph | $K_{m,n}$ | complete bipartite graph |
| $\overline{G}$ | complement graph | $G \cong G'$ | isomorphic | $A^n$ | $n$-th matrix power |
| $w(e), w(G)$ | edge weight, total weight | $C_n$ | cycle graph | $I_n$ | identity matrix |

## 65 · Theorem locator · Lectures 7–13

Theorems are stated with their proofs or definitions in the listed sections. Labels match the slides. **L** = lecture; **s** = slide.

| Slide label | Topic | Source | Section |
|---|---|---|---|
| Theorem 7.1.1 | Function equality | L7 s22 | §27 |
| Proposition (unnumbered); Theorem 7.2.3 | Unique inverse; bijective iff invertible | L7 s34–36 | §28 |
| Theorems 7.3.1, 7.3.2 | Composition with identity; with the inverse | L7 s44, 46 | §29 |
| Theorem (unnumbered); Theorems 7.3.3, 7.3.4 | Associativity; composing injections, surjections | L7 s47, 50, 52 | §29 |
| Propositions (unnumbered) | $+$ and $\cdot$ on $\mathbb{Z}_n$ are well defined | L7 s57–58 | §30 |
| Theorem 5.1.1 | Properties of sums and products | L8 s14 | §31 |
| Theorem 5.2.2; Theorem 5.2.3 | Sum of first $n$ integers; geometric sum | L8 s26, 28 | §32 |
| Proposition 5.3.1; Proposition 5.3.2 | $3 \mid (2^{2n}-1)$; $2n+1 < 2^n$ | L8 s29–30 | §32 |
| Well-Ordering Principle | Non-empty subsets of $\mathbb{Z}_{\geq 0}$ have a least element | L8 s43–44 | §33 |
| Pigeonhole; Dual Pigeonhole | Injections and surjections between finite sets | L9 s6–8 | §35 |
| Theorem Cardinality.1 | Subsets of finite sets are finite | L9 s11 | §35 |
| Theorem 7.4.1 | Same cardinality is an equivalence relation | L9 s14–16 | §36 |
| Proposition 9.1; Lemma 9.2 | Countability via sequences | L9 s33–34 | §37 |
| Theorem 7.4.2 | $(0,1)$ is uncountable | L9 s38–40 | §38 |
| Theorem 7.4.3; Corollary 7.4.4 | Subsets of countable sets; uncountable subsets | L9 s41–43 | §37 |
| Proposition 9.3; Lemma 9.4 | Countably infinite subset; union of two | L9 s44–45 | §37 |
| Theorem 9.1.1; Theorem 9.2.1 | Elements of a list; multiplication rule | L10 s12, 21 | §39 |
| Theorem 9.2.2; Theorem 9.2.3 | Permutations; $r$-permutations | L10 s34, 37 | §39 |
| Theorems 9.3.1, 9.3.2, 9.3.3 | Addition, difference, inclusion/exclusion | L10 s40, 43, 50 | §40 |
| Generalized Pigeonhole Principle | Some value has at least $k+1$ preimages | L10 s62, 65 | §41 |
| Theorem 9.5.1; Theorem 9.5.2 | Formula for $\binom{n}{r}$; repeated objects | L11 s10, 15 | §42 |
| Theorem 9.6.1 | Multisets | L11 s21 | §42 |
| Theorem 9.7.1; Theorem 9.7.2 | Pascal's formula; binomial theorem | L11 s26, 34 | §43 |
| Theorem 9.9.1 | Bayes' theorem | L11 s66 | §45 |
| Theorem 10.1.1; Corollary 10.1.2; Proposition 10.1.3 | Handshake theorem and consequences | L12 s25 | §46 |
| Lemma 10.2.1 | Paths, circuits and connectedness | L12 s37 | §47 |
| Theorems 10.2.2–10.2.4; Corollary 10.2.5 | Euler circuits and trails | L12 s41–43 | §47 |
| Proposition 10.2.6 | Necessary conditions for a Hamiltonian circuit | L12 s52 | §47 |
| Theorem 10.3.2 | Powers of the adjacency matrix count walks | L12 s74 | §48 |
| Theorem 10.4.1; Kuratowski; Euler's formula | Isomorphism; planarity | L12 s81, 83, 86 | §49 |
| Lemma 10.5.1; Theorem 10.5.2 | Leaves; a tree has $n-1$ edges | L13 s7, 10 | §50 |
| Lemma 10.5.3; Theorem 10.5.4; Lemma 10.5.5 | Removing a circuit edge; converse; two paths | L13 s14–15; T11 | §50 |
| Theorem 10.6.1; Theorem 10.6.2 | Full binary trees; leaves and height | L13 s27, 29 | §51 |
| Proposition 10.7.1; Algorithms 10.7.1, 10.7.2 | Spanning trees; Kruskal; Prim | L13 s45, 50, 55 | §52 |